

Notes on

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REGRESSION ANALYSIS AND ECONOMETRICS

M.Sc Statistics

PART II

Econometrics

It is a combination of economics and statistics.

- i. Models of economics are expressed in terms of mathematical equation i.e. equations of the specification.
- ii. Identification of the model in mathematical form.

Regression Analysis

iii. Estimation: We estimate the parameters of the model.

iv. Stat. Inference: Finally we carry out tests on the estimated model in an attempt to judge whether it constitutes a realistic picture of the world.

Econometric refers to application of economic, mathematical and statistical technique for the purpose of testing the hypothesis, estimating and forecasting economic phenomena.

Econometric is subdivided into theoretical and applied econometrics. Theoretical econometric refers to the methods of measurement of economic relationship in general and applied econometrics examines the problems and the findings in particular fields of economics, such as demand theory, production, investment, consumption, and other fields of applied economic research.

Functions: Econometrics has basically three closely interrelated functions. The first is to test economic theories or hypothesis. For example: 'Is consumption directly related to income?', 'quantity demanded of a commodity inversely related to price?'

The second function of econometrics is provide numerical estimates of the coefficients of

economic relationship. These are essential in decision making.

3. The third function of econometrics is the forecasting of economic events or variables, this is necessary in order for policy making makers to take appropriate decisions.

Contribution of Economics in Econometrics:-

Econometric presupposes the existence of a body of economic theories or hypotheses requiring testing. If the variable suggested by economic theory do not provide a satisfactory explanation, the researcher may experiment with alternative formulations and variables suggested by previous tests or opposing theories. In this way econometric research can lead to the acceptance, rejection and reformulation of theories.

Contribution of Mathematics in Econometrics:-

Mathematics is used to express the statements of economic theories in mathematical forms expressing an exact or deterministic functional relationship between the dependent and one or more independent or explanatory variables.

Contribution of Statistics in Econometrics:-

Statistical analysis applies appropriate techniques to estimate the relationship among economic variables by utilizing relevant economic data and evaluating the results.

Exact (Price, Demand) Inexact Relationship

$$D = a + bP$$

$$+ \epsilon I + \delta Y + \Delta W$$

$$D = a + bP + E$$

Error term

Justification for the Inclusion of a Error term:

The inclusion of a ~~random~~ disturbance or error term will ~~be~~ well defined probabilistic properties is required in regression analysis for certain reasons. Since the purpose of theory is to generalize and simplify economic relationships usually include only the most important variables at work. This means that number of variables with slight and irregular effects are not included. The error term can be ^{supposed} ~~represented~~ as representing the net effect of these small and irregular variables. The inclusion of the error term can be justified in order to take into consideration the net effect of possible errors in measuring the variables. Since personal behaviour usually differs in a random way under identical circumstances. The disturbance or error term can be used to capture this inherently random personal behaviour. The error term thus allows for individual random deviations from the exact and deterministic relationship postulated by economic theory and mathematical economics.

The Principle of Least Squares;

The principle of least squares is that the $\hat{\alpha}$, $\hat{\beta}$ values should be chosen so as to make $\sum \hat{e}$ as small as possible. A necessary condition is that the partial derivatives of the sum with respect to $\hat{\alpha}$, $\hat{\beta}$ should both be zero.

Various Stages of Econometrics

Stage I

Econometric Theory



Mathematical Model



Econometric Model

Stage 2.

Collection of Appropriate data



Estimation of Parameters of the model

Stage 3;

Evaluation of model on the basis of economic, statistical and econometric criteria.



Accept theory if compatible with data.



Prediction (Forecasting)

Reject theory if incompatible with data.

Revise theory if incompatible with data.

Confirm the revised theory with new data.

Simple Linear Regression..

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- i, Simple means only one independent variable or two variables (one independent and one dependent variable)
- ii, Linear means power of independent variable or variables is one.
- iii, Regression means dependence of one variable on one or more than one variables.

Model

Let X and Y are two variables, then

$$Y = \alpha + \beta X + \epsilon$$

where

' X , is independent, explanatory variable, Regressor, exogenous or fixed variable.

' Y , is dependent or explained or regressand, endogenous or stochastic variable.

' ϵ , is error term.

Assumptions:- ' X , is a fixed variable.

' ϵ , is random disturbance term, whose dist is Normal i.e. $N(0, \sigma^2)$

Methods of Estimation

A point estimate of a parameter can be obtained by several methods.

i, Least Square Method:-

If $\hat{\alpha}$ and $\hat{\beta}$ are the estimators of α and β , then Sum of Squares of residuals is

$$SSR = \sum_{i=1}^n [Y_i - (\hat{\alpha} + \hat{\beta} X_i)]^2$$

minimizing w.r.t, $\hat{\alpha}$

$$\frac{\partial SSR}{\partial \hat{\alpha}} = 2 \sum_{i=1}^n [Y_i - \hat{\alpha} - \hat{\beta} X_i] (-1)$$

$$\frac{\partial^2 SSR}{\partial \hat{\alpha}^2} = -2 \sum_{i=1}^n (-1) > 0 \quad \therefore \frac{\partial SSR}{\partial \hat{\alpha}} = 0$$

$$\sum_{i=1}^n Y_i = n \hat{\alpha} + \hat{\beta} \sum_{i=1}^n X_i \quad \text{--- (1)}$$

This is Normal eq. of $\hat{\alpha}$

Dividing by n

$$\bar{Y} = \hat{\alpha} + \hat{\beta} \bar{X} \implies \hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

Now minimizing w.r.t $\hat{\beta}$

$$\frac{\partial SSR}{\partial \hat{\beta}} = 2 \sum_{i=1}^n [Y_i - \hat{\alpha} - \hat{\beta} X_i] (-X_i)$$

$$\frac{\partial^2 SSR}{\partial \hat{\beta}^2} = 2 \sum_{i=1}^n X_i^2 > 0 \quad \therefore \frac{\partial SSR}{\partial \hat{\beta}} = 0$$

$$\sum_{i=1}^n X_i Y_i = \hat{\alpha} \sum_{i=1}^n X_i + \hat{\beta} \sum_{i=1}^n X_i^2 \quad \text{--- (2)}$$

This is NE of $\hat{\beta}$

Putting $\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$ in (2)

$$\sum_{i=1}^n X_i Y_i = (\bar{Y} - \hat{\beta} \bar{X}) \sum_{i=1}^n X_i + \hat{\beta} \sum_{i=1}^n X_i^2$$

$$\sum_{i=1}^n X_i Y_i = \frac{1}{n} \sum_{i=1}^n Y_i \sum_{i=1}^n X_i - \hat{\beta} \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 + \hat{\beta} \sum_{i=1}^n X_i^2$$

$$\sum_{i=1}^n X_i Y_i - \frac{1}{n} \sum_{i=1}^n X_i \sum_{i=1}^n Y_i = \hat{\beta} \left[\sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 \right]$$

$$\hat{\beta} = \frac{\sum_{i=1}^n X_i Y_i - \frac{1}{n} \sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{\sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2}$$

or

$$y = Y - \bar{Y}, \quad x = X - \bar{X}$$

$$\hat{\beta} = \frac{n \sum_{i=1}^n X_i Y_i - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} = \frac{\sum xy}{\sum x^2}$$

1. Maximum Likelihood Estimators:-

$$Y_i = \alpha + \beta X_i + \epsilon_i$$

$$\epsilon_i \sim N(0, \sigma^2)$$

$$f(\epsilon_i) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}\epsilon_i^2}$$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\epsilon_i - 0)^2}$$

$$f(\epsilon_1, \dots, \epsilon_n) = \prod_{i=1}^n f(\epsilon_i)$$

$$f(\epsilon) = \frac{1}{\sigma^n (2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n \epsilon_i^2}$$

$$f(\epsilon) = \frac{1}{\sigma^n (2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \alpha - \beta X_i)^2}$$

Taking natural log.

$$\ln f(\epsilon) = -n \ln \sigma - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \alpha - \beta X_i)^2 \quad \text{--- (A)}$$

minimizing w.r.t α

$$\frac{\partial \ln f(\epsilon)}{\partial \alpha} = -\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \alpha - \beta X_i) (-2)$$

$$\frac{\partial^2 \ln f(\epsilon)}{\partial \alpha^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (-1) < 0 \quad \text{i.e function is max}$$

$$\therefore \text{put } \frac{\partial \ln f(\epsilon)}{\partial \alpha} = 0 \implies \sum_{i=1}^n (Y_i - \alpha - \beta X_i) = 0$$

$$\sum_{i=1}^n Y_i = n\alpha + \beta \sum_{i=1}^n X_i \quad \text{--- (1)}$$

Now minimizing (A) w.r.t β

$$\frac{\partial \ln f(\epsilon)}{\partial \beta^*} = -\frac{2}{2\sigma^2} \sum_{i=1}^n (Y_i - \alpha^* - \beta^* X_i) (-X_i)$$

$$\frac{\partial^2 \ln f(\epsilon)}{\partial \beta^{*2}} = +\frac{1}{\sigma^2} \sum_{i=1}^n (+X_i^2) < 0 \Rightarrow \text{function is max}$$

put $\frac{\partial \ln f(\epsilon)}{\partial \beta^*} = 0$

$$\Rightarrow \sum_{i=1}^n X_i Y_i = \alpha^* \sum_{i=1}^n X_i + \beta^* \sum_{i=1}^n X_i^2 \quad \text{--- (2)}$$

Now minimizing wrt σ

$$\frac{\partial \ln f(\epsilon)}{\partial \sigma^*} = -\frac{n}{\sigma} - \frac{1}{2} \left(\frac{-2}{\sigma^3} \right) \sum_{i=1}^n (Y_i - \alpha^* - \beta^* X_i)^2$$

$$\frac{\partial^2 \ln f(\epsilon)}{\partial (\sigma^*)^2} = +\frac{n}{\sigma^2} - 3\sigma^{-4} \sum_{i=1}^n (Y_i - \alpha^* - \beta^* X_i)^2 < 0$$

put $\frac{\partial \ln f(\epsilon)}{\partial \sigma^*} = 0 \Rightarrow \frac{n}{(\sigma^*)^*} = \frac{\sum_{i=1}^n (Y_i - \alpha^* - \beta^* X_i)^2}{(\sigma^*)^3}$

$$\sigma^{*2} = \frac{\sum_{i=1}^n (Y_i - \alpha^* - \beta^* X_i)^2}{n}$$

Solving (1) and (2) simultaneously

$$\alpha^* = \bar{Y} - \beta^* \bar{X}$$

and $\beta^* = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$

It means that α^* and β^* are same estimator as least squares estimators.

Again we have

$$f(\epsilon) = (2\pi\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha^* - \beta^* x_i)^2}$$

taking ln

$$\ln f(\epsilon) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha^* - \beta^* x_i)^2$$

minimizing w.r.t. σ^2

$$\frac{\partial \ln f(\epsilon)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} - \frac{1}{2} \frac{(-1)}{(\sigma^2)^2} \sum_{i=1}^n (y_i - \alpha^* - \beta^* x_i)^2$$

$$\frac{\partial^2 \ln f(\epsilon)}{\partial (\sigma^2)^2} = \frac{(-2n)}{2(\sigma^2)^2} - \frac{2}{2} \frac{(-2)}{(\sigma^2)^3} \sum_{i=1}^n (y_i - \alpha^* - \beta^* x_i)^2 < 0$$

$$\therefore \frac{\partial \ln f(\epsilon)}{\partial \sigma^2} = 0$$

$$\frac{n}{2\sigma^2} = \frac{1}{2\sigma^4} \sum_{i=1}^n (y_i - \alpha^* - \beta^* x_i)^2$$

$$\sigma^{*2} = \frac{1}{n} \sum_{i=1}^n (y_i - \alpha^* - \beta^* x_i)^2$$

which is the MLE of error variance

Properties of Estimators

SLR

In simple linear regression model

$$Y = \alpha + \beta X + \epsilon$$

L.S.E of α is $\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$

Least square estimator of β is $\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$

Maximum likelihood Estimator of σ^2 is

$$\sigma^{*2} = \frac{\sum_{i=1}^n e_i^2}{n}$$

where $e = Y - \hat{\alpha} - \hat{\beta} X$

Properties of Least Square Estimator of β

- (1) $\hat{\beta}$ is linear in Y , values.
- (2) It is unbiased estimator of β .

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \frac{1}{\sum_{i=1}^n x_i^2} \sum_{i=1}^n x_i y_i$$

but $Y = \alpha + \beta X + \epsilon$

Hence

$$\hat{\beta} = \frac{1}{\sum_{i=1}^n x_i^2} \sum_{i=1}^n x_i [\alpha + \beta X + \epsilon]$$

$$= \frac{1}{\sum_{i=1}^n x_i^2} \left[\alpha \sum_{i=1}^n x_i + \beta \sum_{i=1}^n x_i X + \sum_{i=1}^n x_i \epsilon \right]$$

$$\therefore \sum_{i=1}^n x_i = \sum (X - \bar{X}) = 0$$

$$\sum x_i X = \sum x_i^2$$

Therefore
$$\hat{\beta} = \frac{\beta \sum x_i^2}{\sum x_i^2} + \frac{\sum x_i \epsilon}{\sum x_i^2}$$

Applying Expectation

$$E(\hat{\beta}) = \beta + \frac{\sum x_i}{\sum x_i^2} E\epsilon \quad \checkmark$$

$$E(\hat{\beta}) = \beta \quad \because E\epsilon = 0$$

$\Rightarrow$ estimate of β is unbiased.

$$(3) \quad \text{Var}(\hat{\beta}) = E[\hat{\beta} - E(\hat{\beta})]^2 = E[\hat{\beta} - \beta]^2$$

but
$$\hat{\beta} - \beta = \frac{\sum_{i=1}^n x_i \epsilon}{\sum_{i=1}^n x_i^2}$$

$$\therefore \text{Var}(\hat{\beta}) = E\left[\frac{\sum x_i \epsilon}{\sum x_i^2}\right]^2$$

$$= E\left[\frac{\sum_{i=1}^n x_i^2 \epsilon_i^2 + \sum_{i \neq j} x_i x_j \epsilon_i \epsilon_j}{\left(\sum_{i=1}^n x_i^2\right)^2}\right]$$

$$= \frac{\sum_{i=1}^n x_i^2 E(\epsilon_i^2)}{\left(\sum_{i=1}^n x_i^2\right)^2} + \frac{\sum_{i \neq j} x_i x_j E(\epsilon_i \epsilon_j)}{\left(\sum_{i=1}^n x_i^2\right)^2} \quad \begin{matrix} E(\epsilon_i \epsilon_j) = 0 \\ E(\epsilon_i^2) = \sigma^2 \end{matrix}$$

$$\boxed{\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum_{i=1}^n x_i^2}}$$

Since $E\epsilon_i \epsilon_j = 0, E\epsilon_i^2 = \sigma^2$

(4) $\hat{\beta}$ has minimum variance among all linear and unbiased estimators.)

BT II BLUE

considering any other estimator say,

(*)
$$B = \sum_{i=1}^n c_i Y$$

where 'c' is some non stochastic function of X values only.

and
$$Y = \alpha + \beta X + \epsilon$$

$$B = \sum_{i=1}^n c_i (\alpha + \beta X_i + \epsilon_i)$$

$$B = \alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i + \sum_{i=1}^n c_i \epsilon_i$$

$$E(B) = \alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i \quad \therefore E\epsilon = 0$$

$$E(B) = \beta \rightarrow \text{if } \sum c_i = 0, \text{ and } \sum c_i X_i = 1$$

$$\text{Now } \text{Var}(B) = E[B - E(B)]^2$$

$$B - E(B) = \alpha \sum c_i + \beta \sum c_i X_i + \sum c_i \epsilon_i - \alpha \sum c_i - \beta \sum c_i X_i$$

$$B - E(B) = \sum c_i \epsilon_i$$

$$\begin{aligned} \text{Var}(B) &= E\left[\sum_{i=1}^n c_i \epsilon_i\right]^2 = E\left[\sum_{i=1}^n c_i^2 \epsilon_i^2 + \sum_{i \neq j} 2 c_i c_j \epsilon_i \epsilon_j\right] \\ &= \sum_{i=1}^n c_i^2 E\epsilon_i^2 + \sum_{i \neq j} 2 c_i c_j E(\epsilon_i \epsilon_j) \end{aligned}$$

$$\text{Var}(B) = \sum c_i^2 \sigma^2$$

By using Lagrange Multiplier

$$F = \sum c_i^2 - 2\lambda \sum c_i - 2\mu (\sum c_i X_i - 1)$$

$$\frac{\partial F}{\partial c_i} = 2c_i - 2\lambda - 2\mu X_i$$

$$\frac{\partial F}{\partial c_i} = 0 \Rightarrow c_i = \lambda + \mu X_i \quad \text{--- (1)}$$

Applying summation in (1)

$$\sum c_i = n\lambda + \mu \sum_{i=1}^n X_i$$

$$n\lambda + \mu \sum_{i=1}^n X_i = 0 \quad \text{--- I} \quad \therefore \sum C_i = 0 \text{ if } E(B) = \beta$$

NE for λ

Again now multiplying Eq (I) by X_i and applying summation

$$\sum C_i X_i = \lambda \sum X_i + \mu \sum X_i^2$$

$$\lambda \sum_{i=1}^n X_i + \mu \sum_{i=1}^n X_i^2 = 1 \quad \text{--- II} \quad \therefore \sum C_i X_i = 1 \text{ if } E(B) = \beta$$

NE for μ

From I

$$n\lambda = -\mu \sum_{i=1}^n X_i$$

$$\lambda = -\mu \bar{X} \quad \text{putting in II}$$

$$-\mu \bar{X} \sum_{i=1}^n X_i + \mu \sum_{i=1}^n X_i^2 = 1$$

~~$$\mu \left[\sum X_i^2 - n \bar{X}^2 \right] = 1$$~~

$$\mu \left[\sum X_i^2 - n \bar{X}^2 \right] = 1 \Rightarrow \mu \sum x^2 = 1$$

$$\mu = \frac{1}{\sum x^2}, \quad \lambda = \frac{-\bar{X}}{\sum x^2}$$

Now From (I)

$$C_i = \frac{-\bar{X}}{\sum x^2} + \frac{X_i}{\sum x^2} = \frac{(X_i - \bar{X})}{\sum x^2}$$

$$C_i = \frac{x_i}{\sum x^2} \quad \text{For this value of } C_i, \text{ variance is minimum.}$$

Thus estimator $B = \sum C_i Y = \sum \frac{x_i}{\sum x^2} Y$

$$B = \frac{\sum x_i y}{\sum x^2} \quad \text{which is least square estimator.}$$

This least square estimator is linear unbiased and

has minimum variance, so it is best linear unbiased estimator

2nd Method

$$\hat{\beta} = \frac{\sum xy}{\sum x^2} = \sum \frac{x_i}{\sum x_i^2} y_i$$

$$\checkmark \hat{\beta} = \sum W_i y_i$$

$$\text{where } W_i = \frac{x_i}{\sum x_i^2}$$

$$\sum W_i = \frac{\sum x_i}{\sum x_i^2} = 0$$

$$\sum W_i^2 = \frac{\sum x_i^2}{(\sum x_i^2)^2} = \frac{1}{\sum x_i^2}$$

$$W_i x_i = \frac{x_i^2}{\sum x_i^2}$$

$$\sum W_i x_i = 1$$

Now

$$V(\hat{\beta}) = \frac{\sigma^2}{\sum x_i^2} = \sum W_i^2 \sigma^2$$

$$\checkmark \boxed{\text{Var}(\hat{\beta}) = \sum W_i^2 \sigma^2}$$

pg # 12

condition

$$B^* = \sum C_i y_i \Rightarrow E(B^*) = \beta \text{ if } \sum C_i = 0, \sum C_i x_i = 1$$

$$\text{Var}(B^*) = \sum C_i^2 \sigma^2 \text{ kga Add } \sum C_i$$

$$C_i = W_i + d_i \quad \text{--- (A)}$$

$$\sum C_i^2 = \sum W_i^2 + \sum d_i^2 + 2 \sum W_i d_i \quad \text{--- (B)}$$

$$\text{And } \sum_{i=1}^n W_i d_i = \frac{\sum_{i=1}^n x_i d_i}{\sum_{i=1}^n x_i^2}$$

$$\therefore W_i = \frac{x_i}{\sum x_i^2}$$

Multiplying (A) by x_i and applying summation

$$\sum_{i=1}^n x_i C_i = \sum_{i=1}^n x_i d_i + \sum_{i=1}^n x_i W_i$$

but $\sum_{i=1}^n C_i x_i = 1$, if B^* is unbiased estimator of β

$$\sum x = \sum (X - \bar{X}) = 0$$

15.

$$\sum w_i x_i = \frac{\sum x_i^2}{\sum x_i^2} = 1$$

therefore

$$\sum_{i=1}^n x_i d_i = 1 - 1 = 0$$

Putting $\sum_{i=1}^n x_i d_i = 0$ in $\sum_{i=1}^n w_i d_i = \frac{\sum_{i=1}^n x_i d_i}{\sum x_i^2} = \frac{0}{\sum x_i^2} = 0$

We have $\sum_{i=1}^n w_i d_i = 0$, putting this value in (3)

$$\sum_{i=1}^n c_i^2 = \sum_{i=1}^n w_i^2 + \sum_{i=1}^n d_i^2$$

$$5 = 3 + 2$$

$$5 > 3$$

$$3 < 5$$

Multiplying by σ^2

$$\sum_{i=1}^n \sigma^2 c_i^2 = \sum_{i=1}^n \sigma^2 w_i^2 + \sum_{i=1}^n \sigma^2 d_i^2$$

Where $V(\hat{\beta}^*) = \sum_{i=1}^n c_i^2 \sigma^2$, $\text{Var}(\hat{\beta}) = \sum w_i^2 \sigma^2$

Hence $\text{Var}(\hat{\beta}^*) = \text{Var}(\hat{\beta}) + \sum_{i=1}^n \sigma^2 d_i^2$

Because $\sum d_i^2 \sigma^2$ is always positive $\therefore \text{Var}(\hat{\beta}^*) > \text{Var}(\hat{\beta})$

$\therefore \text{Var}(\hat{\beta}) < \text{Var}(\hat{\beta}^*)$ if $\boxed{\sum d_i^2 \sigma^2 \neq 0}$

positive

⑤ Distribution of $\hat{\beta}$

Since ϵ is normally and independently distributed, So Y , being the linear function of Normal variable ϵ is again normal variable.

Now again $\hat{\beta}$ being the linear function of Y also follows normal distribution with mean β and variance $\frac{\sigma^2}{\sum_{i=1}^n x_i^2}$.

Properties of Least Square estimator of α .

Remove Watermark Now

The least square estimator of α is

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

where $\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$

$$\hat{\alpha} = \frac{\sum_{i=1}^n y_i}{n} - \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} \bar{x}$$

$$\hat{\alpha} = \sum_{i=1}^n \left[\frac{1}{n} - W_i \bar{x} \right] y_i \quad \text{where } W_i = \frac{x_i}{\sum_{i=1}^n x_i^2}$$

Proof

① $\hat{\alpha}$ is linear in 'Y' values

② $\hat{\alpha}$ is unbiased estimator of α .

we have

$$\hat{\alpha} = \sum_{i=1}^n \left[\frac{1}{n} - W_i \bar{x} \right] y_i$$

and $y_i = \alpha + \beta x_i + \epsilon_i$

$$\hat{\alpha} = \sum_{i=1}^n \left[\frac{1}{n} - W_i \bar{x} \right] (\alpha + \beta x_i + \epsilon_i) \quad \checkmark$$

$$\sum W_i = \frac{\sum x_i}{\sum x_i^2} \quad \boxed{W_i = \frac{x_i}{\sum_{i=1}^n x_i^2}} \quad \checkmark$$

$$\boxed{\sum_{i=1}^n W_i = 0}$$

$$\therefore \sum_{i=1}^n x_i = \sum_{i=1}^n (x_i - \bar{x}) = 0$$

$$\sum W_i x_i = \frac{\sum x_i^2}{\sum x_i^2}$$

$$\boxed{\sum W_i x_i = 1}$$

$$\sum W_i^2 = \frac{\sum x_i^2}{(\sum x_i^2)^2} = \frac{1}{\sum x_i^2}$$

$$\boxed{\sum W_i^2 = \frac{1}{\sum x_i^2}}$$

NEW

$$\hat{\alpha} = \sum_{i=1}^n \frac{1}{n} \alpha + \beta \frac{\sum_{i=1}^n X_i}{n} + \frac{1}{n} \sum_{i=1}^n \epsilon_i + \alpha \sum_{i=1}^n W_i X_i - \beta \sum_{i=1}^n X_i W_i \bar{X} - \bar{X} \sum_{i=1}^n W_i \epsilon_i$$

$$\hat{\alpha} = \alpha + \beta \bar{X} + \frac{\sum_{i=1}^n \epsilon_i}{n} - \alpha \bar{X} - \beta \bar{X} \sum_{i=1}^n W_i X_i - \bar{X} \sum_{i=1}^n W_i \epsilon_i$$

$$\hat{\alpha} = \alpha + \frac{\sum_{i=1}^n \epsilon_i}{n} - \bar{X} \sum_{i=1}^n W_i \epsilon_i \quad \checkmark \quad \because \sum W_i X_i = 1$$

$$E(\hat{\alpha}) = \alpha + \cancel{\frac{\sum \epsilon_i}{n}} - \bar{X} \sum W_i E(\epsilon_i)$$

$$E(\hat{\alpha}) = \alpha \quad \text{Hence } \hat{\alpha} \text{ is an unbiased estimator}$$

Method I

③ $V(\hat{\alpha}) = E[\hat{\alpha} - E(\hat{\alpha})]^2 = E[\hat{\alpha} - \alpha]^2$

putting

$$\hat{\alpha} - \alpha = \frac{\sum \epsilon_i}{n} - \bar{X} \sum W_i \epsilon_i$$

we get

$$Var(\hat{\alpha}) = E\left[\frac{\sum \epsilon_i}{n} - \bar{X} \sum_{i=1}^n \epsilon_i W_i\right]^2$$

$$= \frac{1}{n^2} E\left(\sum_{i=1}^n \epsilon_i\right)^2 + \bar{X}^2 E\left(\sum_{i=1}^n W_i \epsilon_i\right)^2 - \frac{2\bar{X}}{n} E\left(\sum_{i=1}^n \epsilon_i\right)\left(\sum_{i=1}^n W_i \epsilon_i\right)$$

$$= \frac{1}{n^2} E\left[\sum_{i=1}^n \epsilon_i^2 + \sum_{i \neq j} \sum \epsilon_i \epsilon_j\right] + \bar{X}^2 E\left[\sum_{i=1}^n W_i^2 \epsilon_i^2\right]$$

$$+ \sum_{i \neq j} \sum W_i W_j \epsilon_i \epsilon_j - \frac{2\bar{X}}{n} E\left[\sum_{i=1}^n \epsilon_i W_i + \sum_{i \neq j} \sum \epsilon_i \epsilon_j W_i W_j\right]$$

$$= \frac{1}{n^2} (n\sigma^2 + 0) + \bar{X}^2 \left[\frac{\sigma^2}{\sum_{i=1}^n x_i^2} + 0\right] - \frac{2\bar{X}}{n} [\sigma^2(0) + 0]$$

$$Var(\hat{\alpha}) = \frac{n\sigma^2}{n^2} + \bar{X}^2 \frac{\sigma^2}{\sum_{i=1}^n x_i^2}$$

$$\text{Var}(\hat{\alpha}) = \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^n x_i^2} \right] \sigma^2$$

$$= \left[\frac{\frac{n}{n-1} \sum_{i=1}^n x_i^2 + n \bar{X}^2}{n \sum_{i=1}^n x_i^2} \right] \sigma^2$$

$$\text{Var}(\hat{\alpha}) = \frac{\sum X^2}{n \sum x_i^2} \sigma^2$$

$$\begin{aligned} \sum x^2 &= \sum (X - \bar{X})^2 \\ \sum x^2 &= \sum X^2 - n \bar{X}^2 \\ \sum x^2 + n \bar{X}^2 &= \sum X^2 \end{aligned}$$

Method B11

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

$$\text{Var}(\hat{\alpha}) = \text{Var}(\bar{Y}) + \bar{X}^2 \text{Var}(\hat{\beta}) - 2 \bar{X} \text{Cov}(\bar{Y}, \hat{\beta}) \quad \text{--- (1)}$$

We have $Y = \alpha + \beta X + \epsilon$

Var

$$\text{Var}(Y) = \text{Var}(\epsilon) = \sigma^2 \quad \text{--- (2)}$$

Because α, β are constants and X is fixed.

and $\text{Var}(\bar{Y}) = \frac{\sigma^2}{n}, \text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum_{i=1}^n x_i^2} \quad \text{--- (2)}$

$$\text{Cov}(\bar{Y}, \hat{\beta}) = E[\bar{Y} - E(\bar{Y})][\hat{\beta} - E\hat{\beta}] \quad \text{--- (3)}$$

From general model

$$\bar{Y} = \alpha + \beta \bar{X} + \bar{\epsilon}$$

$$E(\bar{Y}) = \alpha + \beta \bar{X} + E\bar{\epsilon}$$

$$E(\bar{Y}) = \alpha + \beta \bar{X} \quad \because E\bar{\epsilon} = 0$$

$$\boxed{\bar{Y} - E(\bar{Y}) = \bar{\epsilon}} \quad \text{--- (4)}$$

Now $\hat{\beta} = \frac{\sum xY}{\sum x^2} = \frac{\sum x(\alpha + \beta X + \epsilon)}{\sum x^2}$

$$\hat{\beta} = \alpha \frac{\sum x}{\sum x^2} + \beta \frac{\sum x^2}{\sum x^2} + \frac{\sum x \epsilon}{\sum x^2}$$

$$\hat{\beta} = \beta + \frac{\sum x \epsilon}{\sum x^2}$$

$$\hat{\beta} - \beta = \frac{\sum x \epsilon}{\sum x^2} \quad \text{--- (3)}$$

putting (4) and (3) in Eq (2)

$$\text{Cov}(\bar{Y}, \hat{\beta}) = E[\bar{\epsilon}] E\left[\frac{\sum x \epsilon}{\sum x^2}\right]$$

$$\text{Cov}(\bar{Y}, \hat{\beta}) = 0 \quad \because E\bar{\epsilon} = 0$$

Hence now from (1)

$$\text{Var}(\hat{\alpha}) = \frac{\sigma^2}{n} + \frac{\bar{x}^2 \sigma^2}{\sum x^2}$$

$$\text{Var}(\hat{\alpha}) = \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum x^2} \right] \sigma^2$$

Theorem

(4) $\hat{\alpha}$ has minimum variance among all linear and unbiased estimators. First Method

38 II, 87 II,

Proof: Let us consider another linear and unbiased estimator
say, $\hat{\alpha}^* = \sum C_i Y_i$ where C_i is some function of X values.

$$\hat{\alpha}^* = \sum_{i=1}^n C_i (\alpha + \beta X_i + \epsilon_i)$$

$$\hat{\alpha}^* = \alpha \sum_{i=1}^n C_i + \beta \sum_{i=1}^n C_i X_i + \sum_{i=1}^n C_i \epsilon_i$$

$$E(\hat{\alpha}^*) = \alpha \quad \text{if } \sum_{i=1}^n C_i = 1, \sum_{i=1}^n C_i X_i = 0$$

Theorem

$\hat{\alpha}$ is
Best
linear
unbiased
estimator

$\hat{\alpha}$ has
minimum
variance
among
linear
unbiased
estimators

$$V(\alpha^*) = \sum_{i=1}^n C_i^2 V(X_i) = \sum_{i=1}^n C_i^2 \sigma^2 = I \quad \text{linear.}$$

let $C_i = \underbrace{\left(\frac{1}{n} - W_i \bar{X}\right)}_{\text{new constant}} + d_i$ where d_i is some non stochastic value.

Squaring and applying summation

$$\begin{aligned} \sum_{i=1}^n C_i^2 &= \sum_{i=1}^n \left(\frac{1}{n} - W_i \bar{X}\right)^2 + \sum_{i=1}^n d_i^2 + 2 \sum_{i=1}^n \left(\frac{1}{n} - W_i \bar{X}\right) d_i \\ &= \frac{n}{n^2} - \bar{X}^2 \sum_{i=1}^n W_i^2 + 2 \bar{X} \sum_{i=1}^n W_i + \sum_{i=1}^n d_i^2 + 2 \frac{\sum_{i=1}^n d_i}{n} - 2 \bar{X} \sum_{i=1}^n W_i d_i \end{aligned}$$

But $\sum_{i=1}^n W_i = 0$, $\sum_{i=1}^n W_i^2 = \frac{1}{\sum_{i=1}^n x_i^2}$

$$\sum_{i=1}^n C_i^2 = \frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^n x_i^2} + \sum_{i=1}^n d_i^2 + 2 \frac{\sum_{i=1}^n d_i}{n} - 2 \bar{X} \sum_{i=1}^n W_i d_i \quad \text{--- III}$$

Now from II Applying Summation

$$\sum Q_i = \frac{n}{n} - \sum W_i \bar{X} + \sum d_i$$

Where $\sum W_i = \frac{\sum x_i}{\sum x_i^2} = 0$

Also we know $\bar{X}$ is unbiased estimator of $\sum Q_i = 1$, $\sum QX = 0$
So

$$1 = 1 = \sum d_i \implies \sum d_i = 0$$

~~and from II multiplying by W_i and applying summation~~

We know

$$\sum W_i d_i = \frac{\sum x_i d_i}{\sum x_i^2} \quad \text{--- IV}$$

Now multiplying equation II by $\bar{X}$ and applying summation

$$\sum x_i d_i = \frac{\sum x_i}{n} - \bar{X} \sum W_i x_i + \sum x_i d_i \quad \text{--- (IV)}$$

but if α^* is an unbiased estimator of α , then

$$\sum C_i = 1, \quad \sum C_i x_i = 0$$

therefore $\sum C_i x_i = \sum C_i x_i - \bar{X} \sum C_i$

$$= 0 - \bar{X} \sum C_i$$

$$\sum C_i x_i = -\bar{X}$$

$$\sum W_i x_i = \frac{\sum x_i^2}{\sum x_i^2} = 1 \quad \text{and} \quad \sum x_i = 0$$

So from Eq. IV we get

$$-\bar{X} = 0 - \bar{X}(1) + \sum x_i d_i$$

$$-\bar{X} + \bar{X} = \sum x_i d_i$$

$$\sum x_i d_i = 0 \quad \text{putting in III}$$

$$\checkmark \quad \sum W_i d_i = 0$$

Now putting $\sum W_i d_i = 0$, $\sum d_i = 0$, in III, we get

$$\sum_{i=1}^n C_i^2 = \frac{1}{n} + \frac{\bar{X}^2}{\sum x_i^2} + \sum d_i^2$$

Multiplying both sides by σ^2

$$\sum C_i^2 \sigma^2 = \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum x_i^2} \right) \sigma^2 + \sum d_i^2 \sigma^2 \quad 5 = 3 + 2$$

but $\text{Var}(\alpha^*) = \sum C_i^2 \sigma^2$

$$\text{Var}(\hat{\alpha}) = \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x_i^2} \right] \sigma^2$$

We get

$$\text{Var}(\alpha^*) = \text{Var}(\hat{\alpha}) + \sum \sigma_i^2 d_i^2 \rightarrow \text{Ke}$$

Since $\sum d_i^2 \sigma^2 \gg 0$

therefore $\text{Var}(\hat{\alpha}) \leq \text{Var}(\alpha^*)$

It means Least Square estimator has minimum variance among all linear and unbiased estimators.

Second Method also after the next question

⑤ Distribution of $\hat{\alpha}$

$$\hat{\alpha} \sim N\left(\alpha, \frac{\sum x_i^2 \sigma^2}{n \sum x_i^2}\right)$$

and $Z = \frac{\hat{\alpha} - \alpha}{\sigma \sqrt{\frac{\sum x_i^2}{n \sum x_i^2}}}$ is Standard Normal Variate.

Joint Dist of $\hat{\alpha}, \hat{\beta}$ also follows bivariate Normal

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = E[(\hat{\alpha} - \alpha)(\hat{\beta} - \beta)]$$

$$\hat{\alpha} - \alpha = \frac{\sum \epsilon_i}{n} - \frac{\sum x_i \epsilon_i}{\sum x_i^2} \bar{x}$$

$$\hat{\beta} - \beta = \frac{\sum x_i \epsilon_i}{\sum x_i^2}$$

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = E\left[\left(\frac{\sum \epsilon_i}{n} - \frac{\sum x_i \epsilon_i}{\sum x_i^2} \bar{x}\right) \left(\frac{\sum x_i \epsilon_i}{\sum x_i^2}\right)\right]$$

$$= E\left[\frac{\sum \epsilon_i^2 x_i + \sum_{i,j} \bar{x} \epsilon_i \epsilon_j x_i x_j}{n \sum x_i^2}\right] - E\left[\bar{x} \left(\frac{\sum x_i \epsilon_i}{\sum x_i^2}\right)^2\right]$$

$$= \frac{\sum x_i^2 \sigma^2 + 0}{n \sum x_i^2} - \bar{x} E\left[\frac{\sum \epsilon_i^2 \sigma^2 + \sum_{i,j} \epsilon_i \epsilon_j x_i x_j}{\left(\sum x_i^2\right)^2}\right]$$

$$\because \sum x_i = 0, E\epsilon_i^2 = \sigma^2, E\epsilon_i \epsilon_j = 0$$

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = -\bar{x} \sigma^2 \frac{\sum x_i^2}{\left(\sum x_i^2\right)^2}$$

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = -\bar{x} \frac{\sigma^2}{\sum x_i^2} = -\bar{x} \cdot \text{var}(\hat{\beta})$$

Write down the pdf for the following cases.

1, Joint distribution of $\hat{\alpha}, \hat{\beta}$,

2, $\hat{\alpha} - \hat{\beta}$

3, $2\hat{\alpha} + 3\hat{\beta}$

4, $c\hat{\alpha}$

5, $2\hat{\beta}$

6, $\hat{\alpha}_1 - \hat{\alpha}_2$ and $\hat{\beta}_1 - \hat{\beta}_2$ where $\hat{\alpha}_1$ and $\hat{\beta}_1$ are computed from n_1 pairs (x_{1i}, y_{1i}) and $\hat{\alpha}_2, \hat{\beta}_2$ are computed from another independent n_2 pairs (x_{2i}, y_{2i})

Sol: First we calculate $E(\hat{\alpha}), E(\hat{\beta}), \text{Var}(\hat{\alpha}), \text{Var}(\hat{\beta}), \text{Cov}(\hat{\alpha}, \hat{\beta}), \rho_{\hat{\alpha}, \hat{\beta}}$.

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

$$\hat{\beta} = \frac{\sum xy}{\sum x^2} \quad \text{where } x = x - \bar{x}, y = y - \bar{y}$$

$$\hat{\beta} = \frac{\sum xy}{\sum x^2}$$

From General regression model, $y = \alpha + \beta x + \epsilon$

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i (\alpha + \beta x_i + \epsilon_i)}{\sum_{i=1}^n x_i^2} \quad \sum x_i x_i = \sum x_i^2$$

$$\hat{\beta} = \frac{1}{\sum x_i^2} \left[\alpha \sum_{i=1}^n x_i + \beta \sum_{i=1}^n x_i^2 + \sum_{i=1}^n x_i \epsilon_i \right]$$

Because $E(\epsilon_i) = 0, \sum x_i = \sum (x_i - \bar{x}) = 0$

$$E(\hat{\beta}) = \frac{\beta}{\sum x_i^2} \sum x_i^2 + \frac{\sum x_i E(\epsilon_i)}{\sum x_i^2}$$

$$E(\epsilon_i) = 0$$

$$E(\hat{\beta}) = \beta$$

$$V(\hat{\beta}) = E[\hat{\beta} - E(\hat{\beta})]^2 = E[\hat{\beta} - \beta]^2$$

$$\hat{\beta} = \beta + \frac{\sum x_i \epsilon_i}{\sum x_i^2}$$

$$\hat{\beta} - \beta = \frac{\sum x_i \epsilon_i}{\sum x_i^2}$$

Therefore

$$\begin{aligned} \text{Var}(\hat{\beta}) &= E\left[\frac{\sum x_i \epsilon_i}{\sum x_i^2}\right]^2 \\ &= E\left[\frac{\sum_{i=1}^n x_i^2 \epsilon_i^2 + \sum_{i \neq j} x_i x_j \epsilon_i \epsilon_j}{(\sum x_i^2)^2}\right] \end{aligned}$$

$$= \frac{\sum_{i=1}^n x_i^2 E\epsilon_i^2}{(\sum x_i^2)^2} + \frac{\sum_{i \neq j} x_i x_j E\epsilon_i \epsilon_j}{(\sum x_i^2)^2}$$

$$\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum x_i^2} \quad \because E\epsilon_i^2 = \sigma^2, E(\epsilon_i) = 0$$

$$\text{Var}(\hat{\beta}) = \sigma_{\hat{\beta}}^2 = \frac{\sigma^2}{\sum x_i^2}$$

Now

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

$$= \frac{\sum Y}{n} - \frac{\sum xY}{\sum x^2} \bar{X}$$

$$\hat{\alpha} = \sum_{i=1}^n \left[\frac{1}{n} - W_i \bar{X} \right] Y$$

$$\hat{\alpha} = \sum_{i=1}^n \left[\frac{1}{n} - W_i \bar{X} \right] (\alpha + \beta x_i + \epsilon_i) \quad \left| \begin{array}{l} \text{Where } W_i = \frac{x_i}{\sum x_i^2} \\ \sum W_i = \frac{\sum x_i}{\sum x_i^2} = 0 \end{array} \right.$$

$$\begin{aligned} \hat{\alpha} &= \frac{1}{n} \sum \alpha + \beta \frac{\sum x_i}{n} + \frac{1}{n} \sum \epsilon_i \\ &\quad - \alpha \sum W_i \bar{X} - \beta \sum W_i x_i \bar{X} \\ &\quad - \bar{X} \sum W_i \epsilon_i \end{aligned}$$

$$\sum W_i x_i = 1$$

$$\sum W_i^2 = \frac{1}{\sum x_i^2}$$

$$\hat{\alpha} = \alpha + \beta \bar{X} + \frac{\sum \epsilon_i}{n} - \alpha \bar{X}(0) - \beta \bar{X}(1) - \bar{X} \sum W_i \epsilon_i$$

$$\hat{\alpha} = \alpha + \frac{\sum \epsilon_i}{n} - \bar{X} \sum W_i \epsilon_i$$

$$E(\hat{\alpha}) = \alpha + E\epsilon_i - \bar{X} \sum W_i E\epsilon_i$$

$$\therefore E\epsilon_i = 0, \sum W_i = 0$$

$$E(\hat{\alpha}) = \alpha$$

$$\text{Var}(\hat{\alpha}) = E[\hat{\alpha} - E(\hat{\alpha})]^2 = E[\hat{\alpha} - \alpha]^2$$

where $\hat{\alpha} - \alpha = \frac{\sum \epsilon_i}{n} - \bar{X} \sum W_i \epsilon_i$

$$\begin{aligned} \text{Var}(\hat{\alpha}) &= E\left[\frac{\sum \epsilon_i}{n} - \bar{X} \sum W_i \epsilon_i\right]^2 \\ &= \frac{1}{n^2} E\left(\sum \epsilon_i\right)^2 + \bar{X}^2 E\left(\sum W_i \epsilon_i\right)^2 - \frac{2\bar{X}}{n} E\left(\sum \epsilon_i\right)\left(\sum W_i \epsilon_i\right) \\ &= \frac{1}{n^2} E\left[\sum \epsilon_i^2 + \sum_{i \neq j} \epsilon_i \epsilon_j\right] + \bar{X}^2 E\left[\sum W_i^2 \epsilon_i^2 + \sum_{i \neq j} W_i W_j \epsilon_i \epsilon_j\right] \\ &\quad - \frac{2\bar{X}}{n} E\left[\sum \epsilon_i^2 W_i + \sum \epsilon_i \epsilon_j W_j\right] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{n^2} [n\sigma^2 + 0] + \bar{X}^2 [\sum W_i^2 \sigma^2 + 0] \\ &\quad - \frac{2\bar{X}}{n} [E(W_i \sigma^2) + 0] \end{aligned}$$

$$\text{Var}(\hat{\alpha}) = \frac{\sigma^2}{n} + \frac{\bar{X}^2 \sigma^2}{\sum x_i^2}$$

$$\sigma_{\hat{\alpha}}^2 = \text{Var}(\hat{\alpha}) = \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x_i^2} \right] \sigma^2$$

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = E[\hat{\alpha} - \alpha][\hat{\beta} - \beta]$$

$$\hat{\alpha} - \alpha = \frac{\sum \epsilon_i}{n} - \bar{x} \sum w_i \epsilon_i$$

$$\hat{\beta} - \beta = \frac{\sum x_i \epsilon_i}{\sum x_i^2}$$

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = E\left[\frac{\sum \epsilon_i}{n} - \bar{x} \sum w_i \epsilon_i\right] \left[\frac{\sum x_i \epsilon_i}{\sum x_i^2}\right]$$

$$= E\left(\frac{\sum \epsilon_i}{n}\right) \left(\frac{\sum x_i \epsilon_i}{\sum x_i^2}\right) - \bar{x} E\left(\frac{\sum x_i \epsilon_i}{\sum x_i^2}\right)^2$$

$$= E\left[\frac{\sum \epsilon_i^2 x_i + \sum_{i \neq j} \sum x_i \epsilon_i \epsilon_j}{n \sum x_i^2}\right] - \bar{x} E\left[\frac{\sum x_i^2 \epsilon_i^2 + \sum_{i \neq j} 2 x_i x_j \epsilon_i \epsilon_j}{(\sum x_i^2)^2}\right]$$

$$= \frac{\sigma^2 \sum x_i^2 + \sum_{i \neq j} \sum E(\epsilon_i \epsilon_j) x_i}{n \sum x_i^2} - \bar{x} \frac{\sum x_i^2 \sigma^2}{(\sum x_i^2)^2}$$

$$\boxed{\text{Cov}(\hat{\alpha}, \hat{\beta}) = -\frac{\bar{x} \sigma^2}{(\sum x_i^2)^2}} \quad \checkmark$$

$$\therefore E \epsilon_i \epsilon_j = 0$$

$$\sum x_i = 0$$

$$\rho_{\hat{\alpha}, \hat{\beta}} = \frac{\text{Cov}(\hat{\alpha}, \hat{\beta})}{\sigma_{\hat{\alpha}} \cdot \sigma_{\hat{\beta}}}$$

$$\sigma_{\hat{\alpha}} \cdot \sigma_{\hat{\beta}}$$

$$= \frac{\bar{x} \sigma^2}{(\sum x_i^2)^2}$$

$$= \frac{-\frac{\bar{x} \sigma^2}{(\sum x_i^2)^2}}{\sqrt{\left(\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum x_i^2}\right) \sigma^2\right) \left(\frac{\sigma^2}{\sum x_i^2}\right)}}$$

$$= \frac{-\frac{\bar{x} \sigma^2}{\sum x_i^2}}{\sqrt{\left(\frac{\sum x_i^2}{n \sum x_i^2} \sigma^2\right) \left(\frac{\sigma^2}{\sum x_i^2}\right)}}$$

$$\rho_{\hat{\alpha}, \hat{\beta}} = \frac{-\bar{X} \frac{\sigma^2}{\sum x^2}}{\sqrt{\frac{\sum X^2}{n}} \frac{\sigma^2}{\sum x^2}}$$

$$\rho_{\hat{\alpha}, \hat{\beta}} = \frac{-\bar{X} \sqrt{n}}{\sqrt{\sum X^2}}$$

ii) Joint dist of $\hat{\alpha}$ and $\hat{\beta}$ is ①

$$= \frac{1}{2\pi \sqrt{1 - \rho_{\hat{\alpha}, \hat{\beta}}^2}} e^{-\frac{1}{2(1 - \rho_{\hat{\alpha}, \hat{\beta}}^2)} \left[\left(\frac{\hat{\alpha} - \alpha}{\sigma_{\hat{\alpha}}} \right)^2 + \left(\frac{\hat{\beta} - \beta}{\sigma_{\hat{\beta}}} \right)^2 - 2 \rho_{\hat{\alpha}, \hat{\beta}} \left(\frac{\hat{\alpha} - \alpha}{\sigma_{\hat{\alpha}}} \right) \left(\frac{\hat{\beta} - \beta}{\sigma_{\hat{\beta}}} \right) \right]}$$

iii) $\hat{\alpha} - \hat{\beta}$

$$\begin{aligned} E(\hat{\alpha} - \hat{\beta}) &= E(\hat{\alpha}) - E(\hat{\beta}) \\ &= \alpha - \beta \end{aligned}$$

$$E(\hat{\alpha} - \hat{\beta}) = \alpha - \beta$$

$$V(\hat{\alpha} - \hat{\beta}) = V(\hat{\alpha}) + V(\hat{\beta}) - 2 \text{Cov}(\hat{\alpha}, \hat{\beta})$$

$$= \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x^2} \right] \sigma^2 + \frac{\sigma^2}{\sum x^2} - 2 \left(-\frac{\bar{X} \sigma^2}{\sum x^2} \right)$$

$$= \sigma^2 \hat{\alpha} + \sigma^2 \hat{\beta} + 2 \bar{X} \sigma^2 \hat{\beta}$$

$$V(\hat{\alpha} - \hat{\beta}) = \sigma^2 \hat{\alpha} + (1 + 2\bar{X}) \sigma^2 \hat{\beta} = \sigma^2(\hat{\alpha} - \hat{\beta})$$

X 20

23

Remove Watermark Now

The pdf of $(\hat{\alpha} - \beta)$ is

$$f(\hat{\alpha} - \beta) = \frac{1}{\sqrt{2\pi} \sigma(\hat{\alpha} - \beta)} e^{-\frac{1}{2} \left(\frac{(\hat{\alpha} - \beta) - (\alpha - \beta)}{\sigma(\hat{\alpha} - \beta)} \right)^2} \quad \frac{X - \mu}{\sigma}$$

(iii) $2\hat{\alpha} + 3\hat{\beta}$

$$E(2\hat{\alpha} + 3\hat{\beta}) = 2E(\hat{\alpha}) + 3E(\hat{\beta}) \\ = 2\alpha + 3\beta$$

$$E(2\hat{\alpha} + 3\hat{\beta}) = 2\alpha + 3\beta$$

$$V(2\hat{\alpha} + 3\hat{\beta}) = 4V_{\hat{\alpha}} + 9V_{\hat{\beta}} + 12Cov(\hat{\alpha}, \hat{\beta}) \\ = 4 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x^2} \right] \sigma^2 + 9 \frac{\sigma^2}{\sum x^2} + 12 \left(\frac{-\bar{X} \sigma^2}{\sum x^2} \right) \\ = 4\sigma_{\hat{\alpha}}^2 + 9\sigma_{\hat{\beta}}^2 - 12\bar{X}\sigma_{\hat{\beta}}^2$$

$$V(2\hat{\alpha} + 3\hat{\beta}) = 4\sigma_{\hat{\alpha}}^2 + 3\sigma_{\hat{\beta}}^2(3 - 4\bar{X}) = \sigma_{(2\hat{\alpha} + 3\hat{\beta})}^2$$

The p.d.f of $(2\hat{\alpha} + 3\hat{\beta})$ is

$$f(2\hat{\alpha} + 3\hat{\beta}) = \frac{1}{\sqrt{2\pi} \sigma(2\hat{\alpha} + 3\hat{\beta})} e^{-\frac{1}{2} \left(\frac{(2\hat{\alpha} + 3\hat{\beta}) - (2\alpha + 3\beta)}{\sigma(2\hat{\alpha} + 3\hat{\beta})} \right)^2}$$

$$f(2\hat{\alpha} + 3\hat{\beta}) = \frac{1}{\sqrt{2\pi} \sigma(2\hat{\alpha} + 3\hat{\beta})}$$

(iv) $c\hat{\alpha}$

$$E(c\hat{\alpha}) = cE\hat{\alpha} = c\alpha$$

$$E(c\hat{\alpha}) = c\alpha$$

$$e^{-\frac{(x-\mu)^2}{2\sigma^2}} e$$

$$Z = \frac{x-\mu}{\sigma} \quad \Rightarrow Z = \frac{x-E(X)}{\sqrt{\text{Var}(X)}}$$

23

$$\begin{aligned} \text{Var}(c\hat{\alpha}) &= c^2 \text{Var}(\hat{\alpha}) \\ &= c^2 \sigma^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x_i^2} \right] \end{aligned}$$

$$\boxed{\text{Var}(\hat{y}) = \sigma^2(c\hat{\alpha}) = c^2 \sigma^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x_i^2} \right]}$$

$$\begin{aligned} E(\hat{\alpha} - \beta) &= E(\hat{\alpha}) - E(\beta) \\ &= \alpha - \beta \end{aligned}$$

Linear function of normal variable is also normal variable
Hence p.d.f. of $c\hat{\alpha}$ is

$$f(c\hat{\alpha}) = \frac{1}{\sqrt{2\pi} \sigma(c\hat{\alpha})} e^{-\frac{1}{2} \left(\frac{c\hat{\alpha} - c\alpha}{\sigma(c\hat{\alpha})} \right)^2}$$

(v) $\hat{\beta}$

$$E(\hat{\beta}) = E(\hat{\beta}) = \beta$$

$$\boxed{E(\hat{\beta}) = \beta}$$

$$\text{Var}(\hat{\beta}) = \hat{\tau}^2 \text{Var}(\hat{\beta}) = \frac{\hat{\tau}^2 \sigma^2}{\sum x_i^2}$$

$$\boxed{\text{Var}(\hat{\beta}) = \frac{\hat{\tau}^2 \sigma^2}{\sum x_i^2}}$$

Linear function of normal variable is also normal variable, therefore
p.d.f. of $\hat{\beta}$ is

$$f(\hat{\beta}) = \frac{1}{\sqrt{2\pi} \sigma(\hat{\beta})} e^{-\frac{1}{2} \left(\frac{\hat{\beta} - \beta}{\sigma(\hat{\beta})} \right)^2}$$

(vi) $\hat{\alpha}_1, \hat{\alpha}_2$ and $\hat{\beta}_1, \hat{\beta}_2$ where $\hat{\alpha}_1$ and $\hat{\beta}_1$ are computed from n_1 pairs (x_{1i}, y_{1i}) and $\hat{\alpha}_2$ and $\hat{\beta}_2$ are computed from another independent n_2 pairs (x_{2i}, y_{2i})

①

$$E(\hat{\alpha}_1) = \alpha_1$$

$$\sigma_{(\hat{\alpha}_1)}^2 = \left[\frac{1}{n_1} + \frac{\bar{X}_1^2}{\sum x_i^2} \right] \sigma^2$$

$$E(\hat{\alpha}_2) = \alpha_2$$

$$\sigma_{(\hat{\alpha}_2)}^2 = \left[\frac{1}{n_2} + \frac{\bar{X}_2^2}{\sum x_{i2}^2} \right] \sigma_2^2$$

$$E(\hat{\alpha}_1 - \hat{\alpha}_2) = E(\hat{\alpha}_1) - E(\hat{\alpha}_2)$$

$$E(\hat{\alpha}_1 - \hat{\alpha}_2) = \alpha_1 - \alpha_2$$

$$\text{Var}(\hat{\alpha}_1 - \hat{\alpha}_2) = \text{Var}(\hat{\alpha}_1) + \text{Var}(\hat{\alpha}_2)$$

Covariance term vanishes because samples are taken independently

$$\text{Var}(\hat{\alpha}_1 - \hat{\alpha}_2) = \left[\frac{1}{n_1} + \frac{\bar{X}_1^2}{\sum x_{i1}^2} \right] \sigma_1^2 + \left[\frac{1}{n_2} + \frac{\bar{X}_2^2}{\sum x_{i2}^2} \right] \sigma_2^2 = \sigma_{(\hat{\alpha}_1 - \hat{\alpha}_2)}^2$$

Linear function of normal variable is also normal variable, therefore p.d.f of $\hat{\alpha}_1 - \hat{\alpha}_2$ is

$$f(\hat{\alpha}_1 - \hat{\alpha}_2) = \frac{1}{\sqrt{2\pi} \sigma_{(\hat{\alpha}_1 - \hat{\alpha}_2)}} e^{-\frac{1}{2} \left(\frac{(\hat{\alpha}_1 - \hat{\alpha}_2) - (\alpha_1 - \alpha_2)}{\sigma_{(\hat{\alpha}_1 - \hat{\alpha}_2)}} \right)^2}$$

(2)

$$E(\hat{\beta}_1) = \beta_1$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma_1^2}{\sum x_{i1}^2}$$

$$E(\hat{\beta}_2) = \beta_2, \quad \text{Var}(\hat{\beta}_2) = \frac{\sigma_2^2}{\sum x_{i2}^2}$$

$$E(\hat{\beta}_1 - \hat{\beta}_2) = E(\hat{\beta}_1) - E(\hat{\beta}_2)$$

$$E(\hat{\beta}_1 - \hat{\beta}_2) = \beta_1 - \beta_2$$

$$\text{Var}(\hat{\beta}_1 - \hat{\beta}_2) = \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2)$$

Covariance of $\hat{\beta}_1, \hat{\beta}_2$ is equal to zero because two samples are independently taken.

$$\text{Var}(\hat{\beta}_1 - \hat{\beta}_2) = \frac{\sigma_1^2}{\sum x_{1i}^2} + \frac{\sigma_2^2}{\sum x_{2i}^2}$$

$$\sigma(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{\left[\frac{\sigma_1^2}{\sum x_{1i}^2} + \frac{\sigma_2^2}{\sum x_{2i}^2} \right]}$$

Linear function of normal variable is also normal variable. therefore p.d.f of $\hat{\beta}_1 - \hat{\beta}_2$ is

$$f(\hat{\beta}_1 - \hat{\beta}_2) = \frac{1}{\sqrt{2\pi} \sigma(\hat{\beta}_1 - \hat{\beta}_2)} e^{-\frac{1}{2} \left(\frac{(\hat{\beta}_1 - \hat{\beta}_2) - (\beta_1 - \beta_2)}{\sigma(\hat{\beta}_1 - \hat{\beta}_2)} \right)^2}$$

Property (ii) of $\hat{\alpha}$. Second Method

$\hat{\alpha}$ has minimum variance among all linear and unbiased estimators.

Considering any other estimator say $\hat{\alpha}^* = \sum C_i Y$

Where C_i is some non stochastic function of X values only and $Y = \alpha + \beta X + \epsilon$

$$\hat{\alpha}^* = \sum_{i=1}^n C_i (\alpha + \beta X_i + \epsilon_i) = \alpha \sum C_i + \beta \sum C_i X_i + \sum C_i \epsilon_i$$

$$E(\hat{\alpha}^*) = \alpha \quad \text{if} \quad \sum C_i = 1, \quad \sum C_i X_i = 0$$

$$\text{Var}(\hat{\alpha}^*) = \sum C_i^2 \text{Var}(Y)$$

$$\text{Var}(\hat{\alpha}^*) = \sum C_i^2 \sigma^2$$

By using Lagrange multiplier

Where $F = \sum C_i^2$

$$F = \sum C_i^2 - 2\lambda (\sum C_i - 1) - 2\mu \sum C_i X_i$$

$$\frac{\partial F}{\partial C_i} = 2C_i - 2\lambda - 2\mu X_i, \quad \frac{\partial^2 F}{\partial C_i^2} = 2 > 0$$

$$\frac{\partial F}{\partial C_i} = 0 \Rightarrow C_i = \lambda + \mu X_i \quad \text{--- (1)}$$

Normal equations are

$$\sum C_i = n\lambda + \mu \sum X_i$$

$$n\lambda + \mu \sum X_i = 1 \quad \text{--- (2)}$$

$$\begin{cases} \sum C_i = 1, \sum C_i X_i = 0 \\ \text{if } \hat{\alpha} = \dots \end{cases}$$

$$\sum C_i X_i = \lambda \sum X_i + \mu \sum X_i^2$$

$$\lambda \sum X_i + \mu \sum X_i^2 = 0 \quad \text{--- (3)}$$

From (3)

$$\mu = -\frac{\lambda \sum X_i}{\sum X_i^2} \quad \text{--- (4)} \quad \text{pulling in (2)}$$

$$n\lambda - \lambda \frac{(\sum X_i)^2}{\sum X_i^2} = 1 \Rightarrow \lambda \left(n - \frac{(\sum X_i)^2}{\sum X_i^2} \right) = 1, \quad \lambda = \frac{1}{n - \frac{(\sum X_i)^2}{\sum X_i^2}}$$

$$\lambda = \frac{\sum X_i^2}{n \sum X_i^2 - (\sum X_i)^2} = \frac{\sum X_i^2}{n [\sum X_i^2 - n \bar{X}^2]}$$

$$\boxed{\lambda = \frac{\sum X_i^2}{n \sum X_i^2}}$$

From (4)

$$\mu = -\frac{\sum X_i^2}{n \sum X_i^2} \left(\frac{\sum X_i}{\sum X_i^2} \right) = -\frac{\sum X_i}{n \sum X_i^2}$$

$$\boxed{\mu = -\frac{\bar{X}}{\sum X_i^2}}$$

pulling λ, μ in (1)

$$C_i = \frac{\sum X_i^2}{n \sum X_i^2} - \frac{\bar{X}}{\sum X_i^2} X_i$$

Adding and subtracting $1/n$, we get

$$C_i = \frac{1}{n} - \frac{X_i \bar{X}}{\sum X_i^2} + \frac{\bar{X}^2}{\sum X_i^2}$$

$$C_i = \frac{1}{n} - \frac{X_i (X_i - \bar{X})}{\sum X_i^2} \bar{X}$$

$$C_i = \frac{1}{n} - \frac{X_i}{\sum X_i^2} \bar{X}, \quad \text{for this value of } C \text{ we have } C \text{ as minimum.}$$

Thus estimate $\hat{\alpha}^* = \sum C_i Y = \sum \left[\frac{1}{n} - \frac{X_i}{\sum X_i^2} \bar{X} \right] Y$

$$= \frac{\sum Y}{n} - \frac{\sum X_i Y}{\sum X_i^2} \bar{X}$$

$$\hat{\alpha}^* = \bar{Y} - \hat{\beta} \bar{X}$$

$$\begin{aligned} & (\sum x_i)(\sum \epsilon_i) \\ & (\sum x_i)(\sum \epsilon_i^2 + \sum_{i \neq j} \epsilon_i \epsilon_j) \end{aligned}$$

Sample model.

33

9-12-90

SLR Unbiased Estimator of σ^2 $y = \alpha + \beta x + \epsilon$

Simple Linear Regression Model is

$$Y = \alpha + \beta X + \epsilon$$

For the sample of 'n' pairs

$$\beta(x - \bar{x})$$

$$= \beta x$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}, \hat{\beta} = \frac{\sum xy}{\sum x^2}, \hat{y} = \hat{\alpha} + \hat{\beta} x$$

Residual is

$$e = y - \hat{y}$$

$$= y - \hat{\alpha} - \hat{\beta} x$$

$$= y - \bar{y} + \hat{\beta} \bar{x} - \hat{\beta} x$$

$$= (\alpha + \beta x + \epsilon) - (\alpha + \beta \bar{x} + \bar{\epsilon}) - \hat{\beta}(x - \bar{x})$$

$$= \alpha + \beta x + \epsilon - \alpha - \beta \bar{x} - \bar{\epsilon} - \hat{\beta} x$$

$$= \beta x + (\epsilon - \bar{\epsilon}) - \hat{\beta} x$$

$$e = -x(\hat{\beta} - \beta) + (\epsilon - \bar{\epsilon})$$

$$\sum e^2 = (\hat{\beta} - \beta)^2 \sum x^2 + \sum (\epsilon - \bar{\epsilon})^2 - 2(\hat{\beta} - \beta) \sum x(\epsilon - \bar{\epsilon})$$

$$E \sum e^2 = E(\hat{\beta} - \beta)^2 \sum x^2 + E \left[\sum (\epsilon - \bar{\epsilon})^2 - \frac{(\sum \epsilon)^2}{n} \right] - 2E(\hat{\beta} - \beta) \left[\frac{\sum x \epsilon - (\sum x)(\sum \epsilon)}{n} \right]$$

$$= \sigma_{\hat{\beta}}^2 \sum x^2 + n\sigma^2 - 2E \left[\frac{\sum x \epsilon}{\sum x^2} \right] \left[\frac{\sum x \epsilon - \sum x \bar{\epsilon}}{n} \right]$$

$$= \frac{\sigma^2}{\sum x^2} \sum x^2 + n\sigma^2 - \frac{1}{n} E \left[\sum \epsilon^2 + \sum_{i \neq j} \epsilon_i \epsilon_j \right] (\sum x) (\sum \epsilon)$$

$$- \frac{2}{\sum x^2} E \left[\sum x \epsilon^2 + \sum_{i \neq j} x_i x_j \epsilon_i \epsilon_j \right]$$

$$+ \frac{2}{\sum x^2} \bar{x} \left[\sum x_i \epsilon_i^2 + \sum_{i \neq j} x_i x_j \epsilon_i \epsilon_j \right]$$

$$= (n+1)\sigma^2 - \sigma^2 - \frac{2}{\sum x^2} \left[\sigma^2 \sum x^2 + 0 + \bar{x} \left\{ \sum x_i \epsilon_i^2 + 0 \right\} \right]$$

$$E(\sum e^2) = (n+1)\sigma^2 - \sigma^2 - 2\sigma^2 \bar{x}^2 \frac{\sum x_i^2}{\sum x^2} \quad E x_i = 0$$

$$= (n+1)\sigma^2 - \sigma^2 - 2\sigma^2 \bar{x}^2 \frac{\sum x_i^2}{\sum x^2} \quad (n+1-3)\sigma^2$$

$$= (n+1)\sigma^2 - 3\sigma^2 + 0 \quad (n-2)\sigma^2$$

$$E \sum C^2 = (n-2) \sigma^2$$

$$E \left[\frac{\sum e^2}{n-2} \right] = \sigma^2$$

$$\sum C^2 = (n-2) \hat{\sigma}^2$$

$$E(\hat{\sigma}^2) = \sigma^2$$

$$\hat{\sigma}^2 = \frac{\sum e^2}{n-2} = s_{y.x}^2$$

 $\hat{\sigma}^2$

It is unbiased estimator of σ^2

* Maximum Likelihood Estimator of σ^2 is

$$\hat{\sigma}^{*2} = \frac{\sum e^2}{n}$$

Biased Estimator

$$= \left(\frac{\sum e^2}{n-2} \right) \left(\frac{n-2}{n} \right) = \hat{\sigma}^2 \left(1 - \frac{2}{n} \right)$$

$$\checkmark \hat{\sigma}^{*2} = \hat{\sigma}^2 - \frac{2}{n} \hat{\sigma}^2$$

$$* \text{Bias} = E(\hat{\sigma}^{*2}) - E(\hat{\sigma}^2)$$

$$\checkmark E(\hat{\sigma}^{*2}) = \sigma^2 - \frac{2}{n} \sigma^2 \quad \therefore E(\hat{\sigma}^{*2}) = \sigma^2$$

$$\text{Hence Bias} = \sigma^2 - \frac{2}{n} \sigma^2 = \sigma^2$$

$$\text{Bias} = -\frac{2}{n} \sigma^2$$

It means MLE is negatively biased.

Properties of Residual.

$$Y = \alpha + \beta X + \epsilon$$

Normal equation for α is

$$Y = \hat{\alpha} + \hat{\beta} X + \epsilon$$

Sample Model

$$\sum Y = n\hat{\alpha} + \hat{\beta} \sum X + \sum \epsilon \quad (1)$$

Fitted equation is

$$\hat{Y} = \hat{\alpha} + \hat{\beta} X$$

Applying summation

$$\sum \hat{Y} = n\hat{\alpha} + \hat{\beta} \sum X \quad (2)$$

Comparing (1) and (2):

$$\sum Y = \sum \hat{Y} \implies \bar{Y} = \bar{\hat{Y}}$$

Now $e = Y - \hat{Y}$ or $\rightarrow$

$$e = Y - \hat{\alpha} - \hat{\beta} X$$

$$\sum e = \sum Y - \sum \hat{Y}$$

$$= Y - \bar{Y} + \hat{\beta} \bar{X} - \hat{\beta} X$$

$$\boxed{\sum e = 0}$$

$$e = Y - \hat{\beta} X$$

$$\sum e = 0$$

$$(i) \sum eX = \sum XY - \frac{\sum X Y}{\sum X^2} \sum X X$$

$$\sum e = \sum Y - \hat{\beta} \sum X$$

$$= \sum XY - \sum X Y = \sum XY - \sum XY = 0$$

$$= 0$$

$$\boxed{\sum eX = 0}$$

It means Sum of product of residual with independent variable vanish.

$$(iii) \sum e \hat{Y} = \sum e (\hat{\alpha} + \hat{\beta} X)$$

$$= (\sum e) \hat{\alpha} + \hat{\beta} \sum eX$$

$$= 0(\hat{\alpha}) + \hat{\beta}(0)$$

$$\therefore \sum eX = 0, \sum e = 0$$

$$\sum e \hat{Y} = 0$$

$$\begin{aligned} \sum eX &= \sum (Y - \hat{\beta} X) X = \sum XY - \hat{\beta} \sum X^2 \\ &= \sum XY - \hat{\beta} \sum X^2 = \sum XY - \sum XY = 0 \end{aligned}$$

Inference In Simple Linear Regression Model.

Simple linear regression model is

$$Y = \alpha + \beta X + \epsilon$$

Estimate of α is, $\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} \sim N\left(\alpha, \frac{\sum X^2 \sigma^2}{n \sum x^2}\right)$

Estimate of β is, $\hat{\beta} = \frac{\sum xy}{\sum x^2} \sim N\left(\beta, \frac{\sigma^2}{\sum x^2}\right)$

Fitted equation is $\hat{y} = \hat{\alpha} + \hat{\beta} x$

$\sum y_i - \sum (\hat{y} - \bar{y}) = 0$
 $\sum y_i = \sum (\hat{y} - \bar{y}) = 0$

Partitioning of Total Sum of Square.

Residual is

$$e = y - \hat{y} = (y - \bar{y}) - (\hat{y} - \bar{y}) \quad \text{--- (1)}$$

Since $\bar{y} = \bar{\hat{y}}$

$$e = y - \hat{y} \quad \text{or} \quad y = \hat{y} + e$$

$$\sum y^2 = \sum \hat{y}^2 + \sum e^2 + 2 \sum e \hat{y}$$

$$\sum e \hat{y} = \sum e \bar{y} - \bar{y} \sum e = \sum e (\hat{y} - \bar{y})$$

$$\sum e \hat{y} = 0$$

From (1) $e = (y - \bar{y}) - (\hat{y} - \bar{y})^2$

$$\sum (y - \bar{y})^2 = \sum (\hat{y} - \bar{y})^2 + \sum e^2 - 2 \sum (\hat{y} - \bar{y})(e)$$

$$\sum (y - \bar{y})^2 = \sum (\hat{y} - \bar{y})^2 + \sum e^2 \quad \text{2060}$$

$$TSS = SSR + SSE$$

$$\begin{aligned}\sum \hat{y}^2 &= \sum [\hat{\alpha} + \hat{\beta}x - \hat{\alpha} - \hat{\beta}\bar{x}]^2 \\ &= \sum [\hat{\beta}(x - \bar{x})]^2 = \hat{\beta}^2 \sum x^2 \\ &= \frac{(\sum xy)^2}{\sum x^2} = \frac{(\sum xy)^2}{\sum x^2} \\ \sum \hat{y}^2 &= \hat{\beta}^2 \sum xy\end{aligned}$$

Statistical Inference in Simple Linear Regression Model. (Continue)

Simple linear regression model is

$$Y = \alpha + \beta X + \epsilon$$

Distribution of Sum of Square is

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\begin{aligned}\epsilon_{i|\delta} &\sim N(0, \sigma^2) \\ y_{i|\delta} &\sim N(\alpha + \beta x_i, \sigma^2)\end{aligned}$$

So

$$\sum_{i=1}^n \left(\frac{y_i - \bar{y}}{\sigma} \right)^2 \sim \chi_{(n-1)}^2$$

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2 = \sigma^2 \chi_{n-1}^2$$

$$Reg\ SS = \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2$$

where $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$

Since least square estimators $\hat{\alpha}, \hat{\beta}$ are linear functions of y_i s which follows the normal distribution, therefore as a result

$\hat{y}_i \sim$ Normal Distribution.

Where Reg SS follows $\sigma^2 \chi^2$ with $df = E(Reg\ SS)$

$$Reg\ SS = \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2$$

$$\hat{y}_i = \hat{\alpha} + \hat{\beta}x$$

$$\sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma} \right)^2 \sim \chi_n^2$$

$$\sum_{i=1}^n \left(\frac{\bar{x} - \bar{x}}{\sigma} \right)^2 \sim \chi_{n-1}^2$$

$$\hat{y} = \hat{\alpha} + \hat{\beta} \bar{x}$$

$$\hat{y}_i - \hat{\bar{y}} = \hat{\alpha} + \hat{\beta} x_i - \hat{\alpha} - \hat{\beta} \bar{x} = \hat{\beta}(x_i - \bar{x})$$

Squaring and applying summation

$$\sum_{i=1}^n (\hat{y}_i - \hat{\bar{y}})^2 = \hat{\beta}^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \hat{\beta}^2 \sum_{i=1}^n x_i^2$$

$$\text{Reg SS} = \sum_{i=1}^n (\hat{y}_i - \hat{\bar{y}})^2 = \hat{\beta}^2 \sum_{i=1}^n x_i^2 \quad \text{--- ①}$$

$$\text{But } \hat{\beta} = \frac{\sum xy}{\sum x^2} = \frac{\sum x y}{\sum x^2}$$

$$\hat{\beta} = \frac{\sum x [\alpha + \beta x + \epsilon]}{\sum x^2}$$

$$\hat{\beta} = \frac{\sum x \alpha + \beta \sum x x + \sum x \epsilon}{\sum x^2}$$

$$\therefore \sum x = 0 \quad \sum x x = \sum x^2$$

$$\therefore \hat{\beta} = \beta + \frac{\sum x \epsilon}{\sum x^2}$$

Now Reg SS is

$$\text{Reg SS} = \left[\beta + \frac{\sum x \epsilon}{\sum x^2} \right]^2 \sum x^2$$

Applying expectation

$$E(\text{Reg SS}) = E \left[\beta + \frac{\sum x \epsilon}{\sum x^2} \right]^2 \sum x^2$$

Under H_0 , $\beta = 0$ i.e. No relationship between x and y .

$$E(\text{Reg SS}) = E \left[0 + \frac{\sum x \epsilon}{\sum x^2} \right]^2 \sum x^2$$

$$= \frac{1}{\sum x^2} E \left[\sum x_i^2 \epsilon_i^2 + \sum_{i \neq j} x_i x_j \epsilon_i \epsilon_j \right]$$

$$E(\text{Reg SS}) = \frac{1}{\sum x^2} \left[\sigma^2 \sum x^2 + 0 \right]$$

$$E(\text{Reg SS}) = \sigma^2$$

$$E(\sigma^2 Y^2) = \sigma^2 E(Y^2)$$

$$E\left(\frac{\text{Reg SS}}{\sigma^2}\right) = E\left(\frac{\sum (y_i - \hat{y})^2}{\sigma^2}\right) = 1$$

It means that Reg SS is $\sigma^2 \chi^2$ with $df = 1$

Now Sum of Squares of residual is

$$\sum e^2 = \sum (Y - \hat{Y})^2$$

Since both Y and $\hat{Y}$ are normal variables, therefore $\sum (Y - \hat{Y})^2$ follows $\sigma^2 \chi^2$ distribution with $df = E[\sum e^2]$

$$e = Y - \hat{Y} = \cancel{\alpha + \beta X + \epsilon} - \cancel{\alpha + \beta X}$$

$$= Y - \hat{\alpha} - \hat{\beta} X$$

$$e = Y - \bar{Y} + \bar{\beta} \bar{X} - \hat{\beta} X = (\alpha + \beta X + \epsilon) - (\alpha + \beta \bar{X} + \bar{\epsilon}) - \hat{\beta}(X - \bar{X})$$

$$= \alpha + \beta X + \epsilon - \alpha - \beta \bar{X} - \bar{\epsilon} - \hat{\beta}(X - \bar{X})$$

$$e = \beta X + (\epsilon - \bar{\epsilon}) - \hat{\beta} X$$

$$e = -(\hat{\beta} - \beta)X + (\epsilon - \bar{\epsilon})$$

Squaring and applying summation

$$\sum e^2 = \sum X^2 (\hat{\beta} - \beta)^2 + \sum (\epsilon - \bar{\epsilon})^2 - 2(\hat{\beta} - \beta) \sum X(\epsilon - \bar{\epsilon})$$

Applying expectation

$$E(\sum e^2) = E(\hat{\beta} - \beta)^2 \sum X^2 + E\left[\sum (\epsilon - \bar{\epsilon})^2 - \frac{(\sum \epsilon)^2}{n}\right] - 2E(\hat{\beta} - \beta) \left[\sum X\epsilon - \bar{\epsilon} \sum X\right]$$

$$= E(\hat{\beta} - \beta)^2 \sum X^2 + n\sigma^2 - 2E\left[\frac{\sum X\epsilon}{\sum X^2}\right] \left(\sum X\epsilon\right)$$

$$= \frac{\sigma^2}{\sum X^2} \sum X^2 + n\sigma^2 - \frac{2}{\sum X^2} E\left[\sum X^2 e^2 + \sum_{i \neq j} X_i X_j e_i e_j\right]$$

$$= \frac{\sigma^2}{\sum X^2} \sum X^2 + n\sigma^2 - \sigma^2 - 2\sigma^2$$

$$E(\sum e^2) = (n-2)\sigma^2 \Rightarrow E\left(\frac{\sum (y_i - \hat{y})^2}{\sigma^2}\right) = n-2$$

Sum of square of residual follows $\sigma^2 \chi^2$ dist with

$$df = n-2$$

Trace the identity among d.f.s is

$$n-1 = 1 + (n-2)$$

ie $df(TSS) = df(Reg SS) + df(SSE)$

Which is the same as sum of squares identity ie

$$TSS = SSR + SSE$$

Therefore Regression SS and SSE are independent χ^2 variable and the statistic

$$F = \frac{\frac{Reg SS}{1}}{\frac{SSE}{n-2}} \sim F(1, n-2)$$

The larger values of F statistic leads to reject the hypothesis of No relationship between the variables.

Test Procedure For Different Hypothesis

Steps

- (i) Search for parameter
- (ii) Define the estimator of the parameter.
- (iii) Discuss the sampling distribution of the statistic.
- (iv) Construct the test statistic.
- (v) Define the Critical Region.

Test Procedure for: $H_0: \alpha = 0$

871

(i) $\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$ is the estimator of α .

(iii)

$$\hat{\alpha} \sim N\left(\alpha, \frac{\sum X^2}{n \sum x^2} \sigma^2\right)$$

(iv) Test Statistic to be used is

① $Z = \frac{\hat{\alpha} - \alpha}{\sigma \sqrt{\frac{\sum X^2}{n \sum x^2}}}$ is Standard normal variable if σ is known.

② If σ is unknown; then we replace it by its estimator denoted by $\hat{\sigma}$

$$\hat{\sigma}^2 = \frac{\sum e^2}{n-2}$$

Test Statistic to be used is

$$t = \frac{\hat{\alpha} - \alpha}{\hat{\sigma} \sqrt{\frac{\sum X^2}{n \sum x^2}}}$$

③ t statistic also ratio of SNU and χ^2 variable divided by its df and under root.

ie $t = \frac{Z}{\sqrt{\chi^2/df}}$

$$\hat{\sigma}^2 = \frac{\sum e^2}{n-2} \Rightarrow \sum e^2 = \hat{\sigma}^2 \chi^2_{(n-2)}$$

$$\Rightarrow \chi^2_{(n-2)} = \frac{\sum e^2}{\hat{\sigma}^2}$$

$$\frac{\chi^2}{df} = \frac{\sum e^2}{(n-2)\hat{\sigma}^2} = \frac{\hat{\sigma}^2}{\hat{\sigma}^2}$$

now $t = \frac{Z}{\sqrt{\chi^2/df}}$ gives $t = \frac{\hat{\alpha} - \alpha}{\sigma \sqrt{\frac{\sum X^2}{n \sum x^2} \frac{\hat{\sigma}^2}{\hat{\sigma}^2}}}$

$$t = \frac{\hat{\alpha} - \alpha}{\hat{\sigma} \sqrt{\frac{\sum X^2}{n \sum x^2}}} \sim t_{(n-2)}$$

(i) If $H_0: \alpha = 0$. Then C.R is $|t| > t_{(\alpha, n-2)}$

(ii) $H_1: \alpha > 0$ then C.R is $t > t_{(\alpha, n-2)}$

(iii) $H_1: \alpha < 0$ then C.R is $t < t_{(1-\alpha, n-2)}$

Test Procedure For $H_0: \alpha_1 = \alpha_2$ or $\alpha_1 - \alpha_2 = 0$

(i) $\hat{\alpha}_1 - \hat{\alpha}_2 = (\bar{Y}_1 - \hat{\beta}_1 \bar{X}_1) - (\bar{Y}_2 - \hat{\beta}_2 \bar{X}_2)$

(ii) $E(\hat{\alpha}_1 - \hat{\alpha}_2) = E(\hat{\alpha}_1) - E(\hat{\alpha}_2) = \alpha_1 - \alpha_2$

$\sigma^2(\hat{\alpha}_1 - \hat{\alpha}_2) = \sigma^2_{\hat{\alpha}_1} + \sigma^2_{\hat{\alpha}_2} = \frac{\sigma_1^2 \sum X_1^2}{n_1 \sum x_1^2} + \frac{\sigma_2^2 \sum X_2^2}{n_2 \sum x_2^2}$

$(\hat{\alpha}_1 - \hat{\alpha}_2) \sim N\left(\alpha_1 - \alpha_2, \frac{\sigma_1^2 \sum X_1^2}{n_1 \sum x_1^2} + \frac{\sigma_2^2 \sum X_2^2}{n_2 \sum x_2^2}\right)$

(iv) Test statistic to be used is

①

$Z = \frac{(\hat{\alpha}_1 - \hat{\alpha}_2) - (\alpha_1 - \alpha_2)}{\sqrt{\frac{\sigma_1^2 \sum X_1^2}{n_1 \sum x_1^2} + \frac{\sigma_2^2 \sum X_2^2}{n_2 \sum x_2^2}}}$ is SNV if σ_1^2, σ_2^2 both known

② When $\sigma_1 = \sigma_2 = \sigma$

$Z = \frac{(\hat{\alpha}_1 - \hat{\alpha}_2) - (\alpha_1 - \alpha_2)}{\sigma \sqrt{\frac{\sum X_1^2}{n_1 \sum x_1^2} + \frac{\sum X_2^2}{n_2 \sum x_2^2}}}$

③ If σ_1^2, σ_2^2 are unknown then we replace them by their estimators

$\hat{\sigma}_1^2 = \frac{\sum e_1^2}{n_1 - 2}, \quad \sum e_1^2 = \sigma_1^2 \chi_{(n_1-2)}^2$

$\hat{\sigma}_2^2 = \frac{\sum e_2^2}{n_2 - 2}, \quad \sum e_2^2 = \sigma_2^2 \chi_{(n_2-2)}^2$

When $\sigma_1 = \sigma_2 = \sigma$ (say), we shall pool both estimators, then pooled estimator of σ^2 is

$$\hat{\sigma}_p^2 = \frac{\sum e_1^2 + \sum e_2^2}{n_1 + n_2 - 4} \quad (\text{by the additive property of } \chi^2)$$

$$\sum e_1^2 + \sum e_2^2 = \sigma^2 \chi_{(n_1+n_2-4)}^2$$

$$\hat{\sigma}_p^2 = \frac{\sigma^2 \chi^2}{df}$$

•

Test statistic to be used is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\alpha_1 - \alpha_2)}{\hat{\sigma}_p \left[\frac{\sum x_1^2}{n \sum x_1^2} - \frac{\sum x_2^2}{n \sum x_2^2} \right]} \sim t_{n_1+n_2-4}$$

(i) H_0 ; $\alpha_1 - \alpha_2 \neq 0$ CR is $|t| > t_{(\frac{\alpha}{2}, n_1+n_2-4)}$

(ii) H_1 ; $\alpha_1 - \alpha_2 > 0$ CR is $t > t_{(\alpha, n_1+n_2-4)}$

(iii) H_1 ; $\alpha_1 - \alpha_2 < 0$ CR is $t < t_{(1-\alpha, n_1+n_2-4)} \Rightarrow t < -t$

Testing Procedure For H_0 ; $\beta = 0$

872

(i) $\hat{\beta} = \frac{\sum xy}{\sum x^2}$ is the estimator of β .

(ii) $\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum x^2}\right)$

(iii) Test statistic to be used is

①

$$Z = \frac{\hat{\beta} - \beta}{\sqrt{\frac{\sigma^2}{\sum x^2}}} \text{ is SNV if } \sigma \text{ is known.}$$

② If σ is unknown, then we replace it by its estimator denoted by

$$\hat{\sigma}^2 = \frac{\sum e^2}{n-2}$$

test statistic is $t = \frac{\bar{Z}}{\sqrt{\chi^2/df}}$ ✓

$$\hat{\sigma}^2 = \frac{\sum e^2}{n-2} \quad , \quad \sum e^2 = \sigma^2 \chi^2_{(n-2)}$$

$$\chi^2_{(n-2)} = \frac{\sum e^2}{\sigma^2} \Rightarrow \frac{\chi^2_{(n-2)}}{df} = \frac{\sum e^2}{(n-2)\sigma^2}$$

$$\frac{\chi^2_{(n-2)}}{df} = \frac{\hat{\sigma}^2}{\sigma^2}$$

Hence $t = \frac{\hat{\beta} - \beta}{\sigma \sqrt{\frac{1}{\sum x^2}} \cdot \frac{\hat{\sigma}}{\sigma}}$

$$t = \frac{\hat{\beta} - \beta}{\hat{\sigma} \sqrt{\frac{1}{\sum x^2}}} \sim t_{(n-2)}$$

(i) If $H_1; \beta \neq 0$, CR is $|t| > t_{(\frac{\alpha}{2}, n-2)}$

(ii) $H_1; \beta > 0$, CR is $t > t_{(\alpha, n-2)}$

(iii) $H_1; \beta < 0$, CR is $t < t_{(1-\alpha, n-2)}$

Testing Procedure for $H_0; \beta_1 - \beta_2 = c$ (c is some constant)

(i) $\hat{\beta}_1 - \hat{\beta}_2 = \frac{\sum x_1 y_1}{\sum x_1^2} - \frac{\sum x_2 y_2}{\sum x_2^2}$ is the estimator of $\beta_1 - \beta_2$

(ii) $(\hat{\beta}_1 - \hat{\beta}_2) \sim N\left(\beta_1 - \beta_2, \frac{\sigma_1^2}{\sum x_1^2} + \frac{\sigma_2^2}{\sum x_2^2}\right)$

(iii) Test statistic to be used is

$$Z = \frac{(\hat{\beta}_1 - \hat{\beta}_2) - (\beta_1 - \beta_2)}{\sqrt{\frac{\sigma_1^2}{\sum x_1^2} + \frac{\sigma_2^2}{\sum x_2^2}}} = \frac{(\hat{\beta}_1 - \hat{\beta}_2) - c}{\sqrt{\frac{\sigma_1^2}{\sum x_1^2} + \frac{\sigma_2^2}{\sum x_2^2}}}$$

④ If $\sigma_1 = \sigma_2 = \sigma$ and unknown, then we pool estimate of

of σ^2 as

$$\hat{\sigma}_p^2 = \frac{\sum e_i^2 + \sum q_i^2}{n_1 + n_2 - 4}$$

$$\sum q_i^2 + \sum e_i^2 = \sigma^2 \chi_{(n_1+n_2-4)}^2$$

$$\hat{\sigma}_p^2 = \frac{\sigma^2 \chi^2}{df}$$

$$\chi_{df}^2 = \frac{\hat{\sigma}_p^2}{\sigma^2}$$

$$\text{Now } t = \frac{(\hat{\beta}_1 - \hat{\beta}_2) - c}{\hat{\sigma}_p \sqrt{\frac{1}{\sum x_1^2} + \frac{1}{\sum x_2^2}}} \quad \hat{\sigma}_p / \sigma$$

$$t = \frac{(\hat{\beta}_1 - \hat{\beta}_2) - c}{\hat{\sigma}_p \sqrt{\frac{1}{\sum x_1^2} + \frac{1}{\sum x_2^2}}}$$

45

i) $H_1; \beta_1 - \beta_2 \neq c$, CR is $|t| > t_{(\frac{\alpha}{2}, n_1+n_2-4)}$

ii) $H_1; (\beta_1 - \beta_2) > c$, CR is $|t| > t_{(\alpha, n_1+n_2-4)}$

iii) $H_1; \beta_1 - \beta_2 < c$, CR is $t < t_{(-\alpha, n_1+n_2-4)}$

Test Procedure For $H_0: \varepsilon\alpha + c\beta = 0$

ii) $(\varepsilon\hat{\alpha} + c\hat{\beta})$ is the estimator of $\varepsilon\alpha + c\beta$.

iii) $(\varepsilon\hat{\alpha} + c\hat{\beta}) \sim N(\varepsilon\alpha + c\beta, \varepsilon^2\sigma_\alpha^2 + c^2\sigma_\beta^2 + 2\varepsilon c\sigma_{\alpha\beta})$

$$\text{i.e. } E(\varepsilon\hat{\alpha} + c\hat{\beta}) = \varepsilon E(\hat{\alpha}) + c E(\hat{\beta}) = \varepsilon\alpha + c\beta$$

$$\text{Var}(\varepsilon\hat{\alpha} + c\hat{\beta}) = \varepsilon^2 \text{Var}(\hat{\alpha}) + c^2 \text{Var}(\hat{\beta}) + 2\varepsilon c \text{Cov}(\hat{\alpha}, \hat{\beta})$$

$$= \varepsilon^2 \sigma^2 \frac{\sum X^2}{n \sum x^2} + \frac{c^2 \sigma^2}{\sum x^2} + 2\varepsilon c \left[\frac{-\bar{X} \sigma^2}{\sum x^2} \right]$$

$$= \frac{\sigma^2}{\sum x^2} \left[\frac{\varepsilon^2 \sum X^2}{n} + c^2 - 2\varepsilon c \bar{X} \right] = \sigma^2 (\varepsilon\hat{\alpha} + c\hat{\beta})$$

iv, Test statistic to be used is

$$\textcircled{1} \quad Z = \frac{(\hat{\alpha} + c\hat{\beta}) - (\alpha + c\beta)}{\sigma_{(\hat{\alpha} + c\hat{\beta})}} \quad \begin{array}{l} \text{is SNV if } \sigma \\ \text{is known} \end{array}$$

$\textcircled{2}$ if σ is unknown, then it is replaced by its estimator s

$$t = \frac{(\hat{\alpha} + c\hat{\beta}) - (\alpha + c\beta)}{\sqrt{\frac{1}{\sum x^2} \left[\frac{c^2 \sum X^2}{n} + c^2 - 2c\bar{X} \right]}} \sim t_{(n-2)}$$

Critical region is determined as in above cases.

Discuss the distribution of Maximum likelihood estimator of σ^2 and obtain its mean and variance: 88 I

Maximum likelihood estimator of σ^2 is

$$\sigma^{*2} = \frac{\sum e^2}{n}$$

$$E(\sigma^{*2}) = \frac{1}{n} E(\sum e^2)$$

$$\checkmark \quad E(\sum e^2) = (n-2) \sigma^2 \quad (\text{Not write directly})$$

$$E(\sigma^{*2}) = \frac{1}{n} (n-2) \sigma^2$$

$$\boxed{E(\sigma^{*2}) = \frac{(n-2)}{n} \sigma^2}$$

$$\text{Var}(\sigma^{*2}) = \frac{1}{n^2} \text{Var} \sum e^2$$

$$\text{but } \sum e^2 = \sigma^2 \chi_{n-2}^2$$

$$\text{Var}(\hat{\sigma}^2) = \frac{1}{n^2} \text{Var}(\sigma^2 \chi_{n-2}^2)$$

$$= \frac{\sigma^4}{n^2} \text{Var}(\chi_{n-2}^2)$$

$$\left\{ \begin{array}{l} V(\chi_{(n-2)}^2) = 2(n-2) \end{array} \right.$$

$$\boxed{\text{Var}(\hat{\sigma}^2) = \frac{2(n-2)\sigma^4}{n^2}}$$



Simple Linear Regression model is

$$Y = \alpha + \beta X + \epsilon$$

Least Square estimators are

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}, \quad \hat{\beta} = \frac{\sum xy}{\sum x^2}$$

Maximum likelihood estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{\sum \epsilon^2}{n}$$

unbiased estimator of σ^2 is $\hat{\sigma}^2 = \frac{\sum \epsilon^2}{n-2}$

Dist of SSE = $\sum \epsilon^2 \sim \sigma^2 \chi_{(n-2)}^2$

$$\hat{\sigma}^2 = \frac{\sigma^2 \chi^2}{df}$$

$$\text{Mean}(\hat{\sigma}^2) = E\left(\frac{\sum \epsilon^2}{n-2}\right) = \frac{(n-2)}{(n-2)} \sigma^2 = \sigma^2$$

$$\text{Var}(\hat{\sigma}^2) = \text{Var}\left(\frac{\sum \epsilon^2}{n-2}\right) = \left(\frac{1}{n-2}\right)^2 \cdot \text{Var}(\sigma^2 \chi_{(n-2)}^2)$$

$$\text{Var}(\hat{\sigma}^2) = \frac{\sigma^4 \cdot 2(n-2)}{(n-2)^2} = \frac{2\sigma^4}{n-2}$$

Testing Procedure For $H_0: \sigma^2 = \sigma_0^2$

$\hat{\sigma}^2 = \frac{\sum \epsilon^2}{n-2} = \frac{\sigma^2 \chi^2}{df}$ is an unbiased estimator of σ^2

$$\hat{\sigma}^2 \sim \sigma^2 \chi_{(n-2)}^2$$

Test statistic to be used is

$$\chi^2 = \frac{(n-2) \hat{\sigma}^2}{\sigma_0^2}$$

$$\text{or } \frac{\chi^2}{df} = \frac{\hat{\sigma}^2}{\sigma_0^2}$$

if $H_1: \sigma^2 > \sigma_0^2$ CR is $\chi^2 > \chi_{(n-2)}^2$

Central Confidence Intervals

① $(1-\alpha)100\%$ Confidence Interval for μ

$$(1-\alpha) = P\left[-Z_{\frac{\alpha}{2}} \leq \frac{\hat{\mu} - \mu}{SE(\hat{\mu})} \leq Z_{\frac{\alpha}{2}}\right] \quad \text{If } \sigma \text{ is known}$$

Where $SE(\hat{\mu}) = \sqrt{\frac{\sum X^2}{n \sum x^2}} \sigma$

Multiplying inequality by $SE(\hat{\mu})$

$$(1-\alpha) = P\left[-Z_{\frac{\alpha}{2}} SE(\hat{\mu}) \leq \hat{\mu} - \mu \leq Z_{\frac{\alpha}{2}} SE(\hat{\mu})\right]$$

Subtracting $\hat{\mu}$

$$= P\left[-\hat{\mu} - Z_{\frac{\alpha}{2}} SE(\hat{\mu}) \leq \hat{\mu} - \mu - \hat{\mu} \leq Z_{\frac{\alpha}{2}} SE(\hat{\mu}) - \hat{\mu} + Z_{\frac{\alpha}{2}} SE(\hat{\mu})\right]$$

② Multiplying by (-1) and changing sign of inequality

$$(1-\alpha) = P\left[\hat{\mu} - Z_{\frac{\alpha}{2}} SE(\hat{\mu}) \leq \mu \leq \hat{\mu} + Z_{\frac{\alpha}{2}} SE(\hat{\mu})\right]$$

If σ is unknown, then it is replaced by its estimator $\hat{\sigma}$ and table value used is $t_{\frac{\alpha}{2}, n-2}$

Hence C.I for μ if σ is unknown is

$$\hat{\mu} \pm t_{\frac{\alpha}{2}} SE(\hat{\mu})$$

② $(1-\alpha)100\%$ C.I for β

$$1-\alpha = P\left[-Z_{\frac{\alpha}{2}} \leq \frac{\hat{\beta} - \beta}{SE(\hat{\beta})} \leq Z_{\frac{\alpha}{2}}\right] \quad \text{If } \sigma \text{ known}$$

$$SE(\hat{\beta}) = \sqrt{\frac{\sigma^2}{\sum x^2}}$$

$$(1-\alpha) = P\left[-Z_{\frac{\alpha}{2}} SE(\hat{\beta}) \leq \hat{\beta} - \beta \leq Z_{\frac{\alpha}{2}} SE(\hat{\beta})\right]$$

$$(1-\alpha) = P\left[\hat{\beta} - Z_{\frac{\alpha}{2}} SE(\hat{\beta}) \leq \beta \leq \hat{\beta} + Z_{\frac{\alpha}{2}} SE(\hat{\beta})\right]$$

If σ is unknown then

$$SE(\hat{\beta}) = \frac{\hat{\sigma}}{\sqrt{\sum x_i^2}}$$

where $\hat{\sigma} = \sqrt{\frac{\sum y_i^2}{n-2}}$

and C.I is $\hat{\beta} \pm t_{(\frac{\alpha}{2}, n-2)} SE(\hat{\beta})$

-(1- α) 100% confidence interval for $\alpha_1 - \alpha_2$

$$(1-\alpha) = P\left[\text{Table value} < \frac{(\hat{\alpha}_1 - \hat{\alpha}_2) - (\alpha_1 - \alpha_2)}{SE(\hat{\alpha}_1 - \hat{\alpha}_2)} < \text{Table value} \right]$$

$$SE(\hat{\alpha}_1 - \hat{\alpha}_2) = \sqrt{\frac{\sum X_1^2}{n_1 \sum x_1^2} + \frac{\sum X_2^2}{n_2 \sum x_2^2} \hat{\sigma}_p^2}$$

$$\hat{\sigma}_p^2 = \frac{\sum e_1^2 + \sum e_2^2}{n_1 + n_2 - 4}$$

C.I. is

$$(\hat{\alpha}_1 - \hat{\alpha}_2) \pm t_{(\frac{\alpha}{2}, n_1 + n_2 - 4)} SE(\hat{\alpha}_1 - \hat{\alpha}_2)$$

(1- α) 100% C.I for $\beta_1 - \beta_2$ is

$$1) (\hat{\beta}_1 - \hat{\beta}_2) \pm t_{(\frac{\alpha}{2}, n-2)} SE(\hat{\beta}_1 - \hat{\beta}_2)$$

when $\sigma_1 = \sigma_2 = \sigma$ and unknown

2. when σ is known.

$$SE(\hat{\beta}_1 - \hat{\beta}_2) = \hat{\sigma} \sqrt{\frac{1}{\sum x_1^2} + \frac{1}{\sum x_2^2}}$$

$$SE(\hat{\beta}_1 - \hat{\beta}_2) = \sigma \sqrt{\frac{1}{\sum x_1^2} + \frac{1}{\sum x_2^2}}$$

$$(\hat{\beta}_1 - \hat{\beta}_2) \pm z_{\frac{\alpha}{2}} SE(\hat{\beta}_1 - \hat{\beta}_2)$$

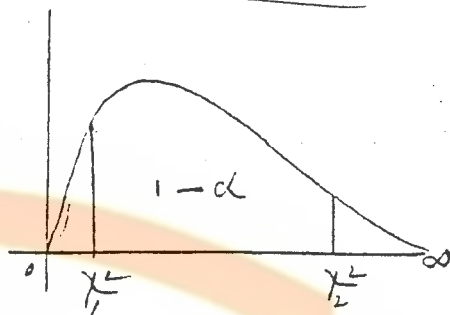
C.I for $\epsilon \alpha + c \beta$

$$(\epsilon \hat{\alpha} + c \hat{\beta}) \pm t_{(\frac{\alpha}{2}, n-2)} SE(\epsilon \hat{\alpha} + c \hat{\beta})$$

$(1-\alpha) 100\%$ confidence interval for σ^2

$$(1-\alpha) = P[\chi_1^2 < \chi^2 < \chi_2^2]$$

$$\chi^2 = \frac{(n-2)\hat{\sigma}^2}{\sigma^2}$$



$$(1-\alpha) = P\left[\chi_1^2 < \frac{(n-2)\hat{\sigma}^2}{\sigma^2} < \chi_2^2\right]$$

$$= P\left[\frac{1}{\chi_1^2} > \frac{\sigma^2}{(n-2)\hat{\sigma}^2} > \frac{1}{\chi_2^2}\right]$$

$$= P\left[\frac{1}{\chi_2^2} < \frac{\sigma^2}{(n-2)\hat{\sigma}^2} < \frac{1}{\chi_1^2}\right]$$

Multiplying by $(n-2)\hat{\sigma}^2$

$$(1-\alpha) = P\left[\frac{(n-2)\hat{\sigma}^2}{\chi_2^2} < \sigma^2 < \frac{(n-2)\hat{\sigma}^2}{\chi_1^2}\right]$$

$(1-\alpha) 100\%$ confidence interval for σ^2 For

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$\text{i.e. } \sigma_1^2 = \sigma_2^2 = \sigma^2$$

we have two sample of size n_1, n_2 pairs of values both follow simple linear regression.

$\sum e_1^2$ and $\sum e_2^2$ are sum of squares of error from two samples respectively.

$$\sum e_1^2 = \sigma_1^2 \chi_{n_1-2}^2, \quad \sum e_2^2 = \sigma_2^2 \chi_{n_2-2}^2$$

By additive property of χ^2 pooled estimate of variance is

$$\hat{\sigma}_p^2 = \frac{\sum e_1^2 + \sum e_2^2}{n_1 + n_2 - 4} = \hat{\sigma}^2 \chi_{(n_1+n_2-4)}^2$$

$$(1-\alpha) = P[\chi^2_L < \chi^2 < \chi^2_U]$$

$$\chi^2_L = \frac{(n_1+n_2-4) \hat{\sigma}_p^2}{\sigma^2}$$

$$(1-\alpha) = P\left[\frac{(n_1+n_2-4) \hat{\sigma}_p^2}{\chi^2_L} < \sigma^2 < \frac{(n_1+n_2-4) \hat{\sigma}_p^2}{\chi^2_U}\right]$$

iii) If σ^2 is ~~known~~ ^{not known} known. i.e. To test $H_0: \sigma_1^2 = \sigma_2^2$

$$\chi^2_1 = \frac{\sum e_1^2}{n_1-2} \quad \frac{\chi^2_1}{df} = \frac{\hat{\sigma}_1^2}{\sigma^2}$$

$$\chi^2_2 = \frac{\sum e_2^2}{n_2-2} \quad \frac{\chi^2_2}{df} = \frac{\hat{\sigma}_2^2}{\sigma^2}$$

Under H_0 ; $F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} \sim F(n_1-2, n_2-2)$

Confidence interval for σ_1^2/σ_2^2

$$P\left[\frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} F_{(\frac{\alpha}{2}, n_2-2, n_1-1)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} (F_{1-\frac{\alpha}{2}, n_2-2, n_1-1})\right]$$

PUACP

H_0 ; $\sigma_1^2 = \sigma_2^2$ Homoscedasticity.

H_0 ; $\sigma_1^2 \neq \sigma_2^2$ Heteroscedasticity.

Test statistic To be Used is

$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2}$$

$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} \sim F(n_1-2, n_2-2)$$

Equality of two Simple Linear Regressions

There are two sets of pairs of values, both follow SLR.

$$\text{i.e. } Y_{1i} = \alpha_1 + \beta_1 X_{1i} + \epsilon_{1i}, \quad i = 1, 2, \dots, n_1$$

$$Y_{2i} = \alpha_2 + \beta_2 X_{2i} + \epsilon_{2i}, \quad i = 1, 2, \dots, n_2$$

Hypothesis to be tested is that

$$H_0: \alpha_1 = \alpha_2 \quad \beta_1 = \beta_2 \quad \text{i.e. Two Regressions are equal}$$

Both hypothesis may be accepted or rejected.

From 'n₁' pairs of values, we have

$$\hat{\alpha}_1 = \bar{Y}_1 - \hat{\beta}_1 \bar{X}_1 \quad \text{and} \quad \hat{\beta}_1 = \frac{\sum x_{1i} y_{1i}}{\sum x_{1i}^2} \quad \text{are estimators of } \alpha_1, \beta_1.$$

Fitted equation is

$$\hat{Y}_1 = \hat{\alpha}_1 + \hat{\beta}_1 X_1 \quad \text{--- (1)}$$

From 'n₂' pairs of values.

$$\text{Estimators of } \alpha_2, \beta_2 \text{ are } \hat{\alpha}_2 = \bar{Y}_2 - \hat{\beta}_2 \bar{X}_2, \quad \hat{\beta}_2 = \frac{\sum x_{2i} y_{2i}}{\sum x_{2i}^2}$$

Fitted regression equation is

$$\hat{Y}_2 = \hat{\alpha}_2 + \hat{\beta}_2 X_2 \quad \text{--- (2)}$$

New SSR from (1) is

$$\begin{aligned} \sum \epsilon_1^2 &= \sum (Y_1 - \hat{Y}_1)^2 = \sum [Y_1 - \hat{\alpha}_1 - \hat{\beta}_1 X_1]^2 \\ &= \sum [Y_1 - \bar{Y}_1 + \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_1 X_1]^2 \\ &= \sum [\underbrace{(Y_1 - \bar{Y}_1)}_{\text{deviation from mean}} - \hat{\beta}_1 \underbrace{(X_1 - \bar{X}_1)}_{\text{deviation from mean}}]^2 \end{aligned}$$

$$\sum \epsilon_1^2 = \sum y_1^2 + \hat{\beta}_1^2 \sum x_1^2 - 2\hat{\beta}_1 \sum x_1 y_1$$

$$= \sum y_1^2 + \hat{\beta}_1^2 \sum x_1^2 - 2\hat{\beta}_1 \sum x_1 y_1$$

$$\sum \epsilon_1^2 = \sum y_1^2 - \hat{\beta}_1 \sum x_1 y_1$$

and

$$\sum \epsilon_1^2 = \sum (Y_1 - \hat{Y}_1)^2 = \sigma_1^2 \chi_{n_1-2}^2 \quad \checkmark$$

Similarly SSR from (2) is

$$\sum \epsilon_2^2 = \sum y_2^2 - \hat{\beta}_2^2 \sum x_2 y_2$$

$$\sum e^2 = \sigma^2 \chi^2_{(n_2-2)}$$

For $\sigma_1 = \sigma_2$ $\sum e^2 + \sum e^2 = \sigma^2 \chi^2_{(n_1+n_2-4)}$ ✓

Now under H_0 , as $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$ i.e. two Regression eqs are the same.

Fitting the simple linear regression by combining the sample is from $n_1 + n_2$ values.

Estimators of α and β are

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}, \quad \hat{\beta} = \frac{\sum xy}{\sum x^2} \quad \text{with } (n_1+n_2) \text{ values.}$$

In this case fitted regression equation is

$$\hat{y} = \hat{\alpha} + \hat{\beta} x$$

Now SS Residual is $\sum e^2 = \sum y^2 - \hat{\beta} \sum xy = \sum (y - \hat{y})^2$

$$\therefore (\sum e^2) = (n_1 + n_2 - 2) \sigma^2$$

$$\sum e^2 = \sigma^2 \chi^2_{(n_1+n_2-2)}$$

Thus

$$(\sum e^2) - (\sum e^2 + \sum e^2) = \sigma^2 \chi^2_{(2)}$$

Hence the statistic

$$F = \frac{(\sum e^2) - (\sum e^2 + \sum e^2)/2}{\frac{\sum e^2}{n_1 + n_2 - 2}} \quad (2, n_1+n_2-2)$$

The larger value of F statistic leads to reject the null hypothesis of equality of two regressions.

Prediction In Simple Linear Regression Model.

One of the main uses of simple linear regression is the prediction in which we want to estimate some value of the dependent variable or we can say that we want to estimate the dependent variable against some fixed value of the independent variable.

Suppose SLR is

$$Y_i = \alpha + \beta X_i + \epsilon_i \quad i = 1, 2, \dots, n$$

Yearly prediction of a factory from 1975-1988

we want to know $\hat{Y}_{1993} = ?$ and its confidence interval when $X = X_0$.

Theoretical value is $Y_0 = \alpha + \beta X_0 + \epsilon_0$ it is known as true value of Y in the prediction period.

where ϵ_0 indicates the value that would be drawn from the disturbance distribution in the prediction period.

$$E(Y_0) = \alpha + \beta X_0 \quad (\text{mean value})$$

we can not estimate ϵ_0 , we prefer to estimate mean value.

We want to know best linear unbiased estimates of

$$E(Y_0) = \alpha + \beta X_0$$

considering any other estimates, say

$$A = \sum C_i Y_i$$

where quantity A is linear in Y -values

$$\text{SLR is } Y_i = \alpha + \beta X_i + \epsilon_i$$

$$A = \sum C_i [\alpha + \beta X_i + \epsilon_i]$$

$$A = \alpha \sum C_i + \beta \sum C_i X_i + \sum C_i \epsilon_i$$

Applying expectation

$$\checkmark E(A) = \alpha \sum C_i + \beta \sum C_i X_i$$

$$E(\epsilon_i) = 0$$

$$E(A) = \alpha + \beta X_0 \quad \text{if } \sum C_i = 1, \sum C_i X_i = X_0$$

$$\text{Var}(X) = E[X - E(X)]^2$$

$$= E(X^2) - \{E(X)\}^2$$

$$\text{Var}(A) = E[A - E(A)]^2$$

$$= E[\sum G_i \epsilon_i]^2$$

$$\text{Var}(A) = E[\sum G_i^2 \epsilon_i^2 + \sum_{i \neq j} 2 G_i G_j \epsilon_i \epsilon_j]$$

$$\text{Var}(A) = \sigma^2 \sum G_i^2$$

Now by using Lagrangian's Multiplier

$$\sum G_i^2 = \sum G_i^2 - 2\mu(\sum G_i - 1) - 2\lambda(\sum G_i X_i - X_0)$$

minimizing w.r.t G_i

$$\frac{\partial \sum G_i^2}{\partial G_i} = 2G_i - 2\mu - 2\lambda(X_i - 0) = 2G_i - 2\mu - 2\lambda X_i$$

$$\frac{\partial^2 \sum G_i^2}{\partial G_i^2} = 2 > 0$$

$$\therefore \frac{\partial \sum G_i^2}{\partial G_i} = 0 \Rightarrow G_i = \mu + \lambda X_i \quad \text{--- (1) } \checkmark$$

Now under conditions $\sum G_i = 1, \sum G_i X_i = X_0$

From (1)

$$\checkmark \sum G_i = n\mu + \lambda \sum X_i$$

$$n\mu + \lambda \sum X_i = 1 \quad \text{--- (2) } \checkmark$$

Again From (1)

$$\sum G_i X_i = \mu \sum X_i + \lambda \sum X_i^2$$

$$\checkmark \mu \sum X_i + \lambda \sum X_i^2 = X_0 \quad \text{--- (3) } \checkmark$$

$$\textcircled{2} \times (\sum X_i)$$

$$n\mu \sum X_i + \lambda (\sum X_i)^2 = \sum X_i$$

$$\textcircled{3} \times n \quad - \quad \frac{n\mu \sum X_i + n\lambda (\sum X_i)^2}{n} = \frac{nX_0}{n}$$

$$\lambda \left[(\sum X_i)^2 - n \sum X_i^2 \right] = \sum X_i - nX_0$$

$$\lambda = \frac{nX_0 - \sum X_i}{n \sum X_i^2 - (\sum X_i)^2}$$

$$\lambda = \frac{X_0 - \bar{X}}{\frac{\sum X_i^2 - (\sum X_i)^2}{n}} = \frac{x_0}{\sum x_i^2}$$

pulling λ in ②

$$n\mu = 1 - \lambda \sum X_i = 1 - \frac{x_0}{\sum x_i^2} \sum X_i$$

Dividing by n on both sides

$$\mu = \frac{1}{n} - \frac{x_0 \bar{X}}{\sum x_i^2}$$

Hence

$$\lambda = \frac{x_0}{\sum x_i^2}, \quad \mu = \frac{1}{n} - \frac{x_0 \bar{X}}{\sum x_i^2}$$

Now by putting the value of λ, μ in ①, we get

$$C_i = \frac{1}{n} - \frac{x_0 \bar{X}}{\sum x_i^2} + \frac{x_0}{\sum x_i^2} X_i = \frac{1}{n} + \frac{x_0 (X_i - \bar{X})}{\sum x_i^2}$$

$$C_i = \frac{1}{n} + \frac{x_0 x_i}{\sum x_i^2}$$

For this value of C_i $\text{Var}(A)$ is minimum.

$$\text{Thus estimator } A = \sum C_i Y_i = \sum \left[\frac{1}{n} + \frac{x_0 x_i}{\sum x_i^2} \right] Y_i$$

$$= \bar{Y} + \frac{\sum x_i y_i}{\sum x_i^2} (x_0 - \bar{X})$$

$$= \bar{Y} + \hat{\beta} x_0 - \hat{\beta} \bar{X} = \bar{Y} - \hat{\beta} \bar{X} + \hat{\beta} x_0$$

$$\therefore \text{So } \hat{Y}_{x=x_0} = A = \hat{\alpha} + \hat{\beta} x_0 \text{ is BLUE of } E(Y_0) = \alpha + \beta x_0$$

$$\sum C_i^2 = C_1^2 + C_2^2 + \dots + C_i^2 + \dots + C_n^2$$

$$\frac{1}{n} \sum C_i^2 = 2 C_i$$

$$2 C_i$$

When X_0 is close to $\bar{X}$, the variance (Y_0) is small, otherwise it is large.

55

Question: Obtain the mean and variance of the point estimator

$$\hat{Y}_0 = \hat{\alpha} + \hat{\beta} X_0$$

C.I. 88I

Sol:

$$\hat{Y}_0 = \hat{\alpha} + \hat{\beta} X_0$$

We know

$$E(\hat{\alpha}) = \alpha, \text{Var}(\hat{\alpha}) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x^2} \right]$$

$$E(\hat{\beta}) = \beta, \text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum x^2}, \text{Cov}(\hat{\alpha}, \hat{\beta}) = -\frac{\bar{X} \sigma^2}{\sum x^2}$$

$$\text{Now } E(\hat{Y}_0) = E(\hat{\alpha}) + X_0 E(\hat{\beta})$$

$$E(\hat{Y}_0) = \alpha + \beta X_0$$

$$\text{If } X = X_0$$

$$E(\hat{Y}_0) = \mu_{Y/X=X_0}$$

$$\text{Var}(\hat{Y}_0) = \text{Var}[\hat{\alpha} + \hat{\beta} X_0]$$

$$= \text{Var}(\hat{\alpha}) + X_0^2 \text{Var}(\hat{\beta}) + 2X_0 \text{Cov}(\hat{\alpha}, \hat{\beta})$$

$$= \sigma^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x^2} \right] + \frac{X_0^2 \sigma^2}{\sum x^2} + 2X_0 \left(-\frac{\bar{X} \sigma^2}{\sum x^2} \right)$$

$$= \sigma^2 \left[\frac{1}{n} + \frac{X_0^2}{\sum x^2} + \frac{\bar{X}^2}{\sum x^2} - \frac{2X_0 \bar{X}}{\sum x^2} \right]$$

$$\text{Var}(\hat{Y}_0) = \sigma^2 \left[\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum x^2} \right]$$

So that the variance of prediction increases the further the X_0 value lies from the mean of the sample values employed to compute $\hat{\alpha}$ and $\hat{\beta}$.

Since $\hat{Y}_0$ is a linear function of $\hat{\alpha}$ and $\hat{\beta}$, which have a bivariate normal dist. It will give a normal distribution with mean $\alpha + \beta X_0$ and variance $\sigma^2 \left[\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum x^2} \right]$

$$\text{Since } \frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim t_{(n-2)}$$

$$\sigma^2 \text{ is replaced by } \hat{\sigma}^2$$

Hence

$$t = \frac{\hat{Y}_0 - E(Y_0/X_0)}{\hat{\sigma} \left[\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum x^2} \right]^{1/2}} \sim t_{(n-2)}$$

$(1-\alpha)100\%$ CI for $E(Y_0/x_0)$

$$(1-\alpha) = P\left[-\text{Table Value} < \frac{\hat{Y}_0 - E(Y_0/x_0)}{SE(\hat{Y}_0)} < +\text{Table Value}\right]$$

$$SE(\hat{Y}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum x^2}}$$

$$(1-\alpha) = P\left[\hat{Y}_0 - SE(\hat{Y}_0)TV < E(Y_0/x_0) < \hat{Y}_0 + SE(\hat{Y}_0)TV\right]$$

Prediction For Individual Y_0

Sometimes we may be more interested in obtaining a confidence interval for the individual Y_0 value associated with the X_0 . To deal with these problems, we proceed as follows.

Since residual is $e_i = Y_i - \hat{Y}_i$

$$e_i = (\alpha + \beta X_i + \epsilon_i) - (\hat{\alpha} + \hat{\beta} X_i)$$

$$\text{For } x = x_0, e_0 = (\alpha + \beta X_0 + \epsilon_0) - (\hat{\alpha} + \hat{\beta} X_0)$$

$$= \alpha + \beta X_0 + \epsilon_0 - \hat{\alpha} - \hat{\beta} X_0$$

$$e_0 = (\alpha - \hat{\alpha}) + (\beta - \hat{\beta}) X_0 + \epsilon_0$$

Applying expectation

$$E(e_0) = E(\alpha - \hat{\alpha}) + E(\beta - \hat{\beta}) X_0 + E(\epsilon_0)$$

$$E(e_0) = -E(\hat{\alpha} - \alpha) - E(\hat{\beta} - \beta) X_0 + E(\epsilon_0)$$

$$\text{Because } E(\hat{\alpha}) = \alpha, E(\hat{\beta}) = \beta, E(\epsilon_0) = 0$$

$$\text{Therefore } E(e_0) = 0$$

$$\text{and } Var(e_0) = E[e_0 - E(e_0)]^2 = E e_0^2$$

$$= E[(\alpha - \hat{\alpha}) + (\beta - \hat{\beta}) X_0 + \epsilon_0]^2$$

$$= E(\alpha - \hat{\alpha})^2 + X_0^2 E(\beta - \hat{\beta})^2 + E \epsilon_0^2$$

$$+ 2 X_0 E(\alpha - \hat{\alpha})(\beta - \hat{\beta}) + 2 E(\alpha - \hat{\alpha}) \epsilon_0$$

$$+ 2 X_0 E(\beta - \hat{\beta}) \epsilon_0$$

Since $(\alpha - \hat{\alpha})$ and $(\beta - \hat{\beta})$ involve $\epsilon_1, \epsilon_2, \dots, \epsilon_n$, since ϵ_0 is independent of $\epsilon_1, \epsilon_2, \dots, \epsilon_n$, so these product with ϵ_0 vanish under expectation.

Now

$$\begin{aligned} \text{Var}(e_0) &= E e_0^2 = \sigma^2 + \text{Var}(\hat{\alpha}) + \text{Var}(\hat{\beta}) X_0^2 + 2X_0 \text{Cov}(\hat{\alpha}, \hat{\beta}) \\ &= \sigma^2 + \left[\frac{1}{n} + \frac{X_0 - \bar{X}}{\sum x_i^2} \right] \sigma^2 + \frac{\sigma^2}{\sum x_i^2} X_0^2 - \frac{2X_0 \bar{X} \sigma^2}{\sum x_i^2} \end{aligned}$$

where $\text{Var}(\hat{\alpha}) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x_i^2} \right]$, $\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum x_i^2}$.

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = -\frac{\bar{X} \sigma^2}{\sum x_i^2}$$

$$\text{Var } e_0 = E e_0^2 = \sigma^2 \left[\frac{1}{n} + \frac{1}{\sum x_i^2} \left\{ X_0^2 + \bar{X}^2 - 2X_0 \bar{X} \right\} \right] \sigma^2$$

$$\text{Var } e_0 = E(e_0^2) = \sigma^2 \left[1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum x_i^2} \right] \quad \text{--- (1)}$$

Since prediction error e_0 is the linear function of the normal variables $(\hat{\alpha}, \hat{\beta} \text{ and } \epsilon_0)$ so e_0 itself follows Normal dist with mean 0 and variance given by (1).

Hence

$$Z = \frac{e_0 - 0}{\sigma \sqrt{\left[\frac{1}{n} + 1 + \frac{(X_0 - \bar{X})^2}{\sum x_i^2} \right]}} \text{ is S.N.V.}$$

If σ is unknown, then

$$\hat{\sigma}_{\text{est}}^2 = \frac{\hat{\sigma}^2}{\sigma^2}$$

and

$$t = \frac{Z}{\sqrt{\hat{\sigma}_{\text{est}}^2}} = \frac{e_0}{\hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum x_i^2}}} \sim t_{(n-2)}$$

$(1-\alpha)100\%$ CI for γ_0 (Individual value of γ for $X=X_0$)

$$(1-\alpha) = P\left[-t_{\frac{\alpha}{2}} \leq t \leq t_{\frac{\alpha}{2}}\right]$$

$$= P\left[-t_{\frac{\alpha}{2}} \leq \frac{Y_0 - \hat{\gamma}_0}{SE} \leq t_{\frac{\alpha}{2}}\right]$$

$$1-\alpha = P\left[\hat{\gamma}_0 - t_{\frac{\alpha}{2}} SE \leq Y_0 \leq \hat{\gamma}_0 + t_{\frac{\alpha}{2}} SE\right]$$

Hence $(1-\alpha)100\%$ Confidence Limits are

$$\hat{\gamma}_0 \pm t_{\frac{\alpha}{2}} SE$$

Exercise (Johnston)

Q 2.3; let $\hat{\alpha} = \sum_{i=1}^n C_i Y_i$ where $Y_i = \alpha + \beta X_i + \epsilon_i$. Using Lagrangian multipliers, find the weights C_i ($i=1, 2, \dots, n$) which will make $\hat{\alpha}$ a best linear unbiased estimate of α and show that

$$C_i = \frac{1}{n} - \frac{\bar{X} X_i}{\sum X_i^2}$$

Sol, Estimator of α is $\hat{\alpha} = \sum_{i=1}^n C_i Y_i$

putting $Y_i = \alpha + \beta X_i + \epsilon_i$

$$\hat{\alpha} = \sum_{i=1}^n C_i (\alpha + \beta X_i + \epsilon_i)$$

$$= \alpha \sum_{i=1}^n C_i + \beta \sum_{i=1}^n C_i X_i + \sum_{i=1}^n C_i \epsilon_i$$

$$E(\hat{\alpha}) = \alpha \quad \text{if} \quad \sum_{i=1}^n C_i = 1, \quad \sum_{i=1}^n C_i X_i = 0$$

$$\text{Var}(\hat{\alpha}) = \sum_{i=1}^n C_i^2 \text{Var}(Y_i) = \sigma^2 \sum_{i=1}^n C_i^2$$

By using Lagrangian multiplier

$$F = \sum C_i^2 - 2\lambda \left(\sum_{i=1}^n C_i - 1 \right) - 2\mu \left(\sum_{i=1}^n C_i X_i \right)$$

$$\frac{\partial F}{\partial C_i} = 2C_i - 2\lambda - 2\mu X_i$$

$$\frac{\partial F}{\partial C_i} = 2 > 0 \quad \therefore \frac{\partial F}{\partial C_i} = 0$$

$$C_i = \lambda + \mu X_i \quad \text{--- ①}$$

Now applying the conditions of unbiasedness i.e. $\sum C_i = 1$, $\sum C_i X_i = 0$

$$\sum_{i=1}^n C_i = n\lambda + \mu \sum_{i=1}^n X_i$$

$$n\lambda + \mu \sum_{i=1}^n X_i = 1 \quad \text{--- ②}$$

$$\sum_{i=1}^n C_i X_i = \lambda \sum_{i=1}^n X_i + \mu \sum_{i=1}^n X_i^2$$

$$\lambda \sum_{i=1}^n X_i + \mu \sum_{i=1}^n X_i^2 = 0 \quad \text{--- ③}$$

From ③

$$\mu = -\lambda \frac{\sum X_i}{\sum X_i^2} \quad \text{--- ④} \quad \text{putting in ②}$$

$$n\lambda - \lambda \frac{(\sum X_i)^2}{\sum X_i^2} = 1, \quad \lambda \left(n - \frac{(\sum X_i)^2}{\sum X_i^2} \right) = 1$$

$$\lambda = \frac{1}{n - \frac{(\sum X_i)^2}{\sum X_i^2}} = \frac{\sum X_i^2}{n \sum X_i^2 - (\sum X_i)^2}$$

$$\therefore \sum X_i^2 = \sum x_i^2 - (\sum x_i)^2$$

$$\boxed{\lambda = \frac{\sum X_i^2}{n \sum x_i^2}}$$

From ④

$$\mu = \frac{-\sum X_i^2}{n \sum x_i^2} \cdot \frac{\sum X_i}{\sum X_i^2} = \frac{-\sum X_i}{n \sum x_i^2}$$

$$\boxed{\mu = -\frac{\bar{X}}{\sum x_i^2}}$$

putting λ, μ in ①

$$C_i = \frac{\sum X_i^2}{n \sum x_i^2} - \frac{\bar{X}}{\sum x_i^2} X_i$$

$$\begin{aligned}
 x_i &= x - \bar{x} \Rightarrow x = x_i + \bar{x} \Rightarrow x^2 = (x_i + \bar{x})^2 \Rightarrow \sum x_i^2 = 0 \\
 \sum x_i^2 &= \sum (x_i + \bar{x})^2 \Rightarrow \sum (x_i^2 + \bar{x}^2 + 2x_i\bar{x}) \\
 \sum x_i^2 &= \sum x_i^2 + n\bar{x}^2 + 2\bar{x}\sum x_i \Rightarrow \sum x_i^2 + n\bar{x}^2 \\
 \text{So, } \frac{\sum x_i^2}{n\sum x_i^2} &= \frac{\sum x_i^2}{n\sum x_i^2} + \frac{n\bar{x}^2}{n\sum x_i^2} \Rightarrow \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum x_i^2} \right] \Leftrightarrow \frac{\sum x_i^2}{n\sum x_i^2}
 \end{aligned}$$

$$\begin{aligned}
 \frac{\sum x_i^2}{n\sum x_i^2} &= \frac{1}{n} + \frac{\sum x_i^2}{n\sum x_i^2} - \frac{1}{n} \\
 &= \frac{1}{n} + \frac{\sum x_i^2 - \sum x_i^2}{n\sum x_i^2} = \frac{1}{n} + \frac{\sum x_i^2 - \sum (x_i - \bar{x})^2}{n\sum x_i^2} \\
 &= \frac{1}{n} + \frac{(\sum x_i^2 - \sum x_i^2 - n\bar{x}^2 + 2n\bar{x}^2)}{n\sum x_i^2}
 \end{aligned}$$

$$\frac{\sum x_i^2}{n\sum x_i^2} = \frac{1}{n} + \frac{\bar{x}^2}{\sum x_i^2}$$

$$C_i = \frac{1}{n} + \frac{\bar{x}^2}{\sum x_i^2} - \frac{\bar{x}x_i}{\sum x_i^2} = \frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{\sum x_i^2}$$

$$C_i = \frac{1}{n} - \frac{x_i\bar{x}}{\sum x_i^2}$$

For this value of C variance is minimum.

Q.2.4; Show that if Z_i are independent quantities from the same population with variance σ^2 , then the sampling variance of $b = \sum_{i=1}^n a_i z_i$ is $\sigma^2 \sum_{i=1}^n a_i^2$. Observations Y_i are related to fixed quantities X_i and the quantities Z_i above, by the relations

$$Y_i = \alpha + \beta X_i + Z_i \quad (i=1, 2, \dots, n)$$

	X_1	X_2	X_3	X_4	X_5	X_6
	1	2	3	4	5	6

an alternative estimate of β is $\frac{1}{3}(Y_6 + Y_5 - Y_2 - Y_1)$. Deduce the sampling variance of this estimate and compare it with the sampling variance of the least-squares estimate.

Sol;

$$\begin{aligned}
 b &= \sum_{i=1}^n a_i z_i \\
 &= a_1 z_1 + a_2 z_2 + \dots + a_n z_n \\
 E(b) &= a_1 E(z_1) + a_2 E(z_2) + \dots + a_n E(z_n) \\
 E(z_i) &= 0
 \end{aligned}$$

$$\begin{aligned}
 \text{As } Z_i &\sim N(0, \sigma^2) \\
 \text{Var}(b) &= E(b^2) - [E(b)]^2 \\
 \text{Var}(b) &= E\left(\sum_{i=1}^n a_i z_i\right)^2 = 0 = \sigma^2 \sum_{i=1}^n a_i^2
 \end{aligned}$$

Now $Y_i = \alpha + \beta X_i + \varepsilon_i$

As $\beta^* = \frac{1}{8} (Y_6 + Y_5 - Y_2 - Y_1)$

$$= \frac{1}{8} [\alpha + \beta X_6 + \varepsilon_6 + \alpha + \beta X_5 + \varepsilon_5 - \alpha - \beta X_2 - \varepsilon_2 - \alpha - \beta X_1 - \varepsilon_1]$$

$$\beta^* = \frac{1}{8} [X_6 + X_5 + X_2 + X_1] \beta + \frac{1}{8} [\varepsilon_6 + \varepsilon_5 - \varepsilon_2 - \varepsilon_1]$$

By putting the values of X_i

$$\beta^* = \frac{1}{8} [6 + 5 - 2 - 1] \beta + \frac{1}{8} [\varepsilon_6 + \varepsilon_5 - \varepsilon_2 - \varepsilon_1]$$

Applying expectation

$$E(\beta^*) = \beta + \frac{1}{8} [E(\varepsilon_6) + E(\varepsilon_5) - E(\varepsilon_2) - E(\varepsilon_1)]$$

$$E(\varepsilon_i) = 0 \quad i=1, 2, \dots, 6. \quad \therefore \varepsilon_i \text{ are}$$

are independent

$$E(\beta^*) = \beta$$

Hence β^* is an unbiased estimator of β .

Now

$$\text{Var}(\beta^*) = E[\beta^* - E(\beta^*)]^2 = E[\beta^* - \beta]^2$$

$$\beta^* = \beta + \frac{1}{8} [\varepsilon_6 + \varepsilon_5 + \varepsilon_2 - \varepsilon_1]$$

$$\beta^* - \beta = \frac{1}{8} [\varepsilon_6 + \varepsilon_5 - \varepsilon_2 - \varepsilon_1]$$

$$\text{Var}(\beta^*) = E\left[\frac{1}{8} (\varepsilon_6 + \varepsilon_5 - \varepsilon_2 - \varepsilon_1)\right]^2$$

$$= \frac{1}{64} \left[E(\varepsilon_6)^2 + E(\varepsilon_5)^2 + E(\varepsilon_2)^2 + E(\varepsilon_1)^2 + 2E(\varepsilon_6 \varepsilon_5) - 2E(\varepsilon_6 \varepsilon_2) - 2E(\varepsilon_6 \varepsilon_1) - 2E(\varepsilon_5 \varepsilon_2) - 2E(\varepsilon_5 \varepsilon_1) + 2E(\varepsilon_2 \varepsilon_1) \right]$$

$$\text{Var}(\varepsilon_1) = E(\varepsilon_1)^2 = \sigma^2 \quad \text{because } E(\varepsilon_i) = 0, \quad E(\varepsilon_i \varepsilon_j) = 0$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

$$\text{Var}(\beta^*) = \frac{1}{64} [\sigma^2 + \sigma^2 + \sigma^2 + \sigma^2] = \frac{\sigma^2}{16}$$

$$\text{Var}(\beta^*) = \frac{\sigma^2}{16}$$

We know that $\text{LSE of } \beta$ is

$$\hat{\beta} = \frac{\sum xy}{\sum x^2}, \quad \text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum x_i^2} = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$$

$$\text{Var}(\hat{\beta}) = \frac{0.2}{17.5}$$

Sampling variance of LSE $\leq \text{Var}(\hat{\beta})^*$ So LSE is more precise.

Q2.5; From a sample of 200 pairs of observations the following quantities were calculated

$$\sum X = 11.34, \quad \sum Y = 20.72, \quad \sum X^2 = 12.16, \quad \sum Y^2 = 84.96, \quad \sum XY = 22.15$$

Estimate the two regression lines and the variance of the estimated regression coefficient of Y on X.

Sol; The linear regression equation of Y on X is

$$\hat{Y} = \hat{\alpha}_1 + \hat{\beta}_1 X$$

$$\hat{\alpha}_1 = \bar{Y} - \hat{\beta}_1 \bar{X}, \quad \hat{\beta}_1 = \frac{\sum xy}{\sum x^2}$$

$$\bar{Y} = \frac{\sum Y}{n} = 0.1036, \quad \bar{X} = \frac{\sum X}{n} = 0.0567$$

$$\sum y^2 = \sum (Y - \bar{Y})^2 = \sum Y^2 - n \bar{Y}^2 = 84.96 - 200(0.1036)^2 = 82.8134$$

$$\sum x^2 = \sum (X - \bar{X})^2 = \sum X^2 - n \bar{X}^2 = 12.16 - 200(0.0567)^2 = 11.5170$$

$$\begin{aligned} \sum xy &= \sum (X - \bar{X})(Y - \bar{Y}) = \sum XY - n \bar{X} \bar{Y} \\ &= 22.15 - 200(0.1036)(0.0567) \end{aligned}$$

$$\sum xy = 20.9552$$

$$\hat{\beta}_1 = \frac{20.9552}{11.5170} = 1.8195$$

$$\hat{\alpha}_1 = 0.1036 - (1.8195)(0.0567) = 0.000434$$

Hence estimated regression eq Y on X is

$$\hat{Y} = 0.000434 + 1.8195 X$$

$$\hat{\sigma}^2 = \frac{\sum e^2}{n-2} = \frac{\sum y^2 - \hat{\beta}_1^2 \sum x^2}{n-2}$$

$$\hat{\sigma}^2 = \frac{1}{198} [82.8134 - (1.8195)^2 (11.5170)]$$

$$\hat{\sigma}^2 = 0.22563$$

$$\text{Var}(\hat{\alpha}_1) = \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum x^2} \right] \hat{\sigma}^2 = \frac{\sum X^2}{n \sum x^2} \hat{\sigma}^2$$

$$= \frac{12.16}{200(11.5170)} (0.22563)$$

$$\text{Var}(\hat{\alpha}_1) = 0.0011914$$

$$\text{Var}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{\sum x^2} = \frac{0.22563}{11.5170} = 0.0196$$

$$\text{Var}(\hat{\beta}_1) = 0.0196$$

Now estimated regression equation X on Y is

$$\hat{X} = \hat{\alpha}_2 + \hat{\beta}_2 Y$$

$$\hat{\alpha}_2 = \bar{X} - \hat{\beta}_2 \bar{Y}, \quad \hat{\beta}_2 = \frac{\sum xy}{\sum y^2}$$

$$\hat{\beta}_2 = \frac{20.9552}{82.8134} = 0.25304$$

$$\hat{\alpha}_2 = 0.0567 - (0.25304)(0.1036) = 0.0305$$

Hence estimated regression equation X on Y is

$$\hat{X} = 0.0305 + 0.25304 Y$$

Q2.7; There are two herds. Calculate the linear regression eq of Y on X for each herd and test whether two lines differ in slope.

Sol; Herd A, $n_1 = 16$, $\sum X_1 = 51.13$, $\sum Y_1 = 117.25$

$$\sum x_1^2 = 1.27, \quad \sum y_1^2 = 4.76, \quad \sum x_1 y_1 = 1.84$$

The linear regression eq of Y on X is

$$\hat{Y}_1 = \hat{\alpha}_1 + \hat{\beta}_1 X_1$$

$$\bar{Y}_1 = \frac{\sum Y_1}{n_1} = 7.33$$

$$\hat{\alpha}_1 = \bar{Y}_1 - \hat{\beta}_1 \bar{X}_1, \quad \hat{\beta}_1 = \frac{\sum x_1 y_1}{\sum x_1^2}$$

$$\bar{X}_1 = \frac{\sum X_1}{n_1} = 3.20$$

$$\hat{\beta}_1 = \frac{1.84}{1.27} = 1.45, \quad \hat{\alpha}_1 = 7.33 - 1.45(3.20) = 2.69$$

$$\hat{Y}_1 = 2.69 + 1.45 X_1$$

Hotel B, $n_2 = 10$, $\sum X_2 = 37.20$, $\sum Y_2 = 78.75$, $\sum x_2^2 = 1.03$,

$$\sum y_2^2 = 2.48, \sum x_2 y_2 = 1.10$$

$$\hat{\beta}_2 = \frac{\sum x_2 y_2}{\sum x_2^2} = \frac{1.10}{1.03} = 1.07, \bar{X}_2 = 3.72, \bar{Y}_2 = 7.875$$

$$\hat{\alpha}_2 = \bar{Y}_2 - \hat{\beta}_2 \bar{X}_2$$

$$\hat{\alpha}_2 = 7.875 - 1.07(3.72) = 3.8946$$

Hence estimated eqn is

$$\hat{Y}_2 = 3.8946 + 1.07 X_2$$

Now we want to test the hypothesis

$$H_0: \beta_1 = \beta_2$$

$$H_1: \beta_1 \neq \beta_2$$

$$\alpha = 0.05$$

$$\text{as } \hat{\beta}_1 \sim NI\left(\beta_1, \frac{\hat{\sigma}_1^2}{\sum x_1^2}\right), \hat{\beta}_2 \sim NI\left(\beta_2, \frac{\hat{\sigma}_2^2}{\sum x_2^2}\right)$$

$$\hat{\beta}_1 - \hat{\beta}_2 \sim NI\left(\beta_1 - \beta_2, \frac{\hat{\sigma}_1^2}{\sum x_1^2} + \frac{\hat{\sigma}_2^2}{\sum x_2^2}\right)$$

$$\text{If } \hat{\sigma}_1^2 = \hat{\sigma}_2^2 = \hat{\sigma}^2$$

then

$$\hat{\sigma}^2 = \frac{\sum e_1^2 + \sum e_2^2}{n_1 + n_2 - 4}$$

$$\sum e_1^2 = \sum y_1^2 - \hat{\beta}_1^2 \sum x_1^2 = 4.78 - (1.45)^2(1.27)$$

$$\sum e_1^2 = 2.11$$

$$\sum e_2^2 = \sum y_2^2 - \hat{\beta}_2^2 \sum x_2^2 = 2.48 - (1.07)^2(1.03)$$

$$\sum e_2^2 = 1.30$$

$$\hat{\sigma}^2 = \frac{2.11 + 1.30}{18 + 10 - 4} = 0.155$$

Test Statistic To be used is

$$t = \frac{(\hat{\beta}_1 - \hat{\beta}_2) - (\beta_1 - \beta_2)}{\hat{\sigma}^2 \sqrt{\frac{1}{\sum x_1^2} + \frac{1}{\sum x_2^2}}}$$

$$t = \frac{1.45 - 1.07}{(0.311) \sqrt{\frac{1}{1.27} + \frac{1}{1.03}}} = 0.728$$

Critical region

$$|t| > t_{(1-\frac{\alpha}{2}, n-2)} = 2.074$$

Conclusion:- Since our calculated value lies in acceptance region, so data provides no evidence to reject the hypothesis. So we accept the hypothesis that the two lines differ in slopes.

Q.2.12; A sample of 20 observations corresponding to the regression model.

$$Y = \alpha + \beta X + \epsilon$$

1985 II

The data is $\Sigma Y = 21.9$, $\Sigma (Y - \bar{Y})^2 = 86.9$, $\Sigma (X - \bar{X})(Y - \bar{Y}) = 106.4$
 $\Sigma X = 186.2$, $\Sigma (X - \bar{X})^2 = 215.4$

Estimate α and β and estimate the variance of your estimates. Estimate the conditional mean value of Y corresponding to the value of X fixed at $X = 10$ and find 95% CI for this mean.

Sol, unbiased estimates of α and β are

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X} \quad , \quad \hat{\beta} = \frac{\Sigma (X - \bar{X})(Y - \bar{Y})}{\Sigma (X - \bar{X})^2}$$

$$\hat{\beta} = \frac{106.4}{215.4} = 0.494 \quad , \quad \hat{\alpha} = 1.09 - (0.494)(9.31) = -3.51$$

Now $\text{Var}(\hat{\alpha}) = \left[\frac{1}{n} + \frac{\bar{X}^2}{\Sigma (X - \bar{X})^2} \right] \hat{\sigma}^2$

An unbiased estimate of σ^2 is $\hat{\sigma}^2 = \frac{\Sigma e^2}{n-2}$

$$\Sigma e^2 = \Sigma Y^2 - \hat{\beta}^2 \Sigma X^2$$

$$\Sigma e^2 = 86.9 - (0.494)^2 (215.4) = 34.3346$$

$$\hat{\sigma}^2 = \frac{34.3346}{18} = 1.9075 \quad , \quad \bar{X} = \frac{\Sigma X}{n} = 9.31$$

$$\text{Var}(\hat{\alpha}) = \left[\frac{1}{20} + \frac{(9.31)^2}{215.4} \right] (1.9075) = 0.8629$$

$$\text{Var}(\hat{\beta}) = \frac{\hat{\sigma}^2}{\Sigma X^2} = \frac{1.9075}{215.4} = 0.00886$$

Estimated linear regression equation of Y on X is

$$\hat{Y} = \hat{\alpha} + \hat{\beta} X$$

$$\hat{Y}_0 = -3.51 + 0.494 X_0$$

For $X_0 = 10$ $\hat{Y}_0 = 1.43$

95% CI for $E(Y|x_0=10)$ is

$$(\hat{\alpha} + \hat{\beta}x_0) \pm t_{\frac{\alpha}{2}, n-2} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum x^2}} (\hat{\sigma})$$

$$1.43 \pm (2.101) \sqrt{\frac{1}{20} + \frac{(10 - 9.31)^2}{215.4}} (1.38)$$

$$t_{0.025, 19} = 2.101$$

$$1.43 \pm 0.6625$$

$$0.7675, 2.0925$$

9, 10, 11

Q2.11) If the sample values of X in the linear model

1989

$$Y_t = \alpha + \beta X_t + u_t$$

have zero mean, show that the covariance of the least-squares estimates of α and β is zero. Hence or otherwise, prove that an unbiased estimator of β can be derived by estimating the equation

$$\hat{Y}_t = \hat{\beta} X_t$$

which is constrained to pass through the origin. What is covariance of this estimator of β ?

Sol;

$$Y_t = \alpha + \beta X_t + \epsilon_t$$

$$\text{Mean}(X_t) = 0$$

$$\epsilon_t \sim N(0, \sigma^2)$$

Now LSE of α is $\hat{\alpha} = \bar{Y}$

$$\hat{\beta} = \frac{\sum XY}{\sum X^2}$$

$$\hat{\alpha} = \frac{1}{n} \sum (\alpha + \beta X_i + \epsilon_i) = \alpha + \beta \bar{X} + \frac{\sum \epsilon_i}{n}$$

$$\hat{\alpha} - \alpha = \frac{\sum \epsilon_i}{n} - (1) \because \bar{X} = 0$$

$$\hat{\beta} = \frac{1}{\sum X^2} \sum X(\alpha + \beta X_i + \epsilon_i)$$

$$= \frac{1}{\sum X^2} [\alpha \sum X + \beta \sum X_i^2 + \sum \epsilon_i X_i]$$

$$\hat{\beta} = \beta + \frac{\sum X_i \epsilon_i}{\sum X^2} \quad \text{--- (2)}$$

$$\hat{\beta} - \beta = \frac{\sum X_i \epsilon_i}{\sum X_i^2}$$

$$\text{New } \text{Cov}(\hat{\alpha}, \hat{\beta}) = E(\hat{\alpha} - \alpha)(\hat{\beta} - \beta)$$

$$= E\left(\frac{1}{n} \sum \epsilon_i\right) \left(\frac{\sum X_i \epsilon_i}{\sum X_i^2}\right)$$

$$= \frac{1}{n} E \left[\frac{\sum \epsilon_i^2 X_i + \sum \epsilon_i \epsilon_j X_i}{n \sum X_i^2} \right]$$

$$= \frac{\sigma^2 \sum X_i}{n \sum X_i^2} \quad \because E \epsilon_i \epsilon_j = 0$$

$$\boxed{\text{Cov}(\hat{\alpha}, \hat{\beta}) = 0}$$

(ii)

$$\hat{y}_t = \hat{\beta} x_t$$

$$\text{NEW } \sum xy = \hat{\beta} \sum x^2 \Rightarrow \hat{\beta} = \frac{\sum xy}{\sum x^2}$$

From (2)

$$E(\hat{\beta}) = \beta \quad \because E \epsilon_i = 0$$

$$\text{Var}(\hat{\beta}) = E(\hat{\beta} - \beta)^2 = E\left(\frac{\sum X_i \epsilon_i}{\sum X_i^2}\right)^2$$

$$= E\left[\frac{\sum X_i^2 \epsilon_i^2 + \sum_{i \neq j} X_i X_j \epsilon_i \epsilon_j}{(\sum X_i^2)^2}\right]$$

$$\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum X_i^2}$$

GENERAL LINEAR REGRESSION

(It is an extension of simple linear regression, and it involves more than one independent variables; then model is)

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_K X_{Ki} + \epsilon_i$$

where K regressors and ϵ $(K+1)$ parameters are included.
 K independent variables ✓

OR

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_K X_{Ki} + \epsilon_i$$

where $(K-1)$ regressors, K parameters are included ✓

Let we have n sets of values (X_i, Y_i) $i = 1, 2, \dots, n$ ✓

$$Y_1 = \beta_1 + \beta_2 X_{21} + \beta_3 X_{31} + \dots + \beta_K X_{K1} + \epsilon_1$$

$$Y_2 = \beta_1 + \beta_2 X_{22} + \beta_3 X_{32} + \dots + \beta_K X_{K2} + \epsilon_2$$

$$Y_n = \beta_1 + \beta_2 X_{2n} + \beta_3 X_{3n} + \dots + \beta_K X_{Kn} + \epsilon_n$$

These equations can be expressed as

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{21} & X_{31} & \dots & X_{K1} \\ 1 & X_{22} & X_{32} & \dots & X_{K2} \\ 1 & X_{23} & X_{33} & \dots & X_{K3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{2n} & X_{3n} & \dots & X_{Kn} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \vdots \\ \beta_K \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

(12)

$$Y = X \cdot \beta + \epsilon$$

$$n \times 1 \quad (n \times K) \quad (K \times 1) \quad n \times 1$$

Assumptions; i) ϵ_i 's $\sim NII(0, \sigma^2)$

ii) The explanatory variables $x_1, x_2, \dots, x_k$ are fixed and distributed without error and they are linearly independent. So matrix X should be of full rank i.e.

$$\text{Rank}(X) = \text{Rank}(X'X) = k \leq n \text{ vector}$$

* In matrix notation; $\underline{\epsilon}$ is normal vector such that

$$E(\underline{\epsilon}) = E \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \underline{0}$$

Variance Covariance matrix is

$$E(\underline{\epsilon}\underline{\epsilon}') = \text{Var}(\underline{\epsilon}) = \begin{pmatrix} \epsilon_1^2 & \epsilon_1\epsilon_2 & \epsilon_1\epsilon_3 & \dots & \epsilon_1\epsilon_n \\ \epsilon_2\epsilon_1 & \epsilon_2^2 & \epsilon_2\epsilon_3 & \dots & \epsilon_2\epsilon_n \\ \epsilon_3\epsilon_1 & \epsilon_3\epsilon_2 & \epsilon_3^2 & \dots & \epsilon_3\epsilon_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \epsilon_n\epsilon_1 & \epsilon_n\epsilon_2 & \epsilon_n\epsilon_3 & \dots & \epsilon_n^2 \end{pmatrix}$$

$$\underline{E(\underline{\epsilon}\underline{\epsilon}')} = \sigma^2 \underline{I}_n \quad \text{as} \quad \underline{E(\epsilon_i^2)} = \sigma^2, \quad \underline{E(\epsilon_i\epsilon_j)} = 0 \quad (i \neq j)$$

(Properties of variance Covariance matrix)

- ① It is a symmetric matrix.
- ② It is a positive definite matrix.

$$E(\underline{\epsilon}\underline{\epsilon}') = \begin{bmatrix} \sigma^2 & 0 & 0 & 0 & \dots & 0 \\ 0 & \sigma^2 & 0 & 0 & \dots & 0 \\ 0 & 0 & \sigma^2 & 0 & \dots & 0 \\ 0 & 0 & 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \sigma^2 \end{bmatrix}$$

Least Square Estimators

73

Remove Watermark Now

$$Y = X\beta + \epsilon$$

$$\hat{Y} = X\hat{\beta}$$

$$Y_i = \beta_1 + \beta_2 X_{i1} + \beta_3 X_{i2} + \dots + \beta_K X_{iK} + \epsilon_i$$

$$\epsilon_{i,b} \sim NID(0, \sigma^2)$$

$$\text{Residual is } e = Y - \hat{Y} = Y - X\hat{\beta}$$

If $\hat{\beta}$ is an estimator of β then S.S. residual is

$$\sum_{i=1}^n e_i^2 = e'e = (Y - X\hat{\beta})'(Y - X\hat{\beta})$$

$$(AB)^t = B^t A^t$$

$$e'e = Y'Y - Y'X\hat{\beta} - \hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta}$$

$$\text{order of } Y'X\hat{\beta} = 1 \times n \times n \times K = 1 \times K$$

$$(X\hat{\beta})^t = \hat{\beta}^t X^t$$

$\hat{\beta}'X'Y$ is transpose of $Y'X\hat{\beta}$ and its order is

It means that second and third terms are equivalent.

$$\hat{\beta}' = \beta'$$

$$e'e = Y'Y - (2Y'X\hat{\beta}) + \hat{\beta}'X'X\hat{\beta}$$

minimizing w.r.t $\hat{\beta}$

$$\frac{\partial e'e}{\partial \hat{\beta}} = 0 - (2Y'X) + 2\hat{\beta}'X'X$$

$$\frac{\partial e'e}{\partial \hat{\beta}} = 0 \text{ gives Normal equation of } \beta$$

$$\text{check order}$$

$$2Y'X = 2\hat{\beta}'X'X$$

Taking transpose

$$X'Y = X'X\hat{\beta}$$

$$2X'Y = 2X'X\hat{\beta}$$

$$X'Y = X'X\hat{\beta}$$

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$PG = R \Rightarrow P^{-1}R$$

$$PGG^{-1} = R\bar{G}$$

$$\bar{P}PG = \bar{P}R$$

$$\hat{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{bmatrix}$$

$K \times 1$

$$X'X = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ X_{21} & X_{22} & X_{23} & \dots & X_{2n} \\ X_{31} & X_{32} & X_{33} & \dots & X_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_{K1} & X_{K2} & X_{K3} & \dots & X_{Kn} \end{bmatrix} \begin{bmatrix} 1 & X_{21} & X_{31} & \dots & X_{K1} \\ 1 & X_{22} & X_{32} & \dots & X_{K2} \\ 1 & X_{23} & X_{33} & \dots & X_{K3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{2n} & X_{3n} & \dots & X_{Kn} \end{bmatrix}$$

$$X'X = \begin{bmatrix} 1 & X_2 & X_3 & \dots & X_K \\ \begin{matrix} \sum_{i=1}^n X_{2i} \\ \sum_{i=1}^n X_{2i}^2 \\ \sum_{i=1}^n X_{2i}X_3 \\ \vdots \\ \sum_{i=1}^n X_{2i}X_K \end{matrix} & \begin{matrix} \sum_{i=1}^n X_{3i} \\ \sum_{i=1}^n X_{3i}^2 \\ \sum_{i=1}^n X_{3i}X_4 \\ \vdots \\ \sum_{i=1}^n X_{3i}X_K \end{matrix} & \dots & \begin{matrix} \sum_{i=1}^n X_{Ki} \\ \sum_{i=1}^n X_{Ki}^2 \\ \sum_{i=1}^n X_{Ki}X_{K+1} \\ \vdots \\ \sum_{i=1}^n X_{Ki}X_n \end{matrix} \end{bmatrix}$$

$X'X$ is of order $K \times n$ $n \times K$ $= K \times K$ ✓

$$X'Y = \begin{bmatrix} \sum Y_i \\ \sum X_{2i}Y_i \\ \sum X_{3i}Y_i \\ \vdots \\ \sum X_{ni}Y_i \end{bmatrix}$$

Assumptions i, X is a matrix of non-stochastic variables with

$$\text{Rank}(X) = K$$

$$(ii) \quad \epsilon \sim N(0, \sigma^2)$$

Maximum likelihood Estimator of β (MLE)

GLR model is

$$Y = X\beta + \epsilon$$

$n \times 1$ $(n \times K)(K \times 1)$ $n \times 1$

Assumption $\epsilon \sim NLL(0, \sigma^2 I)$

$$f(\epsilon_i) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}\epsilon_i^2}$$

$$X \rightarrow n \times K$$

$$\beta \rightarrow K \times 1$$

$$\epsilon \rightarrow n \times 1$$

$$i = 1, 2, \dots, n$$

likelihood function of $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ is

$$\sum \epsilon_i^2 = \epsilon' \epsilon$$

$$L = f(\epsilon) = \frac{1}{\sigma^n (2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum \epsilon_i^2}$$

Joint distribution

$$L = \frac{1}{\sigma^n (2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} \epsilon' \epsilon}$$

$$L = \frac{1}{\sigma^n (2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} [(Y - X\beta)'(Y - X\beta)]}$$

$\because \epsilon = Y - X\beta$

Taking $\ln$

$$\ln L = -n \ln \sigma - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} [Y'Y - \underbrace{Y'X\beta - \beta'X'Y + \beta'X'X\beta}] \quad (line)$$

$\underline{Y'X\beta}$, $\underline{\beta'X'Y}$ are transpose of each other and of same order, it means these two terms are same.

$$\ln L = -n \ln \sigma - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} [Y'Y - \underline{2\beta'X'Y} + \beta'X'X\beta]$$

$$\frac{\partial \ln L}{\partial \beta} = 0 - \frac{1}{2\sigma^2} [-2X'Y + 2X'X\beta]$$

$$\text{putting } \frac{\partial \ln L}{\partial \beta} = 0$$

Estimating equation is

$$\Rightarrow -\frac{1}{2\sigma^2} [-2X'Y + 2X'X\beta] = 0$$

$$\underline{X}'\underline{Y} = (\underline{X}'\underline{X}) \underline{\beta}^*$$

$$\underline{\beta}^* = (\underline{X}'\underline{X})^{-1} \underline{X}'\underline{Y} \quad \text{which is the same as LSE}$$

MLE of σ^2

$$\ln L = -\frac{n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} [\underline{Y}'\underline{Y} - 2\underline{\beta}'\underline{X}'\underline{Y} + \underline{\beta}'\underline{X}'\underline{X}\underline{\beta}]$$

$$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2} \left(\frac{1}{(\sigma^2)^2} \right) [\underline{Y}'\underline{Y} - 2\underline{\beta}'\underline{X}'\underline{Y} + \underline{\beta}'\underline{X}'\underline{X}\underline{\beta}]$$

$$\frac{\partial \ln L}{\partial (\sigma^2)^2} = +\frac{n}{2\sigma^4} + \frac{1}{2} (-2(\sigma^2)^{-3}) [\underline{\epsilon}'\underline{\epsilon}]$$

$$= \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \underline{\epsilon}'\underline{\epsilon} < 0 \implies \frac{\partial \ln L}{\partial \sigma^2} = 0$$

$\implies$ estimating equation is

$$\frac{-n}{2\sigma^2} + \frac{1}{2\sigma^4} \underline{\epsilon}'\underline{\epsilon} = 0$$

$$-n\sigma^{*2} + \underline{\epsilon}'\underline{\epsilon} = 0$$

$$\sigma^{*2} = \frac{\underline{\epsilon}'\underline{\epsilon}}{n}$$

MLE of σ^2

invariant

MLE of σ is $\sigma^* = \sqrt{\frac{\underline{\epsilon}'\underline{\epsilon}}{n}}$ (by using property of MLE)

Properties of Residual

GLR Model is

$$Y_i = \beta_1 + \beta_2 X_{2i} + \dots + \beta_K X_{Ki} + \epsilon_i$$

or

$$Y = XB + \epsilon$$

LSE is $\hat{\beta} = (X'X)^{-1} X'Y$

Fitted equation is

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \dots + \hat{\beta}_K X_{Ki}$$

$$\hat{Y} = X \hat{\beta}$$

Residual is $e = Y - \hat{Y}$

$$\begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}$$

$$= Y_1 + Y_2 + \dots + Y_n$$

$$= \sum Y$$

$$\sum Y = \sum \hat{Y}$$

① Sum of the observed Y -values equal to the sum of the estimated Y -values.

i.e. $\sum Y_i = \sum \hat{Y}_i$

$$a'Y = a'\hat{Y}$$

Proof $\sum \hat{Y}_i = n\hat{\beta}_1 + \hat{\beta}_2 \sum X_{2i} + \dots + \hat{\beta}_K \sum X_{Ki}$ — (1)

Normal equation for β_1 is

$$\sum Y_i = n\hat{\beta}_1 + \hat{\beta}_2 \sum X_{2i} + \dots + \hat{\beta}_K \sum X_{Ki}$$
 — (2)

From (1) and (2)

$$\sum Y_i = \sum \hat{Y}_i$$

$$\sum \hat{Y} = a'\hat{Y}$$

$$= a'X\hat{\beta}$$

$$= \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} X_{21} & \dots & X_{K1} \\ X_{22} & \dots & X_{K2} \\ \vdots & & \vdots \\ X_{2n} & \dots & X_{Kn} \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_K \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 + \hat{\beta}_2 X_{21} + \dots + \hat{\beta}_K X_{K1} \\ \hat{\beta}_1 + \hat{\beta}_2 X_{22} + \dots + \hat{\beta}_K X_{K2} \\ \vdots \\ \hat{\beta}_1 + \hat{\beta}_2 X_{2n} + \dots + \hat{\beta}_K X_{Kn} \end{bmatrix}$$

$$a'\hat{Y} = \sum \hat{Y}_i = n\hat{\beta}_1 + \hat{\beta}_2 \sum X_{2i} + \dots + \hat{\beta}_K \sum X_{Ki}$$
 — (3)

and

$$\sum Y_i = a'Y$$
 — (4)

Now
OR
Matrix method

Hence From (3) (1) $\underline{a}'\underline{y} = \underline{a}'\hat{\underline{y}}$

(II) Sum of residual variates

i.e. $\sum e = 0$ (or) $\underline{a}'\underline{e} = 0$

Proof by definition

$$\underline{e} = \underline{y} - \hat{\underline{y}} = \underline{y} - \underline{X}\hat{\underline{\beta}}$$

Applying summation

$$\underline{a}'\underline{e} = \underline{a}'\underline{y} - \underline{a}'\hat{\underline{y}}$$

since $\underline{a}'\hat{\underline{y}} = \underline{a}'\underline{y}$

therefore $\underline{a}'\underline{e} = \underline{a}'\underline{y} - \underline{a}'\underline{y}$

$$= 0$$

(or)

$$\checkmark e_i = y_i - \hat{y}_i, \quad \sum e_i = \sum y_i - \sum \hat{y}_i = 0$$

(III) Sum of product of residual with each regressor variates

i.e. $\sum e_i x_{ij} = 0 \quad \forall \quad j = 2, 3, \dots, K$

(or) $\sum \underline{e} x_j = 0$

(or) $\underline{e}'\underline{X} = 0$ or $\underline{X}'\underline{e} = 0$

Proof As we know

$$\underline{e} = \underline{y} - \hat{\underline{y}}$$

$$= \underline{y} - \underline{X}\hat{\underline{\beta}}$$

$$\underline{e} = \underline{y} - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{y}$$

$$\underline{X}'\underline{e} = \underline{X}'\underline{y} - \underline{X}'\underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{y}$$

$$= \underline{X}'\underline{y} - \underline{X}'\underline{y}$$

$$\underline{X}'\underline{e} = 0$$

(or)

$$\hat{\underline{\beta}} = (\underline{X}'\underline{X})^{-1}\underline{X}'\underline{y}$$

by using (I), (II)

$$\underline{X}'\underline{e} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_{21} & x_{22} & \dots & x_{2n} \\ x_{31} & x_{32} & \dots & x_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{k1} & x_{k2} & \dots & x_{kn} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} \sum e_i \\ \sum e_i x_{2i} \\ \sum e_i x_{3i} \\ \vdots \\ \sum e_i x_{ki} \end{bmatrix} = 0$$

IV) Sum of product of residual with fitted value vanish
 i.e. $\sum e_i \hat{y}_i = 0$ or $\boxed{e' \hat{y} = 0 \text{ or } \hat{y}' e = 0}$

Proof:

$$\hat{y} = X \hat{\beta}$$

$$\hat{y}' = (X \hat{\beta})'$$

$$\hat{y}' e = (X \hat{\beta})' e = \hat{\beta}' X' e$$

$$\text{As } X' e = 0$$

$$\Rightarrow \boxed{\hat{y}' e = 0}$$

$$(AB)^t = B^t A^t$$

Unbiased Estimator of σ^2

GLR model is

$$Y = X\beta + \epsilon$$

Under the assumption $E(\epsilon) = 0$, $E(\epsilon\epsilon') = \sigma^2 I$

L.S.E. of β is $\hat{\beta} = (X'X)^{-1} X'Y$

$$\hat{y} = X \hat{\beta}$$

Residual is

$$e = Y - \hat{y}$$

$$= Y - X \hat{\beta}$$

$$= Y - X(X'X)^{-1} X'Y$$

$$= X\beta + \epsilon - X(X'X)^{-1} X' \{X\beta + \epsilon\}$$

$$= X\beta + \epsilon - X(X'X)^{-1} X'X\beta - X(X'X)^{-1} X'\epsilon$$

$$= X\beta + \epsilon - X\beta - X(X'X)^{-1} X'\epsilon$$

$$= \epsilon - X(X'X)^{-1} X'\epsilon$$

$$e = [I - X(X'X)^{-1} X'] \epsilon$$

$$e = M\epsilon$$

Where M is symmetric and idempotent matrix of order $(n \times n)$

$$M'M = M$$

SS Residual is

$$M' = M$$

$$e'e = (M\epsilon)'(M\epsilon)$$

$$= \epsilon' M' M \epsilon$$

$$e'e = \epsilon' M \epsilon$$

$$\begin{aligned}
 E(\underline{\underline{e}}'\underline{\underline{e}}) &= E[\underline{\underline{e}}'M\underline{\underline{e}}] \\
 &= E \text{tr}[\underline{\underline{e}}'M\underline{\underline{e}}] \\
 &= E \text{tr}[\underline{\underline{e}}\underline{\underline{e}}'M] \\
 &= \text{trace } E(\underline{\underline{e}}\underline{\underline{e}}'M) \\
 &= \text{tr } M E(\underline{\underline{e}}\underline{\underline{e}}')
 \end{aligned}$$

$$\begin{aligned}
 E(\underline{\underline{e}}'\underline{\underline{e}}) &= \text{tr } M \sigma^2 \\
 &= \text{tr} [I - X(X'X)^{-1}X'] \sigma^2 \\
 &= [\text{tr}(I) - \text{tr}(X(X'X)^{-1}X')] \sigma^2 \\
 &= [n - \text{tr}(X'X(X'X)^{-1})] \sigma^2 \\
 &= [n - \text{tr } I_{K \times K}] \sigma^2
 \end{aligned}$$

$$E(\underline{\underline{e}}'\underline{\underline{e}}) = (n - K) \sigma^2$$

We have K parameters and $(K-1)$ explanatory variables

$$\frac{E(\underline{\underline{e}}'\underline{\underline{e}})}{n-K} = \sigma^2$$

$$\hat{\sigma}^2 = \frac{\underline{\underline{e}}'\underline{\underline{e}}}{n-K}$$

Hence $\hat{\sigma}^2$ is an unbiased estimator of σ^2

$$\text{Since } \underline{\underline{e}}'\underline{\underline{e}} = \underline{\underline{e}}'M\underline{\underline{e}}$$

It is quadratic form in normal vector $\underline{\underline{e}}$.

Hence $\underline{\underline{e}}'\underline{\underline{e}}$ follows $\sigma^2 \chi^2$ dist, with

$$\begin{aligned}
 df &= \text{Rank}[\text{Matrix in QF}] \\
 &= \text{Rank}[M] \\
 &= \text{tr}[M] = \text{tr}[I - X(X'X)^{-1}X'] \\
 &= \text{tr}(I) - \text{tr}(X\{X'X\}^{-1}X') \\
 &= \text{tr } I - \text{tr}(X'X)(X'X)^{-1}
 \end{aligned}$$

$$df = n - \text{tr } I_K$$

$$df = n - K$$

$$\text{MLE } \hat{\sigma}^2 = \frac{\underline{\underline{e}}'\underline{\underline{e}}}{n}, \quad E(\hat{\sigma}^2) = \frac{1}{n} E(\underline{\underline{e}}'\underline{\underline{e}})$$

$$E(\hat{\sigma}^2) = \frac{(n-K)}{n} \sigma^2$$

$$E(\hat{\sigma}^2) \neq \sigma^2$$

So $\hat{\sigma}^2$ is a biased estimator of σ^2

Trace = Sum of diagonal elements

$$\text{ic } \text{tr } A = \sum_i a_{ii}$$

$$\text{tr } AB = \text{tr } BA$$

$$\text{tr } (A+B) = \text{tr } A + \text{tr } B$$

$$E \text{tr } A = \sum_i E a_{ii}$$

$$\begin{array}{c}
 X'X \\
 K \times n \\
 \hline
 n \times K \\
 K \times K
 \end{array}$$

Amount of bias is obtained as

$$E(\hat{\sigma}^{*2}) = \sigma^2 - \frac{k}{n} \sigma^2$$

$$E(\hat{\sigma}^{*2}) - \sigma^2 = -\frac{k}{n} \sigma^2$$

$$\text{Bias} = E(\hat{\sigma}^{*2}) - \sigma^2$$

$$\text{Bias} = -\frac{k}{n} \sigma^2$$

hence $\hat{\sigma}^{*2}$ is negatively biased

Properties of Least Square Estimator $\hat{\beta}$

i) The estimator $\hat{\beta} = (X'X)^{-1}X'Y$ is linear in Y -values.

ii) It is an unbiased estimator

$$\text{i.e. } E(\hat{\beta}) = \beta$$

Proof,

$$\hat{\beta} = (X'X)^{-1}X'Y$$

$$= (X'X)^{-1}X'(X\beta + \epsilon)$$

$$= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\epsilon$$

$$\hat{\beta} = \beta + (X'X)^{-1}X'\epsilon \quad \text{--- 0}$$

$$E(\hat{\beta}) = \beta + (X'X)^{-1}X'E(\epsilon)$$

$$= \beta + 0$$

$$E(\hat{\beta}) = \beta$$

Hence $\hat{\beta} = (X'X)^{-1}X'Y$ is an unbiased estimator of β .

iii) $\text{Var}(\hat{\beta})$

$$\text{Var}(\hat{\beta}) = E[\hat{\beta} - E(\hat{\beta})][\hat{\beta} - E(\hat{\beta})']$$

$$= E[\hat{\beta} - \beta][\hat{\beta} - \beta']$$

$$\text{From 0} \quad \hat{\beta} - \beta = (X'X)^{-1}X'\epsilon$$

Therefore

$$\text{Var}(\hat{\beta}) = E[(X'X)^{-1}X'\epsilon][(X'X)^{-1}X'\epsilon']$$

$$= E[(X'X)^{-1}X'\epsilon\epsilon'X(X'X)^{-1}]$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}_{k \times 1}$$

$$\begin{aligned}\text{Var}(\hat{\beta}) &= (X'X)^{-1} X' E(\epsilon\epsilon') X (X'X)^{-1} \\ &= (X'X)^{-1} X' \sigma^2 X (X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1} X'X (X'X)^{-1}\end{aligned}$$

$$\boxed{\text{Var}(\hat{\beta}) = (X'X)^{-1} \sigma^2}$$

The $\text{Var}(\hat{\beta})$ may be obtained by taking the i th term from principal diagonal of $(X'X)^{-1}$ and multiplying by σ^2 .

(IV) ~~Method~~ Method By Using Lagrange's Multiplier

L.S.E $\hat{\beta}$ has minimum variance among all linear and unbiased estimators.

Proof: Considering any other estimator say

$$\beta^* = \underline{a}'Y \text{ of } \beta$$

$$= \underline{a}'(X\beta + \epsilon)$$

$$\underline{B} = \underline{a}'X\beta + \underline{a}'\epsilon$$

$$E(\underline{B}) = \underline{a}'X\beta + \underline{a}'E(\epsilon)$$

$$E(\underline{B}^*) = \underline{a}'X\beta$$

$$E(\underline{B}^*) = \beta \quad \text{if} \quad \underline{a}'X = I_{k \times k}$$

$$\text{Var}(\underline{B}^*) = E[(\underline{B}^* - E(\underline{B}^*))(\underline{B}^* - E(\underline{B}^*))'] = E[\underline{B}^* - E(\underline{B}^*)][\underline{B}^* - E(\underline{B}^*)']$$

$$= E[\underline{a}'\epsilon][\underline{a}'\epsilon']$$

$$= E[\underline{a}'\epsilon][\epsilon'\underline{a}] = E(\underline{a}'\epsilon\epsilon'\underline{a})$$

$$= \underline{a}' E(\epsilon\epsilon') \underline{a} = \underline{a}' \sigma^2 \underline{a}$$

$$\boxed{\text{Var}(\underline{B}^*) = \underline{a}' \underline{a} \sigma^2}$$

Using Lagrange's multiplier, we have to minimize $\underline{a}' \underline{a}$

$$M = \underline{a}' \underline{a} - 2\lambda(\underline{a}'X - I_k) \quad \text{or} \quad \underline{a}' \underline{a} - 2(\underline{a}'X - I_k)\lambda$$

$$\frac{\partial M}{\partial \underline{a}} = 2\underline{a} - 2X'\lambda$$

λ is of order $k \times 1$

$$X'X\lambda = k \times k$$

$$(\underline{a}'X)\lambda = k \times k$$

$$\frac{\partial M}{\partial a} = 0 \Rightarrow$$

$$a = X'X^{-1}$$

①

$$a' = \lambda X'$$

$$a'X = \lambda X'X$$

Under the condition of unbiasedness $a'X = I_k$

$$I_k = (\lambda X'X)$$

$$\lambda = (X'X)^{-1}$$

putting the value of λ in ①

$$a = X(X'X)^{-1}$$

$$a' = (X'X)^{-1} X'$$

For this value of a , $\text{Var}(\hat{\beta})$ is minimum

$\text{Var}(\hat{\beta}^*) = a'a\sigma^2$ is minimum

Thus the estimator

$$\hat{\beta}^* = a'Y = (X'X)^{-1}X'Y$$

which is the LSE, therefore

LSE $\hat{\beta}$ has minimum variance among all linear and unbiased estimators. That is LSE is best linear unbiased estimator (BLUE).

Question; Let c be $(K \times 1)$ vector of known constant. If b is LSE of β . prove that $c'\hat{\beta}$ is BLUE of $c'\beta$.

where model is $Y = X\beta + u$

OR

In GLR model. Show that linear function of L.S.E is BLUE of corresponding linear function of parameter.

Solution, GLR model is

$$Y = X\beta + \epsilon$$

$$\epsilon \sim NI(0, \sigma^2 I)$$

We want to predict Y values when

$$X_1 = x_{10}, X_2 = x_{20}, \dots, X_K = x_{K0}$$

$$\hat{Y} = \hat{\beta}_1 + \hat{\beta}_2 x_{20} + \dots + \hat{\beta}_K x_{K0}$$

$$\hat{Y} = c'\hat{\beta}$$

$$\text{where } c' = [1 \quad x_{20} \quad \dots \quad x_{K0}]$$

Then $\underline{c}'\hat{\beta}$ is the predicted or estimated value for given values of explanatory variable. $\underline{c}'\hat{\beta}$ is linear function in OLS for given $\underline{c}$.

$$E(\underline{c}'\hat{\beta}) = \underline{c}'E(\hat{\beta}) \\ = \underline{c}'\beta$$

Hence $\underline{c}'\hat{\beta}$ is an unbiased estimator of $\underline{c}'\beta$.

$$\text{Now } \text{Var}(\underline{c}'\hat{\beta}) = E[(\underline{c}'\hat{\beta} - E(\underline{c}'\hat{\beta}))(\underline{c}'\hat{\beta} - E(\underline{c}'\hat{\beta}))']$$

$$= E[(\underline{c}'\hat{\beta} - \underline{c}'\beta)(\underline{c}'\hat{\beta} - \underline{c}'\beta)']$$

$$= E[\underline{c}'(\hat{\beta} - \beta)(\hat{\beta} - \beta)']$$

$$= \underline{c}'E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']\underline{c}$$

$$= \underline{c}'\text{Var}(\hat{\beta})\underline{c} = \underline{c}'(X'X)^{-1}\sigma^2\underline{c}$$

$$\boxed{\text{Var}(\underline{c}'\hat{\beta}) = \underline{c}'(X'X)^{-1}\underline{c}\sigma^2} \quad \text{--- ①}$$

Now considering another linear estimator say

$$\underline{b} = \underline{a}'\underline{y}$$

$$= \underline{a}'[X\beta + \underline{\epsilon}]$$

$$\underline{b} = \underline{a}'X\beta + \underline{a}'\underline{\epsilon}$$

$$E(\underline{b}) = \underline{a}'X\beta + \underline{a}'E(\underline{\epsilon})$$

$$E(\underline{b}) = \underline{a}'X\beta$$

$$E(\underline{b}) = \underline{c}'\beta$$

$$\text{if } \underline{a}'X = \underline{c}'$$

$\underline{a}$ is of

order

$1 \times n$

X

$n \times k$

$\underline{c}'$

$1 \times k$

$$\text{Var}(\underline{b}) = E[(\underline{b} - E(\underline{b}))(\underline{b} - E(\underline{b}))']$$

$$= E[\underline{a}'\underline{\epsilon}(\underline{a}'\underline{\epsilon})']$$

$$= E[\underline{a}'\underline{\epsilon}\underline{\epsilon}'\underline{a}]$$

$$= \underline{a}'E(\underline{\epsilon}\underline{\epsilon}')\underline{a}$$

$$= \underline{a}'\sigma^2\underline{a}$$

$$\text{Var}(\underline{b}) = \underline{a}'\underline{a}\sigma^2 \quad \text{--- ②}$$

Comparing ① and ②

$$\begin{aligned} \text{Var}(b) - \text{Var}(\hat{\beta}) &= \underline{a}'\underline{a}\sigma^2 - \underline{c}'(\underline{X}'\underline{X})^{-1}\underline{c}\sigma^2 \\ &= \underline{a}'\underline{a}\sigma^2 - (\underline{a}'\underline{X})(\underline{X}'\underline{X})^{-1}(\underline{X}'\underline{a})\sigma^2 \\ &= \underline{a}'[1 - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}']\underline{a}\sigma^2 \\ &= \underline{a}'\underline{M}\underline{a}\sigma^2 \end{aligned}$$

Where $\underline{M}$ is symmetric i.e. $\underline{M} = \underline{M}'$ and idempotent i.e. $\underline{M}\underline{M} = \underline{M}$

$\underline{a}'\underline{M}\underline{a}$ is p.d. matrix of order (1×1) i.e. $\underline{a}'\underline{M}\underline{a} \geq 0$

So $\text{Var}(b) - \text{Var}(\hat{\beta}) \geq 0$

$$\text{Var}(\hat{\beta}) \leq \text{Var}(b)$$

Thus $\hat{\beta}$ is BLUE of β .

After Equation ② We can proceed by using Lagrange's multiplier as

$$S = \underline{a}'\underline{a} - 2\lambda'(\underline{X}'\underline{a} - \underline{c}) \text{ or } \underline{a}'\underline{a} - 2(\underline{X}'\underline{X} - \underline{c}')\lambda$$

$$\frac{\partial S}{\partial \underline{a}} = 2\underline{a} - 2\underline{X}\lambda'$$

$$\frac{\partial S}{\partial \underline{a}} = 0 \implies \underline{a} = \underline{X}\lambda' \text{ or } \underline{a}' = \lambda\underline{X}'$$

$$\underline{a}'\underline{X} = \lambda(\underline{X}'\underline{X})$$

$$\underline{c}' = \lambda(\underline{X}'\underline{X})$$

$$\lambda = \underline{c}'(\underline{X}'\underline{X})^{-1}$$

Now

$$\underline{a} = \underline{X}(\underline{X}'\underline{X})^{-1}\underline{c}$$

For this value of $\underline{a}$ $\text{Var}(b)$ is minimum and estimator will be BLUE.

$$\text{Thus } b = \underline{a}'\underline{Y} = \underline{c}'(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{Y}$$

$$b = \hat{\beta} \text{ is BLUE of } \beta$$

We conclude that

① If $\underline{c}' = [1 \ 0 \ 0 \ \dots \ 0]$

then $\hat{\beta} = \hat{\beta}_1$

In general if $\underline{c}' = [0 \ 0 \ 0 \ \dots \ 1 \ \dots \ 0]$

at

then $\underline{c}'\hat{\beta} = \hat{\beta}_i$

It means that $\hat{\beta}_i$ is BLUE of β_i

ii) if $\underline{c}' = [1 \ 2 \ 3 \ \dots \ k]$

The linear function of L.S.E $\underline{c}'\hat{\beta}$ is BLUE of corresponding linear function of parameters.

iii, Prediction; We want to forecast some future value

Predicted value of Y for $x_2 = x_{20}, x_3 = x_{30}, \dots, x_k = x_{k0}$ is

$$\hat{y}_0 = \hat{\beta}_1 + \hat{\beta}_2 x_{20} + \hat{\beta}_3 x_{30} + \dots + \hat{\beta}_k x_{k0}$$

$$\hat{y}_0 = \underline{c}'\hat{\beta} \quad \text{if } \underline{c}' = [1 \ x_{20} \ x_{30} \ \dots \ x_{k0}]$$

$$\hat{y}_0 = \underline{c}'\hat{\beta} \text{ is BLUE of } \underline{c}'\beta = E(Y/x_2=x_{20} \dots x_k=x_{k0})$$

Gauss-Markov's Theorem;

Statement; In GLR model $Y = X\beta + \epsilon$

the LSE $\hat{\beta} = (X'X)^{-1}X'Y$ is BLUE with variance

$$\text{Var}(\hat{\beta}) = (X'X)^{-1}\sigma^2$$

Proof; Let us define a linear estimator of β

$$b = C'Y$$

$$= C'[X\beta + \epsilon]$$

$$b = C'X\beta + C'\epsilon$$

$$E(b) = C'X\beta + C'E\epsilon$$

$$E(b) = \beta \quad \text{if } C'X = I_K$$

$$\text{Var}(b) = C' \text{Var}(Y) C$$

$$= C'\sigma^2 C$$

$$\text{Var}(b) = (C'C)\sigma^2$$

LSE is

$$\hat{\beta} = (X'X)^{-1}X'Y$$

$$= a'Y$$

$$= a'(X\beta + \epsilon) = a'X\beta + a'\epsilon$$

$$E(\hat{\beta}) = a'X\beta + 0$$

$$E(\hat{\beta}) = \beta \quad \text{if } a'X = I, \quad \text{Var}(\hat{\beta}) = a'a\sigma^2$$

If $C' = a'$ then both estimators are same but we are assuming that they are not same.

$$C' = a' + d' \quad \checkmark$$

where d' is some matrix quantity.

$$\text{now } C'C = (a' + d')(a + d)$$

$$C'C = a'a + a'd + d'a + d'd$$

$$\text{As } C'X = I$$

$$I = (a' + d')X$$

$$I = a'X + d'X$$

$$I = (X'X)^{-1}X'X + d'X$$

$$I = I + d'X$$

but

$$a' = (X'X)^{-1}X'$$

$$\text{So } D'X = 0$$

$$(D'X)' = X'D = 0$$

- Now from above $\underline{a'D = (X'X)^{-1} X'D = 0}$ also $\underline{\underline{a'a = 0}}$

- Therefore $\underline{c'c = a'a + d'd}$

Now $\text{Var}(c) = (a'a + d'd) \sigma^2$

$$= \underline{a'a \sigma^2 + d'd \sigma^2}$$

$$\text{Var}(c) = \text{Var}(\hat{\beta}) + \underline{d'd \sigma^2}$$

$d'd$ is non-negative quantity matrix, therefore for any value of c

$$\underline{\text{Var}(\hat{\beta}_i) \leq \text{Var}(c_i)} \quad \text{Var}(\underline{\hat{\beta}}) \leq \text{Var}(\underline{c})$$

So LSE are BLUE

Result: If X is normal vector $X \sim N(0, \Sigma)$

where Σ is positive definite symmetric matrix then

$$\underline{X' \Sigma^{-1} X} \sim \chi^2 \text{ with } n \text{ degree of freedom}$$

Results:

① If X is a standard normal vector then $X'X$ follow χ^2 -variate with $df = \text{Rank}(I)$

I is defined as $X'IX$ $I_{n \times n}$

② If X is a normal vector then the quadratic form $X'AX$ (where A is positive definite) follows $\sigma^2 \chi^2$ dist with

$$df = \text{Rank}(A)$$

③ Two quadratic forms in normal vector X , $X'AX$, $X'BX$ are independent χ^2 variate if $AB=0$ or $BA=0$

④ If matrix A is p.d. symmetric and idempotent then

$$\text{Rank}(A) = \text{tr}(A)$$

⑤ The quadratic form $X'AX$ with " A " a symmetric, idempotent matrix of order n and " LX " be an m elements vector are said to be independently distributed if

$$LA = 0$$

Statistical Inference In GLR Model.

The GLR model is $Y = X\beta + \epsilon$ (k parameters)

LSE $\hat{\beta} = (X'X)^{-1}X'Y$ is BLUE of β

Residual is

$$\begin{aligned}
 e &= Y - \hat{Y} \\
 &= Y - X\hat{\beta} \\
 &= Y - X(X'X)^{-1}X'Y \\
 &= X\beta + \epsilon - X(X'X)^{-1}X'(X\beta + \epsilon) \\
 &= X\beta + \epsilon - X(X'X)^{-1}X'X\beta - X(X'X)^{-1}X'\epsilon \\
 &= X\beta + \epsilon - X\beta - X(X'X)^{-1}X'\epsilon \\
 e &= \epsilon - X(X'X)^{-1}X'\epsilon \\
 &= [I - X(X'X)^{-1}X']\epsilon \\
 e &= M\epsilon
 \end{aligned}$$

Where M is symmetric and idempotent matrix of order (n, n)

$$\begin{aligned}
 e'e &= (M\epsilon)'(M\epsilon) \\
 &= \epsilon'M'M\epsilon
 \end{aligned}$$

$$e'e = \epsilon'M\epsilon$$

$$\begin{aligned}
 E(e'e) &= E[\epsilon'M\epsilon] = E \text{tr}[\epsilon'M\epsilon] \\
 &= E \text{tr}[\epsilon'\epsilon M] = \text{tr}(M) E(\epsilon'\epsilon) \\
 &= \text{tr}[I - X(X'X)^{-1}X'] \sigma^2
 \end{aligned}$$

$$E(\epsilon'\epsilon) = (n - k) \sigma^2$$

$$\hat{\sigma}^2 = \frac{e'e}{n - k}$$

As $e'e = \epsilon'M\epsilon$ is quadratic form in normal vector ϵ
therefore $\epsilon'M\epsilon \sim \sigma^2 \chi^2$ where

$$\begin{aligned}
 df &= \text{Rank}(\text{matrix in GF}) = \text{Rank}(M) \\
 &= \text{Rank}[I_n - X(X'X)^{-1}X'] \\
 &= \text{tr}[I_n - X(X'X)^{-1}X'] \\
 &= \text{tr}(I_n) - \text{tr}(X(X'X)^{-1}X') = \text{tr} I - \text{tr} X'X(X'X)^{-1}
 \end{aligned}$$

$$df = n - k$$

$$\therefore \mathbf{e}'\mathbf{e} = \sigma^2 \chi_{(n-k)}^2$$

$$\text{and } \hat{\sigma}^2 = \frac{\mathbf{e}'\mathbf{e}}{(n-k)} = \frac{\sigma^2 \chi_{(n-k)}^2}{(n-k)}$$

Since $\mathbf{e} \sim NI(0, \sigma^2 \mathbf{I})$

So $\mathbf{y}$ also is a normal vector with

$$E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta} \quad \text{and} \quad V(\mathbf{y}) = \sigma^2 \mathbf{I}$$

And $\boldsymbol{\beta}$ is a linear function of normal vector $\mathbf{y}$

$\therefore \hat{\boldsymbol{\beta}}$ also follows $N(\boldsymbol{\beta}, (\mathbf{X}'\mathbf{X})^{-1}\sigma^2)$

Now $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and quadratic form $\mathbf{e}'\mathbf{M}\mathbf{e}$ are distributed independently if $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = 0$

which is true as $\mathbf{X}'\mathbf{M} = 0$

$$\text{i.e. } \mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

$$\mathbf{X}'\mathbf{M} = \mathbf{X}' - \mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = 0$$

As $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, (\mathbf{X}'\mathbf{X})^{-1}\sigma^2)$

So $\hat{\beta}_i \sim N(\beta_i, a_{ii}\sigma^2)$

Thus $Z = \frac{\hat{\beta}_i - \beta_i}{\sigma \sqrt{a_{ii}}}$ is SNV

If σ is unknown then

$$t = \frac{Z}{\sqrt{\chi^2/df}}$$

$$= \frac{\hat{\beta}_i - \beta_i}{\sigma \sqrt{a_{ii}}} \div \sqrt{\frac{\hat{\sigma}^2}{\sigma^2}}$$

$$t = \frac{\hat{\beta}_i - \beta_i}{\hat{\sigma} \sqrt{a_{ii}}} \sim t_{(n-k)}$$

Decomposition (or) Partitioning of TSS

The general linear regression model is

$$Y = X\beta + \epsilon$$

$$E\epsilon = 0 \quad E\epsilon\epsilon' = \sigma^2 I$$

The estimated value is $\hat{Y} = X\hat{\beta}$

$$\text{L.S.E is } \hat{\beta} = (X'X)^{-1}X'Y$$

Residual is

$$e = Y - \hat{Y}$$

$$Y = e + \hat{Y}$$

$$\text{TSS (uncorrected)} = \sum_{i=1}^n Y_i^2 = Y'Y$$

$$Y'Y = (e + \hat{Y})'(e + \hat{Y})$$

$$Y'Y = e'e + e'\hat{Y} + \hat{Y}'e + \hat{Y}'\hat{Y}$$

$$\text{As } e'\hat{Y} = 0, \hat{Y}'e = 0$$

$$\therefore Y'Y = e'e + \hat{Y}'\hat{Y}$$

$$\hat{Y}'\hat{Y} = \sum_{i=1}^n \hat{Y}_i^2 = (\text{Reg SS}) \text{ uncorrected or explained SS}$$

$e'e$ is Residual SS or SSE or unexplained S.S.

Reg. S.S

$$\begin{aligned} \text{Reg SS} &= \hat{Y}'\hat{Y} = \hat{\beta}'X'X\hat{\beta} \\ &= [(X'X)^{-1}X'Y]'X'X[(X'X)^{-1}X'Y] \\ &= Y'X(X'X)^{-1}X'Y \end{aligned}$$

$$\text{Reg SS} = Y'X(X'X)^{-1}X'Y$$

It is quadratic form in normal vector Y , therefore it follows χ^2 dist, with

$$df = \text{Rank}[X(X'X)^{-1}X']$$

$$= \text{tr}[X(X'X)^{-1}X']$$

$$= \text{tr}[(X'X)(X'X)^{-1}] = \text{tr}[I_n]$$

$$df = K$$

$$\underline{a} = [1 \ 1 \ \dots \ 1]$$

$$\text{Correction factor (Cf)} = \frac{(\sum Y_i)^2}{n} = \frac{(\underline{a}'Y)^2}{n} \text{ with } df = 1$$

$$\text{Reg SS (corrected)} = \underline{\hat{Y}}' \underline{\hat{Y}} - \frac{1}{n} (\underline{Q}' \underline{Y})^2$$

$\sim \sigma^2 \chi^2_{\text{dist}} \text{ df} = n-1$

C.f

$$\underline{Q}' \underline{Y} = \underline{Q}' (\underline{X} \underline{\beta} + \underline{\epsilon}) = \underline{Q}' \underline{X} \underline{\beta} + \underline{Q}' \underline{\epsilon}$$

$$(\underline{Q}' \underline{Y})^2 = (\underline{Q}' \underline{X} \underline{\beta} + \underline{Q}' \underline{\epsilon}) (\underline{Q}' \underline{X} \underline{\beta} + \underline{Q}' \underline{\epsilon})$$

$$= (\underline{\beta}' \underline{X}' \underline{Q} + \underline{\epsilon}' \underline{Q}) (\underline{Q}' \underline{X} \underline{\beta} + \underline{Q}' \underline{\epsilon})$$

$$(\underline{Q}' \underline{Y})^2 = \underline{\beta}' \underline{X}' \underline{Q} \underline{Q}' \underline{X} \underline{\beta} + \underline{\beta}' \underline{X}' \underline{Q} \underline{Q}' \underline{\epsilon} + \underline{\epsilon}' \underline{Q} \underline{Q}' \underline{X} \underline{\beta} + \underline{\epsilon}' \underline{Q} \underline{Q}' \underline{\epsilon}$$

under H_0 , $\underline{\beta} = 0$.

$$(\underline{Q}' \underline{Y})^2 = \underline{\epsilon}' \underline{Q} \underline{Q}' \underline{\epsilon}$$

It is quadratic form in normal vector $\underline{\epsilon}$ therefore it follows $\sigma^2 \chi^2$ distribution with

$$\text{df} = \text{Rank} [\underline{Q} \underline{Q}'] = \text{Rank} [\underline{Q}' \underline{Q}]$$

$$\text{df} = 1$$

$\underline{Q}'$
1xn

Total S.S

$$\begin{aligned} \text{TSS (corrected)} &= \underline{Y}' \underline{Y} - \frac{(\underline{Q}' \underline{Y})^2}{n} \\ &= (\underline{X} \underline{\beta} + \underline{\epsilon})' (\underline{X} \underline{\beta} + \underline{\epsilon}) - \frac{(\underline{Q}' \underline{Y})^2}{n} \end{aligned}$$

under H_0 , $\underline{\beta} = 0$

$$\boxed{\text{TSS} = \underline{\epsilon}' \underline{I} \underline{\epsilon} - \frac{(\underline{Q}' \underline{Y})^2}{n}}$$

(1xn) (nxn) (nx1)

$\underline{\epsilon}' \underline{I} \underline{\epsilon}$ is QF in normal vector $\underline{\epsilon}$, so it follows $\sigma^2 \chi^2$ dist

with $\text{df} = \text{Rank} (\underline{I}_n) = \text{tr } \underline{I}_n$

$$\text{df} = n$$

Thus TSS follows $\sigma^2 \chi^2$ dist with $\text{df} = n-1$

SSE

Now Residual Sum of Squares is

$$\underline{\epsilon}' \underline{\epsilon} = (\underline{Y} - \underline{\hat{Y}})' (\underline{Y} - \underline{\hat{Y}})$$

$$= (\underline{Y} - \underline{X} \underline{\hat{\beta}})' (\underline{Y} - \underline{X} \underline{\hat{\beta}})$$

$$\boxed{\underline{\epsilon}' \underline{\epsilon} = \underline{\epsilon}' \underline{M} \underline{\epsilon}}$$

df

$$\text{where } \underline{M} = [\underline{I} - \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}']$$

$$\text{df} = \text{Rank} (\underline{M}) = \text{Rank} [\underline{I}_n - \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}']$$

$$= \text{tr} [\underline{I}_n - \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}']$$

$$\begin{aligned} df &= \text{tr } I_n - \text{tr } X(X'X)^{-1}X' \\ &= n - \text{tr } (X'X)(X'X)^{-1} = n - \text{tr } I_k \\ df &= n - k \end{aligned}$$

So $\epsilon'\epsilon$ follows $\sigma^2 \chi^2$ dist with $df = n - k$

Reg SS and SSE are distributed independently if multiplication of matrices in Q.F.s $Y'X(X'X)^{-1}X'Y$ and $\epsilon'M\epsilon$ or $Y'MY'$ is zero.

$$\begin{aligned} M &= I - X(X'X)^{-1}X' \\ X(X'X)^{-1}X'M &= X(X'X)^{-1}X' - X(X'X)^{-1}X'X(X'X)^{-1}X' \\ &= X(X'X)^{-1}X' - X(X'X)^{-1}X' = 0 \end{aligned}$$

Both Q.F.s are distributed independently, therefore the statistic

$$F = \frac{\text{Reg SS}/k-1}{\text{SSE}/n-k} \sim F(k-1, n-k)$$

① Testing of Single Parameters

Suppose we want to test the hypothesis

$$H_0: \beta = \beta_0$$

Consider a statistic

$$Q = (\hat{\beta} - \beta_0)'(X'X)(\hat{\beta} - \beta_0)$$

$$\begin{aligned} \text{L.S.E is } \hat{\beta} &= (X'X)^{-1}X'Y \\ &= (X'X)^{-1}X'(X\beta + \epsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\epsilon \\ \hat{\beta} &= \beta + (X'X)^{-1}X'\epsilon \end{aligned}$$

$$(\hat{\beta} - \beta_0) = (X'X)^{-1}X'\epsilon$$

$$(\hat{\beta} - \beta_0)' = \epsilon'X(X'X)^{-1}$$

$$\Rightarrow Q = \epsilon'X(X'X)^{-1}(X'X)(X'X)^{-1}X'\epsilon$$

$$Q = \epsilon'X(X'X)^{-1}X'\epsilon$$

$$\text{Now let } M = I - X(X'X)^{-1}X' \quad \text{or} \quad X(X'X)^{-1}X' = I - M$$

$$\Rightarrow Q = \epsilon'(I - M)\epsilon$$

It is quadratic form in normal vector ϵ , therefore it follows χ^2 dist

$$\text{df} = \text{Rank}(I-M)$$

$$= \text{tr}(I-M) = \text{tr}(I - L + X(X'X)^{-1}X')$$

$$= \text{tr}(X(X'X)^{-1}X') = \text{tr}(X'X(X'X)^{-1})$$

$$= \text{tr}(I_K)$$

$$\text{df} = K$$

$$\text{and } SSE = \epsilon'\epsilon = \sigma^2 \chi^2_{(n-k)}$$

$$= \epsilon'M\epsilon$$

where $(K-1)$
regressors and
 K parameters.

Hence the statistic

$$F = \frac{Q/K-1}{SSE/n-K-1} \sim F_{(K-1, n-K-1)}$$

Testing Set of Hypothesis

$$H_0 : \beta_1 = 0$$

$$H_0 : \beta_1 + \beta_2 + \dots + \beta_K = a$$

$$H_0 : \beta_3 = \beta_4$$

$$H_0 : \sum_{i=1}^K \beta_i = b$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 1 & 1 & 1 & 1 & \dots & 1 \\ 0 & 0 & 1 & -1 & \dots & 0 \\ 1 & 2 & 3 & 4 & \dots & K \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \vdots \\ \beta_K \end{bmatrix} = \begin{bmatrix} 0 \\ a \\ 0 \\ b \end{bmatrix}$$

$$H_0 : R\beta = \tau$$

$$\tau \rightarrow q \times 1$$

S.E of $R\hat{\beta}$ is $R\hat{\sigma}$

$$\beta \rightarrow K \times 1$$

we know that $\hat{\beta} \sim N(\beta, (X'X)^{-1}\sigma^2)$

$$K \rightarrow K \times K$$

$\therefore R\hat{\beta}$ also follow normal distribution with mean

$$E(R\hat{\beta}) = R\beta$$

$$\begin{aligned} \text{Var}(R\hat{\beta}) &= R \text{Var}(\hat{\beta}) R' \\ &= R(X'X)^{-1}R' \sigma^2 \end{aligned}$$

So $R\hat{\beta} \sim N(R\beta, R(X'X)^{-1}\sigma^2 R')$

$(R\hat{\beta} - R\beta) \sim N(0, R(X'X)^{-1}R'\sigma^2)$

Since $\hat{\beta}$ is normal vector, therefore the Q.F

$(R\hat{\beta} - R\beta)'(R(X'X)^{-1}R')^{-1}(R\hat{\beta} - R\beta)$

will follow $\sigma^2 \chi^2$ distribution with q d.f

Thus the statistic

$$\frac{(R\hat{\beta} - R\beta)'[R(X'X)^{-1}R']^{-1}(R\hat{\beta} - R\beta) / q}{\hat{\sigma}^2 / n - k} \sim F(q, n - k)$$

③ $H_0: C'\beta = 0$

$C'\hat{\beta}$ is the estimator of $C'\beta$

ie $E(C'\hat{\beta}) = C'E(\hat{\beta}) = C'\beta$

$\text{Var}(C'\hat{\beta}) = C'\text{Var}(\hat{\beta})C$

$= C'(X'X)^{-1}\sigma^2 C$

$\text{Var}(C'\hat{\beta}) = C'(X'X)^{-1}C\sigma^2$, Since $\hat{\beta} \sim N(\beta, (X'X)^{-1}\sigma^2)$

Thus the statistic

$C'\hat{\beta} \sim N(C'\beta, C'(X'X)^{-1}C\sigma^2)$

$t = \frac{C'\hat{\beta} - C'\beta}{\hat{\sigma}^2 \sqrt{C'(X'X)^{-1}C}} \sim t_{(n-k)}$

Question;

$$X = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 2 \\ -1 \\ 1 \end{bmatrix}$$

- Obtain linear regression and verify the properties of residual.
- Test the significance of the regression.
- Construct 95% CI for $X' = [1 \ 1 \ 1]$

Sol: GLR model is $Y = X\beta + \epsilon$

② LSE of β is $\hat{\beta} = (X'X)^{-1}X'Y$

$$X'X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$X'X = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 6 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$(X'X)^{-1} = \frac{1}{|X'X|} \text{Adj}(X'X)$$

$$|X'X| = 6(18-4) - 2(6-2) + 1(4-6) = 84 - 8 - 2 = 74$$

$$a_{11} = 18-4 = 14, \quad a_{12} = (-1)^{1+2}(6-2) = -4, \quad a_{13} = (-1)^{1+3}(4-6) = -2$$

$$a_{21} = (-1)^{2+1}(4) = -4, \quad a_{22} = 18-1 = 17, \quad a_{23} = (-1)^{2+3}(12-2) = -10$$

$$a_{31} = 4-6 = -2, \quad a_{32} = 12-2 = 10, \quad a_{33} = 36-4 = 32$$

Matrix of co factors is

$$\text{Adj}(X'X) = \begin{bmatrix} 14 & -4 & -2 \\ -4 & 17 & -10 \\ -2 & 10 & 32 \end{bmatrix}$$

$$\text{Now } (X'X)^{-1} = \frac{1}{74} \begin{bmatrix} 14 & -4 & -2 \\ -4 & 17 & -10 \\ -2 & 10 & 32 \end{bmatrix}$$

$$X'Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 1 \\ 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$$

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$= \frac{1}{74} \begin{bmatrix} 14 & -4 & -2 \\ -4 & 17 & -10 \\ -2 & -10 & 32 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$$

$$\hat{\beta} = \frac{1}{74} \begin{bmatrix} 40 \\ 15 \\ 100 \end{bmatrix}$$

Hence fitted regression is

$$\hat{Y} = \frac{40}{74} + \frac{15}{74} X_1 + \frac{100}{74} X_2$$

Properties of Residual.

① $\sum Y_i = \sum \hat{Y}_i$ or $\underline{a}'Y = \underline{a}'\hat{Y}$ where $\underline{a}' = [1 \ 1 \ 1 \ 1 \ 1 \ 1]$

$$\hat{Y} = X\hat{\beta} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{40}{74} \\ \frac{15}{74} \\ \frac{100}{74} \end{bmatrix} = \begin{bmatrix} 25/74 \\ 140/74 \\ 55/74 \\ 170/74 \\ -60/74 \\ 40/74 \end{bmatrix}$$

$$\sum Y_i = \underline{a}'Y = [1 \ 1 \ 1 \ 1 \ 1 \ 1] \begin{bmatrix} 0 \\ 2 \\ 1 \\ 2 \\ -1 \\ 1 \end{bmatrix} = [5]$$

$$\sum \hat{Y}_i = \underline{a}'\hat{Y} = [1 \ 1 \ 1 \ 1 \ 1 \ 1] \begin{bmatrix} 25/74 \\ 140/74 \\ 55/74 \\ 170/74 \\ -60/74 \\ 40/74 \end{bmatrix} = \left[\frac{370}{74} \right] = [5]$$

$$\Rightarrow \sum Y_i = \sum \hat{Y}_i$$

② Sum of residual vanish i.e. $\sum e_i = 0$ or $\underline{a}'e = 0$

$$e = Y - \hat{Y} = \begin{bmatrix} -25/74 \\ 0/74 \\ 14/74 \\ -22/74 \\ -14/74 \\ 57/74 \end{bmatrix}$$

$$\sum a = a'e = [1 \ 1 \ 1 \ 1 \ 1] \begin{bmatrix} -25 \\ 8 \\ 19 \\ -22 \\ -14 \\ 34 \end{bmatrix} \frac{1}{74} = 0/74 = 0$$

③ Sum of product of residual with each regressor vanishes. i.e.
 $\sum e x_{2i} = 0$ and $\sum e x_{1i} = 0$ $i = 1, 2, \dots, 6$

$$\sum e x_{2i} = \frac{1}{74} \begin{bmatrix} -25 \\ 8 \\ 19 \\ -22 \\ -14 \\ 34 \end{bmatrix} \quad \sum e x_{2i} = \frac{1}{74} [-25 \ 8 \ 19 \ -22 \ -14 \ 34] \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{74} [0 + 8 + 0 - 22 + 14 + 0]$$

$$\sum e x_{2i} = 0$$

$$\sum e x_{1i} = \frac{1}{74} [-25 \ 8 \ 19 \ -22 \ -14 \ 34] \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \\ 2 \\ 0 \end{bmatrix}$$

$$\sum e x_{1i} = \frac{1}{74} [25 + 0 + 19 - 44 - 28] = 0$$

④ Sum of product of residual with fitted value vanishes
 i.e. $\sum e \hat{y} = 0$ or $\hat{y}'e = 0$

$$\sum e \hat{y} = [-25 \ 8 \ 19 \ -22 \ -14 \ 34] \frac{1}{74} \begin{bmatrix} 25/74 \\ 140/74 \\ 55/74 \\ 170/74 \\ -60/74 \\ 40/74 \end{bmatrix} = \begin{bmatrix} -625/74^2 \\ 1120/74^2 \\ 1045/74^2 \\ -3740/74^2 \\ 840/74^2 \\ 1360/74^2 \end{bmatrix} = 0$$

(b)

$$H_0: \beta = 0, \quad H_1: \beta \neq 0$$

$$\alpha = 0.05$$

$$F = \frac{\text{Reg SS} / (K-1)}{e'e / (n-K)}$$

H_0 will be Rejected if $F_c > F_{0.05}(K-1, n-K)$

$$K=3, \quad n=6$$

$$F_{(2,3)} = 9.55$$

$$\text{Reg SS} = \hat{\beta}' X' Y - c.f$$

$$= \frac{1}{74} \begin{bmatrix} 60 & 15 & 100 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} - c.f = \frac{775}{74} - c.f$$

$$c.f = \frac{(\sum Y)^2}{n} = \frac{(5)^2}{6} = \frac{25}{6}$$

$$\text{Reg SS} = \frac{775}{74} - \frac{25}{6} = 6.3063$$

$$\boxed{\text{Reg SS} = 6.3063}$$

$$Y'Y = \begin{bmatrix} 0 & 2 & 1 & 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 1 \\ 2 \\ -1 \\ 1 \end{bmatrix} = 11$$

$$e'e = \sum Y^2 - \hat{\beta}'(X'Y)$$

$$= Y'Y - \hat{\beta}'(X'Y)$$

$$e'e = 11 - \frac{775}{74}$$

$$\boxed{e'e = 0.5270}$$

$$TSS = Y'Y - \frac{(\sum Y)^2}{n} = 6.8333$$

$$\text{now } F = \frac{6.3063/2}{0.5270/3} = 17.9496 > 9.55$$

Hence we reject H_0 and conclude that

$$\beta \neq 0$$

c) Now we compute $\hat{Y}$ for $X'_0 = [1 \ 1]$ and $V(\hat{Y})$

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2, \quad X'_0 = [X_1 \ X_2]$$

$$= \frac{40}{74} + \frac{15}{74} + \frac{100}{74}$$

$$\hat{Y} = \frac{155}{74} = 1.95946$$

$$\frac{155}{74}$$

$$\begin{aligned} \text{Var}(\hat{Y}) &= \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2) \\ &= \text{Var}(\hat{\beta}_0) + X_1^2 \text{Var}(\hat{\beta}_1) + X_2^2 \text{Var}(\hat{\beta}_2) + 2 X_1 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &\quad + 2 X_2 \text{Cov}(\hat{\beta}_0, \hat{\beta}_2) + 2 X_1 X_2 \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \end{aligned}$$

$$\text{Var}(\hat{\beta}) = (X'X)^{-1} \hat{\sigma}^2$$

$$\hat{\sigma}^2 = \frac{e'e}{n-k} = \frac{0.5270}{3} = 0.17567$$

$$\text{Var}(\hat{\beta}) = \frac{0.17567}{74} \begin{bmatrix} 14 & -4 & -2 \\ -4 & 17 & -10 \\ -2 & -10 & 32 \end{bmatrix} = \begin{bmatrix} 0.033235 & -0.009416 & -0.004778 \\ -0.009416 & 0.040357 & -0.023739 \\ -0.004778 & -0.023739 & 0.075965 \end{bmatrix}$$

PUACP

$$\begin{aligned} \text{Var}(\hat{Y}) &= (0.033235) + 1^2(0.040357) + 1^2(0.075965) + 2(1)(-0.009416) \\ &\quad + 2(1)(-0.004778) + 2(1)(1)(-0.023739) \end{aligned}$$

$$\text{Var}(\hat{Y}) = 0.073591, \quad \text{SE}(\hat{Y}) = 0.271277$$

95% CI for $\hat{Y}$ is

$$\hat{Y} - t_{\left(\frac{\alpha}{2}, n-k\right)} \text{SE}(\hat{Y}) < Y < \hat{Y} + t_{\left(\frac{\alpha}{2}, n-k\right)} \text{SE}(\hat{Y})$$

$$t_{(0.025, 3)} = 3.182$$

$$1.95946 \pm (3.182)(0.271277)$$

$$1.0463 < Y < 2.8727$$

Coefficient of Determination

$$R^2 = \frac{\text{Reg SS}}{\text{TSS}}$$

$$R^2 = 1 - \frac{\text{SSE}}{\text{TSS}}$$

$$= 1 - \frac{0.5270}{6.8333}$$

$$R^2 = 0.92$$

G.L.R. model (Partitioned form)

Testing some parameters of G.L.R model

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \epsilon_i$$

(K=5) p:

from description of the model we know the significance of the variance and Y_i not depends on X_2 and X_4

$$Y_i = (\beta_1 + \beta_3 X_{3i} + \beta_5 X_{5i}) + (\beta_2 X_{2i} + \beta_4 X_{4i}) + \epsilon_i \quad i=1, 2, \dots, n$$

If it is assumed that Y does not depend upon X_2 and X_4 , then we can test this assumption by ~~G.L.R~~ partitioning the G.L.R model as

$$\underline{Y} = \underline{X}_1 \underline{\beta}_1 + \underline{X}_2 \underline{\beta}_2 + \underline{\epsilon}$$

Where $\underline{\beta}_1$ is of order $(H \times 1)$, $\underline{X}_1$ is of order $(n \times H)$

$$\underline{X}_1 = \begin{bmatrix} 1 & X_{31} & X_{51} \\ 1 & X_{32} & X_{52} \\ \vdots & \vdots & \vdots \\ 1 & X_{3n} & X_{5n} \end{bmatrix}, \quad \underline{X}_2 = \begin{bmatrix} X_{21} & X_{41} \\ X_{22} & X_{42} \\ \vdots & \vdots \\ X_{2n} & X_{4n} \end{bmatrix}$$

$\underline{X}_2$ is of order $n \times (K-H)$ and $\underline{\beta}_2$ is of order $(K-H) \times 1$

$$H_0: \quad \underline{\beta}_2 = \begin{bmatrix} \beta_2 \\ \beta_4 \end{bmatrix} = 0$$

Least Square Estimation

If $\hat{\beta}_1$ and $\hat{\beta}_2$ are the LSE of β_1 and β_2 , then we can obtain these estimators by minimizing S.S.E. w.r.t. $\hat{\beta}_1, \hat{\beta}_2$.

$$\begin{aligned} e'e = S &= (Y - X_1 \hat{\beta}_1 - X_2 \hat{\beta}_2)'(Y - X_1 \hat{\beta}_1 - X_2 \hat{\beta}_2) \\ &= Y'Y - Y'X_1 \hat{\beta}_1 - Y'X_2 \hat{\beta}_2 - \hat{\beta}_1' X_1' Y + \hat{\beta}_1' X_1' X_1 \hat{\beta}_1 + \hat{\beta}_1' X_1' X_2 \hat{\beta}_2 \\ &\quad - \hat{\beta}_2' X_2' Y + \hat{\beta}_2' X_2' X_1 \hat{\beta}_1 + \hat{\beta}_2' X_2' X_2 \hat{\beta}_2 \end{aligned}$$

$$\frac{\partial S}{\partial \hat{\beta}_1} = -2X_1'Y + 2X_1'X_1\hat{\beta}_1 + 2X_1'X_2\hat{\beta}_2$$

$$\frac{\partial S}{\partial \hat{\beta}_1} = 0 \implies X_1'Y = X_1'X_1\hat{\beta}_1 + X_1'X_2\hat{\beta}_2 \quad \text{--- ① NE for } \hat{\beta}_1$$

$$\frac{\partial S}{\partial \hat{\beta}_2} = -2X_2'Y + 2X_2'X_1\hat{\beta}_1 + 2X_2'X_2\hat{\beta}_2$$

$$\frac{\partial S}{\partial \hat{\beta}_2} = 0 \implies X_2'Y = X_2'X_1\hat{\beta}_1 + X_2'X_2\hat{\beta}_2 \quad \text{--- ② NE for } \hat{\beta}_2$$

$$\text{From ①} \quad \hat{\beta}_1 = (X_1'X_1)^{-1}(X_1'Y - X_1'X_2\hat{\beta}_2) \quad \text{--- ③}$$

pulling in ②

$$X_2'Y = X_2'X_1(X_1'X_1)^{-1}(X_1'Y - X_1'X_2\hat{\beta}_2) + X_2'X_2\hat{\beta}_2$$

$$X_2'Y - X_2'X_1(X_1'X_1)^{-1}X_1'Y = [X_2'X_2 - X_2'X_1(X_1'X_1)^{-1}X_1'X_2]\hat{\beta}_2$$

$$X_2'[I_{(H-K)} - X_1(X_1'X_1)^{-1}X_1']Y = X_2'[I - X_1(X_1'X_1)^{-1}X_1']X_2\hat{\beta}_2$$

let $M_1 = I - X_1(X_1'X_1)^{-1}X_1'$

So $X_2'M_1Y = X_2'M_1X_2\hat{\beta}_2$

$$\boxed{\hat{\beta}_2 = (X_2'M_1X_2)^{-1}X_2'M_1Y} \implies \boxed{\hat{\beta}_2 = \bar{D}'X_2M_1Y}$$

pulling $\hat{\beta}_2$ in ③

$$\hat{\beta}_1 = (X_1'X_1)^{-1}X_1'Y - (X_1'X_1)^{-1}X_1'X_2(\bar{D}'X_2M_1Y)$$

$$\boxed{\hat{\beta}_1 = (X_1'X_1)^{-1}X_1'Y - (X_1'X_1)^{-1}X_1'X_2\bar{D}'X_2M_1Y}$$

Similarly

$$\hat{\beta}_1 = (X_1' M_2 X_1)^{-1} X_1' M_2 Y$$

where $M_2 = I - X_2(X_2' X_2)^{-1} X_2'$ and let $E = X_1' M_2 X_1$, then

$$\hat{\beta}_1 = E^{-1} E X_1' M_2 Y$$

and

$$\hat{\beta}_2 = (X_2' X_2)^{-1} X_2' Y - (X_2' X_2)^{-1} X_2' X_1 E^{-1} X_1' M_2 Y$$

30/05/2014

Question: For the partitioned GLR model $Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$,Find then mean, variance and Covariance of the L.S.E., $\hat{\beta}_1, \hat{\beta}_2$ **Solution:**

$$Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$$

$$\text{L.S.E. are } \hat{\beta}_1 = E^{-1} (X_1' M_2 Y) \quad \text{--- (1)}$$

$$E^{-1} = (X_1' M_2 X_1)^{-1}$$

$$\text{where } \hat{\beta}_2 = D^{-1} (X_2' M_1 Y) \quad \text{--- (2)}$$

$$D^{-1} = (X_2' M_1 X_2)^{-1}$$

$$M_1 = I - X_1 (X_1' X_1)^{-1} X_1' \quad \text{and} \quad M_2 = I - X_2 (X_2' X_2)^{-1} X_2'$$

Now from (1)

$$\hat{\beta}_1 = E^{-1} X_1' M_2 (X_1 \beta_1 + X_2 \beta_2 + \epsilon)$$

$$= E^{-1} (X_1' M_2 X_1) \beta_1 + E^{-1} X_1' M_2 X_2 \beta_2 + E^{-1} X_1' M_2 \epsilon$$

$$E(\hat{\beta}_1) = E^{-1} E \beta_1 + E^{-1} X_1' [I - X_2 (X_2' X_2)^{-1} X_2'] X_2 \beta_2 + E^{-1} X_1' M_2 \epsilon$$

$$= E^{-1} E \beta_1 + E^{-1} [X_1' X_2 - X_1' X_2] \beta_2 + E^{-1} X_1' M_2 \epsilon$$

$$\hat{\beta}_1 = E^{-1} E \beta_1 + E^{-1} X_1' M_2 \epsilon$$

$$\hat{\beta}_1 = \beta_1 + E^{-1} X_1' M_2 \epsilon \quad \text{--- (3)}$$

$$E(\hat{\beta}_1) = \beta_1$$

$$\therefore E(\epsilon) = 0$$

$$\begin{aligned} \text{Var}(\hat{\beta}_1) &= E(\hat{\beta}_1 - \beta_1)(\hat{\beta}_1 - \beta_1)' \\ &= E[E^{-1} X_1' M_2 \epsilon][E^{-1} X_1' M_2 \epsilon]' \\ &= E[E^{-1} X_1' M_2 \epsilon \epsilon' M_2 X_1 E^{-1}] \end{aligned}$$

$$= E(\epsilon \epsilon') [E^{-1} X_1' M_2 X_1 E^{-1}] = \sigma^2 [E^{-1} X_1' M_2 X_1 E^{-1}]$$

since M_2 is a idempotent matrix so that $M_2 M_2' = M_2 = M_2'$

$$\text{Var}(\hat{\beta}_1) = \sigma^2 E^{-1} X_1' M_2 X_1 E^{-1}$$

$$\text{but } E = X_1' M_2 X_1$$

$$\text{Var}(\hat{\beta}_1) = \sigma^2 E^{-1}$$

Now from ②

$$\hat{\beta}_2 = \bar{D}' X_2' M_1 Y$$

$$= \bar{D}' X_2' M_1 (X_1 \beta_1 + X_2 \beta_2 + \epsilon)$$

$$\hat{\beta}_2 = \bar{D}' X_2' M_1 X_1 \beta_1 + \bar{D}' X_2' M_1 X_2 \beta_2 + \bar{D}' X_2' M_1 \epsilon$$

$$\begin{aligned} \bar{D}' X_2' M_1 X_1 \beta_1 &= \bar{D}' \left[X_2' (I - X_1 (X_1' X_1)^{-1} X_1') \right] X_1 \beta_1 \\ &= \bar{D}' \left[X_2' X_1 - X_2' X_1 (X_1' X_1)^{-1} (X_1' X_1) \right] \beta_1 \end{aligned}$$

$$\bar{D}' X_2' M_1 X_1 \beta_1 = 0$$

So

$$\hat{\beta}_2 = \bar{D}' D \beta_2 + \bar{D}' X_2' M_1 \epsilon \quad \because D = X_2' M_1 X_2$$

$$\hat{\beta}_2 = \beta_2 + \bar{D}' X_2' M_1 \epsilon \quad \text{--- (4)}$$

$$E(\hat{\beta}_2) = \beta_2$$

$$\text{Var}(\hat{\beta}_2) = E[(\hat{\beta}_2 - \beta_2)(\hat{\beta}_2 - \beta_2)']$$

$$\text{from ④} \quad \hat{\beta}_2 - \beta_2 = \bar{D}' X_2' M_1 \epsilon$$

$$\text{Var}(\hat{\beta}_2) = E[\bar{D}' X_2' M_1 \epsilon][\bar{D}' X_2' M_1 \epsilon]'$$

$$= E[\bar{D}' X_2' M_1 \epsilon \epsilon' M_1' X_2 \bar{D}']$$

$$= \bar{D}' X_2' M_1 E(\epsilon \epsilon') M_1' X_2 \bar{D}'$$

$$= \sigma^2 [\bar{D}' X_2' M_1 M_1' X_2 \bar{D}']$$

$$\because M_1 M_1' = M_1$$

$$= \sigma^2 \bar{D}' X_2' M_1 X_2 \bar{D}'$$

$$= \sigma^2 \bar{D}' D \bar{D}'$$

$$\text{Var}(\hat{\beta}_2) = \sigma^2 \bar{D}'$$

$$\begin{aligned}
 \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) &= E[(\hat{\beta}_1 - \beta_1)(\hat{\beta}_2 - \beta_2)'] \\
 &= E[(E'X_1M_2\varepsilon)(D'X_2M_1\varepsilon)'] \\
 &= E[E'X_1M_2\varepsilon\varepsilon'M_1'X_2D'] \\
 &= E'X_1M_2E(\varepsilon\varepsilon')M_1'X_2D' \\
 &= E'X_1M_2\sigma^2M_1'X_2D'
 \end{aligned}$$

$$\text{Cov}(\hat{\beta}_1, \hat{\beta}_2) = E'X_1M_2M_1'X_2D'\sigma^2$$

where, D, E, M_1, M_2 are symmetric matrices

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Q. 5.1, Test the hypotheses (N.B. Plural) $\beta_1=1, \beta_2=1, \beta_3=-2$ in the regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

given the following sums of squares and products of deviations from means for 24 observations.

$$\sum y^2 = 60, \quad \sum x_1^2 = 10, \quad \sum x_2^2 = 30, \quad \sum x_3^2 = 20$$

$$\sum yx_1 = 7, \quad \sum yx_2 = -7, \quad \sum yx_3 = -26$$

$$\sum x_1 x_2 = 10, \quad \sum x_1 x_3 = 5, \quad \sum x_2 x_3 = 15$$

(b) Test also the hypothesis that $\beta_1 + \beta_2 + \beta_3 = 0$. How does this differ from the hypothesis that $[\beta_1 \quad \beta_2 \quad \beta_3] = [1 \quad 1 \quad -2]$? (c) Test the latter hypothesis.

Sol:

Estimator of β is $\hat{\beta} = (X'X)^{-1} X'y$

$$\text{Var}(\hat{\beta}) = (X'X)^{-1} \hat{\sigma}^2$$

$$\hat{\sigma}^2 = \frac{e'e}{n-k}$$

$$(X'X) = \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 & \sum x_1 x_3 \\ \sum x_1 x_2 & \sum x_2^2 & \sum x_2 x_3 \\ \sum x_1 x_3 & \sum x_2 x_3 & \sum x_3^2 \end{bmatrix} = \begin{bmatrix} 10 & 10 & 5 \\ 10 & 30 & 15 \\ 5 & 15 & 20 \end{bmatrix}$$

$$(X'X)^{-1} = \begin{bmatrix} 0.15 & -0.05 & 0 \\ -0.05 & 0.07 & -0.04 \\ 0 & -0.04 & 0.08 \end{bmatrix},$$

$$X'y = \begin{bmatrix} \sum x_1 y \\ \sum x_2 y \\ \sum x_3 y \end{bmatrix} = \begin{bmatrix} 7 \\ -7 \\ -26 \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} 0.15 & -0.05 & 0 \\ -0.05 & 0.07 & -0.04 \\ 0 & -0.04 & 0.08 \end{bmatrix} \begin{bmatrix} 7 \\ -7 \\ -26 \end{bmatrix} = \begin{bmatrix} 1.40 \\ 0.20 \\ -1.8 \end{bmatrix}$$

$$TSS = \sum y^2 = 60$$

$$\begin{aligned} \text{Reg SS} &= \hat{\beta}' x' y \\ &= [1.40 \quad 0.20 \quad -1.60] \begin{bmatrix} 7 \\ -7 \\ -26 \end{bmatrix} \end{aligned}$$

$$K = 4$$

$$\text{Reg SS} = 46.8$$

$$\text{Reg SS} = 55.2$$

$$\text{SSE} = \text{TSS} - \text{Reg SS}$$

$$\underline{e}'\underline{e} = 60 - 46.8 = 13.2$$

$$\hat{\sigma}^2 = \frac{\underline{e}'\underline{e}}{n-K} = \frac{13.2}{24-4} = 0.6286$$

$$\ast \quad df = 24 - 4 = 20$$

$$\text{Var}(\hat{\beta}) = (0.6286) \begin{bmatrix} 0.15 & -0.05 & 0 \\ -0.05 & 0.7 & -0.04 \\ 0 & -0.04 & 0.08 \end{bmatrix} = \begin{bmatrix} 0.0943 & -0.0314 & 0 \\ 0.044 & -0.221 & \\ 0.0503 & & \end{bmatrix}$$

(a)

(b)

$$H_0: \beta_1 = 1, \quad H_1: \beta_1 \neq 1$$

$$\alpha = 0.05$$

$$t = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1)}}$$

Critical region:

$$|t| > t_c$$

$$t_c = t_{\frac{\alpha}{2}, n-K} = 2.08$$

$$|t| = \frac{|1.40 - 1|}{\sqrt{0.0943}} = 1.30 < 2.08 \quad \therefore H_0 \text{ is accepted}$$

(ii)

$$H_0: \beta_2 = 1, \quad H_1: \beta_2 \neq 1$$

$$\alpha = 0.05$$

$$t = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{\text{Var}(\hat{\beta}_2)}}$$

Critical region $|t| > 2.28$

$$t_c = \frac{0.20 - 1}{\sqrt{0.041}} = -3.81$$

$$|t| = 3.81 > 2.28 \quad \therefore H_0 \text{ is rejected}$$

(iii)

$$H_0 : \beta_3 = -2.0 \quad H_1 : \beta_3 \neq -2.0$$

$$\alpha = 0.05$$

$$t = \frac{\hat{\beta}_3 - \beta_3}{\sqrt{\text{Var}(\hat{\beta}_3)}}$$

$$\text{C.R. } |t_c| > 2.08$$

$$t_c = \frac{-1.80 + 2.0}{\sqrt{0.0503}}$$

$$|t_c| = 0.89 < 2.08 \quad \therefore H_0 \text{ is rejected}$$

(b)

$$H_0 : \beta_1 + \beta_2 + \beta_3 = 0 \quad \text{or} \quad C'\beta = 0 \quad \text{where } C' = [1 \ 1 \ 1]$$

$$H_1 : \beta_1 + \beta_2 + \beta_3 \neq 0 \quad \text{or} \quad C'\beta \neq 0$$

$$\alpha = 0.05$$

$$t = \frac{C'\hat{\beta} - C'\beta}{\sigma \sqrt{C'(X'X)^{-1}C}} \sim t_{(n-k)}$$

$$\text{C.R. reject } H_0 \text{ if } t_c < -2.08 \text{ or } t_c > 2.08$$

$$C'(X'X)^{-1}C = [1 \ 1 \ 1] \begin{bmatrix} 0.15 & -0.05 & 0 \\ -0.05 & 0.07 & -0.04 \\ 0 & -0.04 & 0.08 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= [0.10 \quad -0.02 \quad 0.04] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0.12$$

$$\hat{\sigma} = 0.7728$$

$$t = \frac{-0.2 - 0}{(0.7728)\sqrt{0.12}} = -0.73$$

t_c falls in acceptance region. H_0 is accepted.

The hypothesis $(\beta_1 \ \beta_2 \ \beta_3) = (1 \ 1 \ -2)$ is the special case of $\beta_1 + \beta_2 + \beta_3 = 0$

(C) $H_0; \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

$$H_1: R\beta \neq t$$

$$R\beta = t$$

$$\alpha = 0.05$$

$$F = \frac{(R\hat{\beta} - R\beta)' [R(x'x)^{-1}R']^{-1} (R\hat{\beta} - R\beta) / v}{\hat{\sigma}^2 / n - k}$$

$$n = 24, k = 3, v = 3$$

$$(R\hat{\beta} - R\beta) = \begin{bmatrix} 1.40 \\ 0.20 \\ -1.80 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0.40 \\ -0.80 \\ 0.20 \end{bmatrix}$$

$$R(x'x)^{-1}R' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.15 & -0.05 & 0 \\ -0.05 & 0.07 & -0.04 \\ 0 & -0.04 & 0.08 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.15 & -0.05 & 0 \\ -0.05 & 0.07 & -0.04 \\ 0 & -0.04 & 0.08 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R(x'x)^{-1}R' = \begin{bmatrix} 0.15 & -0.05 & 0 \\ -0.05 & 0.07 & -0.04 \\ 0 & -0.04 & 0.08 \end{bmatrix}$$

$$(R(x'x)^{-1}R')^{-1} = \begin{bmatrix} 10 & 10 & 5 \\ 10 & 30 & 15 \\ 5 & 15 & 20 \end{bmatrix}$$

$$(R\hat{\beta} - R\beta)' [R(x'x)^{-1}R']^{-1} (R\hat{\beta} - R\beta)$$

$$= \begin{bmatrix} 0.40 & -0.80 & 0.20 \end{bmatrix} \begin{bmatrix} 10 & 10 & 5 \\ 10 & 30 & 15 \\ 5 & 15 & 20 \end{bmatrix} \begin{bmatrix} 0.40 \\ -0.80 \\ 0.20 \end{bmatrix}$$

$$= 0.4$$

$$F = \frac{0.4/3}{152/21} = 0.2121$$

$$CR \quad F > F_{(\alpha, q, n-k)}$$

$$F_{(0.05, 3, 21)} = 3.67$$

$$(3, 20)$$

$\therefore F < F_c \therefore H_0$ is accepted.

5.2 The following sums were obtained from 10 sets of observations on Y , X_1 and X_2 :

$$\sum Y = 20, \sum X_1 = 30, \sum X_2 = 40$$

$$\sum Y^2 = 88.2, \sum X_1^2 = 92, \sum X_2^2 = 163$$

$$\sum X_1 Y = 59, \sum X_2 Y = 83, \sum X_1 X_2 = 119$$

Estimate the regression of Y on X_1 and X_2 and test the hypothesis that the coefficient of β_2 is zero.

Sol: Estimator of β is $\hat{\beta} = (X'X)^{-1} X'Y$

$$(X'X) = \begin{bmatrix} n & \sum X_1 & \sum X_2 \\ \sum X_1 & \sum X_1^2 & \sum X_1 X_2 \\ \sum X_2 & \sum X_1 X_2 & \sum X_2^2 \end{bmatrix} = \begin{bmatrix} 10 & 30 & 40 \\ 30 & 92 & 119 \\ 40 & 119 & 163 \end{bmatrix}$$

$$(X'X)^{-1} = \begin{bmatrix} 16.7 & -2.60 & -2.20 \\ -2.60 & 0.60 & 0.20 \\ -2.20 & 0.20 & 0.40 \end{bmatrix}, \quad X'Y = \begin{bmatrix} \sum Y \\ \sum X_1 Y \\ \sum X_2 Y \end{bmatrix} = \begin{bmatrix} 20 \\ 59 \\ 83 \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} -13.0 \\ 1 \\ 3 \end{bmatrix}$$

Hence estimated regression of Y on X_1 and X_2 is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$$

$$\hat{Y} = -13.0 + X_1 + 3X_2$$

$$H_0: \beta_2 = 0, \quad H_1: \beta_2 \neq 0$$

$$\alpha = 0.05$$

$$t = \frac{\hat{\beta}_2 - \mu_2}{SE(\hat{\beta}_2)}$$

$$CR \quad t > t_{\left(\frac{\alpha}{2}, n-k\right)} = 2.365 \text{ or } t < -2.365$$

$$(0.025, 7)$$

$$Var(\hat{\beta}_1) = (X'X)^{-1} \hat{\sigma}^2$$

$$\hat{\sigma}^2 = \frac{e'e}{n-k}$$

$$e'e = TSS - Reg SS$$

$$TSS = \sum Y^2 - \frac{(\sum Y)^2}{n} = 88.2 - \frac{(20)^2}{10} = 48.2$$

$$Reg SS = \hat{\beta}'(X'Y) - C.F. = 63 - \frac{(20)^2}{10} = 23.0$$

$$\hat{\beta}'X'Y = \begin{bmatrix} -13 & 1 & 3 \end{bmatrix} \begin{pmatrix} 20 \\ 59 \\ 88 \end{pmatrix} = 63$$

$$e'e = 48.2 - 23.0 = 25.2$$

$$\hat{\sigma}^2 = \frac{25.2}{7} = 3.60$$

$$Var(\hat{\beta}_2) = (a_2 a_2) \hat{\sigma}^2 = (0.40)(3.60) = 1.44$$

$$t = \frac{3-0}{\sqrt{1.44}} = 2.50 > 2.365 \therefore H_0 \text{ is rejected i.e. } \beta_2 = 0 \text{ is rejected}$$

5.7; consider the following regression model in deviation form 86(I)

$$Y_t = \beta_1 x_{1t} + \beta_2 x_{2t} + u_t$$

with sample data

$$n = 100, \quad \sum y^2 = \frac{493}{3}, \quad \sum x_1^2 = 30, \quad \sum x_2^2 = 3$$

$$\sum x_1 y = 30, \quad \sum x_2 y = 20, \quad \sum x_1 x_2 = 0$$

2) Compute the ordinary LS estimates of β_1, β_2 and R^2 .

3) Test the hypothesis $H_0: \beta_2 = 7$ against $\beta_2 \neq 7$

4) Test the hypothesis $H_0: \beta_1 = \beta_2 = 0$ against $H_1: \beta_1 \neq 0 \text{ or } \beta_2 \neq 0$

2) Test the hypothesis $H_0: \beta_2 = 7$ against $H_1: \beta_2 \neq 7$ 81 I
 Sol, 95% C.I. of σ^2

(a) Estimator of β is $\hat{\beta} = (X'X)^{-1} X'y$

$$(X'X) = \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 \\ \sum x_1 x_2 & \sum x_2^2 \end{bmatrix} \quad X'y = \begin{bmatrix} \sum x_1 y \\ \sum x_2 y \end{bmatrix} = \begin{bmatrix} 30 \\ 20 \end{bmatrix}$$

$$(X'X) = \begin{bmatrix} 30 & 0 \\ 0 & 3 \end{bmatrix}$$

$$(X'X)^{-1} = \frac{\text{Adj}(X'X)}{|X'X|} = \frac{\begin{bmatrix} 3 & 0 \\ 0 & 30 \end{bmatrix}}{90} = \begin{bmatrix} \frac{1}{30} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} \frac{1}{30} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 30 \\ 20 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{20}{3} \end{bmatrix}$$

$$\hat{\beta}_1 = 1, \hat{\beta}_2 = \frac{20}{3}$$

$$R^2 = \frac{\text{Reg SS}}{\text{TSS}}$$

$$\text{TSS} = \sum y^2 = \frac{493}{3}$$

$$\text{Reg SS} = \hat{\beta}' X'y = \begin{bmatrix} 1 & \frac{20}{3} \end{bmatrix} \begin{bmatrix} 30 \\ 20 \end{bmatrix} = \frac{490}{3}$$

$$R^2 = \frac{490/3}{493/3} = \frac{490}{493}$$

$$K=3, n=100$$

$$\hat{\sigma}^2 = \frac{e'e}{n-K} = \frac{1}{97}$$

$$e'e = \text{TSS} - \text{Reg SS} = 1$$

$$\text{Var}(\hat{\beta}) = (X'X)^{-1} \hat{\sigma}^2 = \begin{bmatrix} \frac{1}{30} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \left(\frac{1}{97} \right) = \begin{bmatrix} \frac{1}{2940} & 0 \\ 0 & \frac{1}{29} \end{bmatrix}$$

$$\hat{\sigma}^2 = \frac{1}{97} \neq \frac{1}{8}$$

(b)

$$H_0: \beta_2 = 7, H_1: \beta_2 \neq 7$$

$$\alpha = 0.05$$

$$t = \frac{\hat{\beta}_2 - \beta_2}{\text{SE}(\hat{\beta}_2)}$$

$$\text{CR } |t| > t_{(0.025, 97)} = 2.306$$

$$= \frac{\frac{20}{3} - 7}{\sqrt{\frac{1}{29}}} = -1.63, |t| = 1.63 < 2.306$$

$\therefore H_0$ is accepted.

(d) $H_0: \beta_2 - 7\beta_1 = 0$ $H_1: \beta_2 - 7\beta_1 \neq 0$

$$\alpha = 0.05$$

$$t = \frac{(\hat{\beta}_2 - 7\hat{\beta}_1) - (\beta_2 - 7\beta_1)}{\text{Var}(\hat{\beta}_2 - 7\hat{\beta}_1)} \sim t_{(n-k)}$$

$$\begin{aligned} \text{Var}(\hat{\beta}_2 - 7\hat{\beta}_1) &= \text{Var}(\hat{\beta}_2) + 49 \text{Var}(\hat{\beta}_1) - 14 \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \\ &= \frac{1}{24} + \frac{49}{240} = \frac{54}{240} \end{aligned}$$

$$t = \frac{(\frac{2.2}{2} - 7) - 0}{\sqrt{\frac{54}{240}}} = -0.6722$$

C.R. $t < -2.306$ or $t > 2.306$

$\therefore t = -0.6722$ falls in acceptance region $\therefore H_0$ is accepted.

(e) $H_0: \beta_1 = \beta_2 = 0$ $H_1: \beta_1 \neq 0$ or $\beta_2 \neq 0$

$$\alpha = 0.05$$

$$F = \frac{\text{Reg SS}/k-1}{\text{SSE}/n-k} \sim F_{(k-1, n-k)}$$

under H_0

$$y = \beta x_1 + \beta_2 x_2 + u = \beta(x_1 + x_2) + u$$

$$y = \beta w + u$$

$$\hat{\beta} = \frac{\sum w y}{\sum w^2} = \frac{30}{30} = 1.6667$$

$$\sum w y = \sum x_1 y + \sum x_2 y = 30$$

$$\begin{aligned} \sum w^2 &= \sum (x_1 + x_2)^2 = \sum x_1^2 + \sum x_2^2 + 2 \sum x_1 x_2 \\ &= 30 + 30 + 2(0) = 60 \end{aligned}$$

$$SS \text{ due to reg of Reg} = \hat{\beta} \sum w y = (1.6667)(30) = 50.00$$

$$TSS = 493/3$$

$$\text{Reg SS} = 490/3$$

$$SSE = TSS - \text{Reg SS} = 1$$

$$F = \frac{490/2}{1/97} = \frac{(490)(97)}{(2)} = 792167$$

95% i)

$$\frac{\hat{\sigma}^2}{\chi^2 / df} < \sigma^2 < \frac{\hat{\sigma}^2}{\chi^2 / df}$$

$$\chi^2_{\frac{1}{df}} = \chi^2_{(1-\frac{\alpha}{2}, 11-K)} = \chi^2_{(0.975, 97)} = 1.303; \quad \chi^2_{\frac{1}{df}} = \chi^2_{(\frac{\alpha}{2}, 11-K)} = 0.7381$$

$$\frac{197}{1.303} < \sigma^2 < \frac{197}{0.7381}$$

$$0.0039 < \sigma^2 < 0.0139$$

5.13; Let the regression equation be partitioned as

$$Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$$

let b_1 and b_2 be the usual least squares estimators. Suppose that $E(\epsilon) = X_1 Y$, that is, the mean vector of the disturbances is a linear combination of some of the regressors. Prove that b_1 is biased but b_2 is unbiased. Obtain $V(b_1)$, $V(b_2)$.

Sol; Partitioned model is

$$Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$$

least square estimators for β_1 and β_2 are

$$b_1 = E^{-1}(X_1' M_2 Y) \quad \text{--- (1) where } E = X_1' M_2 X_1, \quad M_1 = I - X_1(X_1' X_1)^{-1} X_1'$$

$$b_2 = D^{-1}(X_2' M_1 Y) \quad \text{--- (2) } D = X_2' M_1 X_2, \quad M_2 = I - X_2(X_2' X_2)^{-1} X_2'$$

Now from (1)

$$b_1 = E^{-1} X_1' M_2 (X_1 \beta_1 + X_2 \beta_2 + \epsilon)$$

$$= E^{-1} X_1' M_2 X_1 \beta_1 + E^{-1} X_1' M_2 X_2 \beta_2 + E^{-1} X_1' M_2 \epsilon$$

$$b_1 = E^{-1} E \beta_1 + E^{-1} X_1' M_2 \epsilon$$

$$\because M_2 X_1 = 0$$

$$b_1 = \beta_1 + E^{-1} X_1' M_2 \epsilon$$

$$E(b_1) = \beta_1 + E^{-1} X_1' M_2 E \epsilon$$

$$= \beta_1 + E^{-1} X_1' M_2 X_1 Y$$

$$\because E \epsilon = X_1 Y$$

$$= \beta_1 + E^{-1} E Y$$

$$E(b_1) = \beta_1 + Y$$

$\therefore E(b_1) \neq \beta_1$, therefore b_1 is a biased estimator.

$$\begin{aligned}
 \text{Var}(b_1) &= E[(\hat{\beta}_1 - E(\hat{\beta}_1))(\hat{\beta}_1 - E(\hat{\beta}_1))'] \quad \text{where } E(b_1) = E(\hat{\beta}_1) \\
 &= E[\hat{\beta}_1 + E'X_1M_2\varepsilon - \beta_1 - \gamma][\hat{\beta}_1 + E'X_1M_2\varepsilon - \beta_1 - \gamma]' \quad \hat{\beta}_1 = b_1 \\
 &= E[E'X_1M_2\varepsilon - \gamma][E'X_1M_2\varepsilon - \gamma]' \\
 &= E[E'X_1M_2\varepsilon\varepsilon'M_2X_1E' - E'X_1M_2\varepsilon\gamma - \gamma\varepsilon'M_2X_1E' + \gamma\gamma'] \\
 &= E[E'X_1M_2E(\varepsilon\varepsilon')M_2X_1E' - E'X_1M_2E(\varepsilon)\gamma - \gamma\varepsilon'M_2X_1E' + \gamma\gamma'] \\
 &= \sigma^2 E'E'X_1M_2X_1E' - E'E'X_1M_2X_1\gamma\gamma' - \gamma\gamma'X_1M_2X_1E' + \gamma\gamma' \\
 &= \sigma^2 E'E' - \gamma\gamma' - \gamma\gamma' + \gamma\gamma'
 \end{aligned}$$

$$\text{Var}(b_1) = \sigma^2 E'E' - \gamma\gamma'$$

Now from Eq (2)

$$\begin{aligned}
 b_2 &= D'X_2M_1Y \\
 &= D'X_2M_1(X_1\beta_1 + X_2\beta_2 + \varepsilon) \\
 &= D'X_2M_1X_1\beta_1 + D'X_2M_1X_2\beta_2 + D'X_2M_1\varepsilon \\
 b_2 &= \beta_2 + D'X_2M_1\varepsilon
 \end{aligned}$$

$$E(b_2) = \beta_2 + D'X_2M_1X_1\gamma \quad \text{but } M_1X_1 = X_1 - X(X'X)^{-1}X' = 0$$

$$E(b_2) = \beta_2$$

Hence b_2 is an unbiased estimator of β_2

$$\begin{aligned}
 \text{Now } V(\beta_2) &= E[(b_2 - \beta_2)(b_2 - \beta_2)]' \\
 &= E[D'X_2M_1\varepsilon][D'X_2M_1\varepsilon]' \\
 &= E[D'X_2M_1\varepsilon\varepsilon'M_1X_2D'] \\
 &= D'X_2M_1X_2D'\sigma^2
 \end{aligned}$$

$$M_2M_1 = M_1$$

$$\text{Var}(\beta_2) = D'\sigma^2$$

Stepwise Regression SS II

In the first step of stepwise regression we want to estimate some parameter of the regression model and then in second step, to estimate remaining parameter we use residuals obtained in first step.

Let $Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$ is partitioned model.

In first step β_1 is estimated by regressing Y on X_1 alone.

So stepwise regression estimator of β_1 is

$$\checkmark \quad \tilde{\beta}_1 = (X_1' X_1)^{-1} X_1' Y \quad \text{--- (1) } \checkmark$$

Now the residual of regressing Y on X_1 alone is

$$\begin{aligned} \tilde{Y} &= Y - X_1 \tilde{\beta}_1 \\ &= Y - X_1 (X_1' X_1)^{-1} X_1' Y \\ &= [I - X_1 (X_1' X_1)^{-1} X_1'] Y \end{aligned}$$

$$\checkmark \quad \tilde{Y} = M_1 Y \quad \checkmark$$

Now stepwise regression estimate of β_2 is obtained by regressing $\tilde{Y}$ on X_2 alone

$$\begin{aligned} \tilde{\beta}_2 &= (X_2' X_2)^{-1} X_2' \tilde{Y} \\ \tilde{\beta}_2 &= (X_2' X_2)^{-1} X_2' M_1 Y \quad \text{--- (2) } \end{aligned}$$

Comparing with the full regression estimates of β_1 and β_2

$$\hat{\beta}_1 = (X_1' X_1)^{-1} X_1' Y - (X_1' X_1)^{-1} X_1' X_2 \bar{D}' X_2 M_1 Y \quad \text{--- (3) } \otimes$$

$$\hat{\beta}_2 = \bar{D}' X_2 M_1 Y \quad \text{--- (4) } \checkmark$$

From (1) and (3)

$$\tilde{\beta}_1 = \hat{\beta}_1 + (X_1' X_1)^{-1} X_1' X_2 \bar{D}' X_2 M_1 Y \quad \checkmark$$

$$\tilde{\beta}_1 = \hat{\beta}_1 + (X_1' X_1)^{-1} X_1' X_2 \hat{\beta}_2$$

From (2) and (4)

$$\tilde{\beta}_2 = (X_2' X_2)^{-1} \bar{D}' \hat{\beta}_2$$

Thus $\tilde{\beta}_1 = \hat{\beta}_1$ if $X_1' X_2 = 0$, $\Rightarrow$

$\otimes$ Similarly $\tilde{\beta}_2 = \hat{\beta}_2$ if $X_1' X_2 = 0$

It means stepwise regression estimates and full regression estimates coincides if $\underbrace{X_1' X_2 = 0}$ i.e. $\boxed{X_1' X_2 = 0}$

Equal

In Long

Q: (a) obtain the mean and variance of the stepwise regression estimates in partitioned model. Also find $\text{Cov}(\bar{\beta}_1, \bar{\beta}_2)$

$$Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$$

(b) Show that if we regress $\bar{Y}$ on $\bar{X}_2$ or even Y on $\bar{X}_2$, we would obtain the full regression estimate of β_2 , where $\bar{X}_2$ is the residual obtained from the auxiliary regression of X_2 on X_1 .

Sol: (a) Partitioned model is

$$Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$$

Stepwise regression estimator of β_1 is

$$\begin{aligned}\bar{\beta}_1 &= (X_1' X_1)^{-1} X_1' Y \\ &= (X_1' X_1)^{-1} X_1' (X_1 \beta_1 + X_2 \beta_2 + \epsilon) \\ &= (X_1' X_1)^{-1} X_1' X_1 \beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2 + (X_1' X_1)^{-1} X_1' \epsilon \\ &= \beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2 + (X_1' X_1)^{-1} X_1' \epsilon\end{aligned}$$

$$E(\bar{\beta}_1) = \beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2 + (X_1' X_1)^{-1} X_1' E \epsilon$$

$$E(\bar{\beta}_1) = \beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2$$

So $\bar{\beta}_1$ is a biased estimator of β_1 . It is unbiased estimator of $X_1' X_2 = 0$

Now

$$\text{Var}(\bar{\beta}_1) = E[(\bar{\beta}_1 - E(\bar{\beta}_1))(\bar{\beta}_1 - E(\bar{\beta}_1))']$$

$$\begin{aligned}\bar{\beta}_1 - E(\bar{\beta}_1) &= \beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2 + (X_1' X_1)^{-1} X_1' \epsilon - \beta_1 - (X_1' X_1)^{-1} X_1' X_2 \beta_2 \\ &= (X_1' X_1)^{-1} X_1' \epsilon\end{aligned}$$

$$\begin{aligned}\text{Var}(\bar{\beta}_1) &= E[(X_1' X_1)^{-1} X_1' \epsilon][X_1' X_1)^{-1} X_1' \epsilon'] \\ &= E[(X_1' X_1)^{-1} X_1' \epsilon \epsilon' X_1 (X_1' X_1)^{-1}] \\ &= E[(X_1' X_1)^{-1} X_1' E(\epsilon \epsilon') X_1 (X_1' X_1)^{-1}] \\ &= (X_1' X_1)^{-1} X_1' \sigma^2 I_n X_1 (X_1' X_1)^{-1} \\ &= (X_1' X_1)^{-1} X_1' \lambda_1 (X_1' X_1)^{-1} \sigma^2\end{aligned}$$

$$\text{Var}(\bar{\beta}_1) = (X_1' X_1)^{-1} \sigma^2$$

Now

$$\bar{\beta}_2 = (X_2' X_2)^{-1} X_2' M_1 Y$$

$$= (X_2' X_2)^{-1} X_2' M_1 (X_1 \beta_1 + X_2 \beta_2 + \epsilon)$$

$$\bar{\beta}_2 = (X_2' X_2)^{-1} X_2' M_1 X_1 \beta_1 + (X_2' X_2)^{-1} X_2' M_1 X_2 \beta_2 + (X_2' X_2)^{-1} X_2' M_1 \epsilon$$

$$\text{Now } M_1 X_1 = [I - X_1 (X_1' X_1)^{-1} X_1'] X_1$$

$$= X_1 - X_1 (X_1' X_1)^{-1} X_1' X_1 = 0$$

$$\begin{aligned}
 \Rightarrow \quad \bar{\beta}_2 &= (X_2' X_2)^{-1} X_2' M_1 X_2 \beta_2 + (X_2' X_2)^{-1} X_2' M_1 \epsilon \\
 E(\bar{\beta}_2) &= (X_2' X_2)^{-1} X_2' M_1 X_2 \beta_2 + (X_2' X_2)^{-1} X_2' M_1 E(\epsilon) \\
 &= (X_2' X_2)^{-1} X_2' X_2 \beta_2 \\
 &= (X_2' X_2)^{-1} X_2' [I - X_1 (X_1' X_1)^{-1} X_1'] X_2 \beta_2 \\
 &= (X_2' X_2)^{-1} (X_2' X_2) \beta_2 - (X_2' X_2)^{-1} X_2' X_1 (X_1' X_1)^{-1} X_1' X_2 \beta_2 \\
 E(\bar{\beta}_2) &= \beta_2 - (X_2' X_2)^{-1} X_2' X_1 (X_1' X_1)^{-1} X_1' X_2 \beta_2
 \end{aligned}$$

Hence $\bar{\beta}_2$ is not an unbiased estimator of β_2 . It is unbiased if $X_1' X_2 = 0$.

$$\begin{aligned}
 \text{Now } \text{Var}(\bar{\beta}_2) &= E[(\bar{\beta}_2 - E(\bar{\beta}_2))(\bar{\beta}_2 - E(\bar{\beta}_2))'] \\
 &= E[(X_2' X_2)^{-1} X_2' M_1 \epsilon][X_2' X_2)^{-1} X_2' M_1 \epsilon]' \\
 &= E[(X_2' X_2)^{-1} X_2' M_1 \epsilon \epsilon' M_1' X_2 (X_2' X_2)^{-1}] \\
 &= (X_2' X_2)^{-1} (X_2' M_1 X_2) (X_2' X_2)^{-1} \sigma^2 \\
 \text{Var}(\bar{\beta}_2) &= (X_2' X_2)^{-1} D (X_2' X_2)^{-1} \sigma^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(\bar{\beta}_1, \bar{\beta}_2) &= E[(\bar{\beta}_1 - E(\bar{\beta}_1))(\bar{\beta}_2 - E(\bar{\beta}_2))'] \\
 &= E[(X_1' X_1)^{-1} X_1' \epsilon][X_2' X_2)^{-1} X_2' M_1 \epsilon]' \\
 &= E[(X_1' X_1)^{-1} X_1' \epsilon \epsilon' M_1' X_2 (X_2' X_2)^{-1}] \\
 &= \sigma^2 [(X_1' X_1)^{-1} X_1' M_1' X_2 (X_2' X_2)^{-1}] \quad M_1' = M_1 \\
 &= \sigma^2 [(X_1' X_1)^{-1} X_1' [I - X_1 (X_1' X_1)^{-1} X_1'] X_2 (X_2' X_2)^{-1}] \\
 &= \sigma^2 [(X_1' X_1)^{-1} \{X_1' - X_1'\} X_2 (X_2' X_2)^{-1}]
 \end{aligned}$$

$$\text{Cov}(\bar{\beta}_1, \bar{\beta}_2) = 0$$

(b) In case we regress $\bar{Y}$ on $\bar{X}_2$

$$(\bar{X}_2' \bar{X}_2)^{-1} \bar{X}_2' \bar{Y}$$

where $\bar{X}_2$ is the residual from the auxiliary regression of X_2 on X_1 i.e.

$$\bar{X}_2 = X_2 - X_1 B_2$$

As B_2 is estimated by regressing X_2 on X_1 ,

$$B_2 = (X_1' X_1)^{-1} X_1' X_2$$

$$\text{So } \bar{X}_2 = X_2 - X_1 (X_1' X_1)^{-1} X_1' X_2$$

$$= (I - X_1 (X_1' X_1)^{-1} X_1') X_2$$

$$\bar{X}_2 = M_1 X_2, \quad \bar{Y} = M_1 Y$$

$$\text{Now } (\bar{X}_2' \bar{X}_2)^{-1} \bar{X}_2' \bar{Y} = (X_2' M_1 M_1 X_2)^{-1} X_2' M_1 M_1 Y \\ = (X_2' M_1 X_2)^{-1} X_2' M_1 Y$$

$$(\bar{X}_2' \bar{X}_2)^{-1} \bar{X}_2' \bar{Y} = \bar{D}' X_2' M_1 Y = \hat{\beta}_2$$

If we regress Y on $\bar{X}_2$

$$(\bar{X}_2' \bar{X}_2)^{-1} \bar{X}_2' Y = (X_2' M_1 M_1 X_2)^{-1} X_2' M_1 Y \\ = \bar{D}' X_2' M_1 Y \\ = \hat{\beta}_2$$

$$M_1 M_1' = M_1$$

Procedure For testing $H_0: \beta_2 = \beta_{20}$

GLR model in partitioned form is

$$Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$$

Total number of parameters are K . let in β_1 there are H parameters and in β_2 $(K-H)$ parameters.

To test this hypothesis we should use the sampling distribution of β_2 . Consider the quadratic form.

$$Q_2 = (\hat{\beta}_2 - \beta_{20})' D (\hat{\beta}_2 - \beta_{20})$$

$$\hat{\beta}_2 = \bar{D}' X_2' M_1 Y$$

$$= \bar{D}' X_2' M_1 (X_1 \beta_1 + X_2 \beta_2 + \epsilon)$$

$$= \bar{D}' X_2' M_1 X_1 \beta_1 + \bar{D}' X_2' M_1 X_2 \beta_2 + \bar{D}' X_2' M_1 \epsilon$$

$$= \bar{D}' D \beta_2 + \bar{D}' X_2' M_1 \epsilon$$

$$\because M_1 X_1 = 0, \quad D = X_2' M_1 X_2$$

$$\hat{\beta}_2 = \beta_2 + \bar{D}' X_2' M_1 \epsilon$$

$$\hat{\beta}_2 - \beta_2 = \bar{D}' X_2' M_1 \epsilon$$

$$\Rightarrow Q_2 = \epsilon' M_1' X_2 \bar{D}' D \bar{D}' X_2' M_1 \epsilon$$

$$= \epsilon' M_1 X_2 \bar{D}' X_2' M_1 \epsilon$$

$$Q_2 = \epsilon' P \epsilon$$

Where $P = M_1 X_2 \bar{D}' X_2' M_1$ is an idempotent matrix.

Since ϵ is a normal vector, P is also symmetric and idempotent matrix, so it indicates that Q_2 is quadratic in normal vector ϵ i.e. Q_2 follows χ^2 dist, with

$$df = \text{Rank}[P]$$

$$df = \text{tr}[P]$$

$$\begin{aligned} df &= \text{tr} [M X_2' D^{-1} X_2 M_1] \\ &= \text{tr} [X_2' M_1 M_1 X_2 D^{-1}] \\ &= \text{tr} [D D^{-1}] \end{aligned}$$

$$df = K - H$$

$\therefore M_1$ is idempotent

$$D = X_2' M_1 X_2$$

$(K-H) \times H$
 $H \times H$
 $H \times (K-H)$

Also

$$SSE = e'e$$

$$\begin{aligned} e &= (Y - \hat{Y}) = Y - X\hat{\beta} \\ &= Y - X(X'X)^{-1}X'Y \\ &= [I - X(X'X)^{-1}X']Y \\ &= MY \end{aligned}$$

$$e = M(X\beta + \epsilon) = MX\beta + M\epsilon$$

$$e = M\epsilon$$

$$\therefore MX = [I - X(X'X)^{-1}X']X = 0$$

$$e'e = e'M'e$$

$$\therefore MM = M$$

$$e'e = e'M'e$$

It is also Q.F in normal vector e , therefore it follows χ^2 dist, with

$$\begin{aligned} df &= \text{Rank}(M) = \text{tr}(M) \\ &= \text{tr}[I - X(X'X)^{-1}X'] \\ &= \text{tr} I_n - \text{tr}[X(X'X)^{-1}X'] \\ &= n - \text{tr}[X'X(X'X)^{-1}] \end{aligned}$$

$X'X$
 $K \times H$
 $H \times K$
 $(K+K)$

$$df = n - k$$

Two quadratic forms are independently distributed if Product of matrices in Q.F.s is zero i.e. $PM = 0$

We can show that $PM = 0$, as

$$M = M_1 - P$$

$$PM = PM_1 - PP$$

Since P is idempotent matrix

$$= M_1' X_2 D^{-1} X_2 M_1 M_1 - P$$

$$PP = P, \quad M_1 M_1 = M_1$$

$$PM = P - P = 0$$

It means that two Q.F.s are independently distributed, hence the statistic

$$F = \frac{Q_2 / k - H}{SSE / n - k} \sim F(k - H, n - k)$$

Special Case 1,

If $H=1$ i.e. β_1 contains only intercept parameter.

We write the standard hypothesis which specifies some elements of β are zero. If the hypothesis in fact is

$$H_0: \beta_2 = 0$$

then for the model $Y = X_1 \beta_1 + X_2 \beta_2 + \epsilon$

where Y is of order $n \times 1$, X_1 is of order $n \times H$

β_1 is of order $H \times 1$, X_2 is of order $n \times (K-H)$

β_2 is of order $(K-H) \times 1$

Quadratic form is

$$Q_2 = (\hat{\beta}_2 - \beta_2)' D (\hat{\beta}_2 - \beta_2)$$

under H_0

$$Q_2 = \hat{\beta}_2' D \hat{\beta}_2$$

$$= (D' X_2 M_1 Y)' D (D' X_2 M_1 Y)$$

$$= Y' M_1 X_2 D' D X_2 M_1 Y$$

$$Q_2 = Y' M_1 X_2 D' X_2 M_1 Y \quad \text{--- (1)}$$

Now

$$SS_{Reg} = \hat{\beta}' X' Y - c.f$$

$$= [\hat{\beta}_1' \quad \hat{\beta}_2'] \begin{bmatrix} X_1' Y \\ X_2' Y \end{bmatrix} - \frac{(\sum Y)^2}{n}$$

$$\sum Y = \alpha' Y$$

$$\alpha = (1 \dots 1)$$

$$= \hat{\beta}_1' X_1' Y + \hat{\beta}_2' X_2' Y - \frac{(\sum Y)^2}{n}$$

$$= \left[(X_1' X_1)^{-1} X_1' Y - (X_1' X_1)^{-1} X_1' X_2 D' X_2 M_1 Y \right]' X_1' Y + \left[D' X_2 M_1 Y \right]' X_2' Y$$

$$- Y' M_1 X_2 D' X_2 X_1 (X_1' X_1)^{-1} X_1' Y - c.f$$

$$= Y' X_1 (X_1' X_1)^{-1} X_1' Y - \underbrace{Y' M_1 X_2 D' X_2 X_1 (X_1' X_1)^{-1} X_1' Y}_{\uparrow} + Y' M_1 X_2 D' X_2 Y - c.f$$

$$= Y' X_1 (X_1' X_1)^{-1} X_1' Y + Y' M_1 X_2 D' X_2 [I - X_1 (X_1' X_1)^{-1} X_1'] Y - c.f$$

$$SS_{Reg} = Y' X_1 (X_1' X_1)^{-1} X_1' Y + Y' M_1 X_2 D' X_2 M_1 Y - c.f \quad \text{--- (1A)}$$

If we consider $H_0: \beta_2 = 0$ and $H=1$ i.e. β_1 contains only intercept parameter and

$$X_1' Y = \sum Y$$

$X_1' = [1 \ 1 \ 1 \dots 1]$, $X_1' X_1 = n$, then β_2 contains all

the regression coefficients, and by testing $H_0: \beta_2 = 0$, means we are testing the significance of regression.

Now

$$SS_{Reg} = (\sum Y) \frac{1}{n} (\sum Y) + Y' M_1 X_2 D' X_2 M_1 Y - c.f$$

$$S.S.Reg = \frac{(\sum Y)^2}{n} + Y' M_1 X_2 D' X_2' M_1 Y - \frac{(\sum Y)^2}{n}$$

$$S.S.R = Y' M_1 X_2 D' X_2' M_1 Y = Q_2 \quad \text{--- (2)}$$

Hence the statistic to test $H_0: \beta_2 = 0$ is

where
 $H=1$

$$F = \frac{Reg SS / k-1}{SSE / n-k} \quad \text{--- } F(k-1, n-k)$$

Special Case 2,

If $H \neq 1$ i.e. β_1 contains at least one regression coefficient and matrix X_1 includes values of at least one explanatory variable and we want to test the same hypothesis $H_0: \beta_2 = 0$.

It means we want to test the significance of some regression coefficient. If we had used only X_1 in the regression model, we would have estimated

$$\tilde{\beta}_1 = (X_1' X_1)^{-1} X_1' Y$$

Now SS due to regressing Y on X_1 alone is

$$(SSR)_n = n \tilde{\beta}_1' X_1' Y - C_4$$

$$(SSR)_n = Y' X_1 (X_1' X_1)^{-1} X_1' Y - \frac{(\sum Y)^2}{n} \quad \text{--- (3)}$$

Difference of equation (1A) and (3) gives

$$\Delta SSR = SSR - (SSR)_n$$

$$\Delta SSR = Y' M_1 X_2 D' X_2' M_1 Y = Q_2$$

Thus the statistic for testing the hypothesis $H_0: \beta_2 = 0$ is

$$F = \frac{\Delta SSR / k-H}{SSE / n-k} \quad \text{--- } F(k-H, n-k)$$

EXTRANEOUS INFORMATION

Sometimes the investigators may have some knowledge regarding to the parameters of the model say ratios of some co-efficients, values of some co-efficients, values of some linear combinations of the coefficients or the signs of some coefficients. This knowledge may derive from economic theory from earlier empirical investigations, or from other samples we refer to this type of knowledge as extraneous information i.e., comes from outside the sample itself.

(The use of extraneous information results gain in efficiency in some modern statistical theory common the Bayesian approach to statistical inference is employed, extraneous and sample informations are combined to obtain the knowledge of the "pop-parameters".)

USE OF EXTRANEOUS INFORMATION

GLR model is

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_K X_K + \epsilon$$

Restriction $Y = X\beta + \epsilon$

$$\beta_1 + \beta_2 + \beta_3 + \dots + \beta_K = 0$$

$$\beta_2 = -a \quad \text{where } a > 0$$

$$\beta_1 = \beta_4 \Rightarrow \beta_1 - \beta_4 = 0$$

These can be written in matrix form as

$$\begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 0 & 0 & \dots & 0 \\ 1 & 0 & 0 & -1 & \dots & 0 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \vdots \\ \beta_K \end{bmatrix} = \begin{bmatrix} 0 \\ -a \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

generally $R\beta = \epsilon$

which is an exact restriction for parameters) Using this information we have to estimate unknown parameters.

$$\sum \epsilon_i^2 = \epsilon' \epsilon$$

$$\sum e^2 = e' e$$

124

To find LSE under condition or restriction

$$\underset{(J \times k)}{R} \underset{(k \times 1)}{\beta} = \underset{(J \times 1)}{\epsilon} \quad \text{--- (1)}$$

Now we find restricted estimator. If β^* is the estimator, then SSE using Lagrange multiplier is obtained as

$$SSE = (Y - X\beta^*)'(Y - X\beta^*) - 2\lambda'(R\beta^* - \epsilon) \quad \lambda' \rightarrow (1 \times J)$$

$$= Y'Y - 2\beta^{*'}X'Y + \beta^{*'}(X'X)\beta^* - 2\lambda'(R\beta^* - \epsilon)$$

$$\frac{\partial SSE}{\partial \beta^*} = -2X'Y + 2X'X\beta^* - 2R'\lambda$$

$$R'\lambda$$

$$k \times J$$

$$J \times 1$$

$$\frac{\partial SSE}{\partial \beta^*} = 0 \Rightarrow$$

$$R'\lambda = X'X\beta^* - X'Y$$

$$X'X\beta^* = [X'Y + R'\lambda]$$

$$\beta^* = (X'X)^{-1}[X'Y + R'\lambda]$$

$$\beta^* = (X'X)^{-1}X'Y + (X'X)^{-1}R'\lambda$$

$$\beta^* = \hat{\beta} + (X'X)^{-1}R'\lambda \quad \text{--- (A)}$$

or

$$R\beta^* = R\hat{\beta} + R(X'X)^{-1}R'\lambda$$

$$\epsilon = R\hat{\beta} + R(X'X)^{-1}R'\lambda$$

$$(\epsilon - R\hat{\beta}) = R(X'X)^{-1}R'\lambda$$

$$\lambda = [R(X'X)^{-1}R']^{-1}(\epsilon - R\hat{\beta})$$

putting λ in (A) we have

$$\boxed{\beta_R^* = \hat{\beta} + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(\epsilon - R\hat{\beta})} \quad \text{--- (2)}$$

this estimator is called restricted estimator.

If we do not apply any restriction, then

$$(\epsilon - R\hat{\beta}) = 0$$

and equation (2) then becomes

$$\beta_R^* = \hat{\beta}$$

PROPERTIES OF RESTRICTED ESTIMATOR

Remove Watermark Now

1) It is a linear estimator.

2) It is an unbiased estimator i.e. $E(\beta^*) = \beta$

$$\beta_R^* = \hat{\beta} + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(\xi - R\hat{\beta})$$

$$E(\beta^*) = E(\hat{\beta}) + E\left\{[(X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}(\xi - R\hat{\beta})]\right\}$$

$$E(\hat{\beta}) = \beta$$

$$= \beta + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(\xi - RE(\hat{\beta}))$$

$$= \beta + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(\xi - R\beta)$$

Because $R\beta = \xi$

$$R\beta = \xi$$

$$\therefore E(\beta^*) = \beta + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(0)$$

$$E(\beta^*) = \beta$$

3)

$$Var(\beta_R^*) = E[\beta^* - E(\beta^*)][\beta^* - E(\beta^*)]'$$

$$= E[\beta^* - \beta][\beta^* - \beta]'$$

$$\beta^* - \hat{\beta} = (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(\xi - R\hat{\beta})$$

$$\therefore \hat{\beta} = (X'X)^{-1}X'Y$$

$$= (X'X)^{-1}X'(X\beta + \xi)$$

$$= (X'X)^{-1}(X'X)\beta + (X'X)^{-1}X'\xi$$

$$\hat{\beta} = \beta + (X'X)^{-1}X'\xi$$

$$\text{now } \beta^* - \hat{\beta} = (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}[\xi - R\{\beta + (X'X)^{-1}X'\xi\}]$$

$$\beta^* - \hat{\beta} - (X'X)^{-1}X'\xi = (X'X)^{-1}[R\{R(X'X)^{-1}R'\}^{-1}][\xi - R\beta]$$

$$- (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}X'\xi$$

$$\beta^* - \hat{\beta} = (X'X)^{-1}X'\xi - (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}X'\xi$$

$$\therefore R\beta = \xi$$

$$(\beta^* - \beta) = [I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R](X'X)^{-1}X'\xi$$

$$\text{Var}(\hat{\beta}) = E\left[(I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R)(X'X)^{-1}X'\epsilon\right]$$

$$\left[(I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R)(X'X)^{-1}X'\epsilon\right]'$$

$$= E\left[I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R\right](X'X)^{-1}X'\epsilon\epsilon'X(X'X)^{-1}\left[I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R\right]'$$

$$= \left[I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R\right](X'X)^{-1}(X'X)\sigma^2(X'X)^{-1}\left[I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R\right]'$$

$$= \left[I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R\right]V(\hat{\beta})\left[I - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}R\right]'$$

$$= IVI - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV$$

$$\therefore (X'X)^{-1}\sigma^2 = V$$

$$- VR'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$+ (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV R'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$= V - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV - VR'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$+ \frac{(X'X)^{-1}\sigma^2}{\sigma^2} R'\{R(X'X)^{-1}R'\}^{-1}RV R'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$= V - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV - VR'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$+ \frac{V}{\sigma^2} R'\{R(X'X)^{-1}R'\}^{-1}RV R'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$= V - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV + \frac{VR'}{\sigma^2}\{R(X'X)^{-1}R'\}^{-1}RV R'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$- VR'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$= \text{// // // // } \epsilon R \sigma^2 (X'X)^{-1} R'$$

$$= V - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV + VR'\{R(X'X)^{-1}R'\}^{-1}RV R'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$- VR'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$= V - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV + VR'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$- VR'\{R(X'X)^{-1}R'\}^{-1}R(X'X)^{-1}$$

$$V(\hat{\beta}) = V_{\text{var}}(\hat{\beta}) - (X'X)^{-1}R'\{R(X'X)^{-1}R'\}^{-1}RV \frac{\sigma^2}{\sigma^2} -$$

$$\text{Var}(\hat{\beta}^*) = Y - (X'X)^{-1} \sigma^2 R' [R(X'X)^{-1} \sigma^2 R']^{-1} RV$$

$$\text{Var}(\hat{\beta}^*) = Y - VR'[RVR']^{-1}RV$$

$$\text{Var}(\hat{\beta}) - \text{Var}(\hat{\beta}^*) = VR'[RVR']^{-1}RV$$

Variance Covariance matrix is always symmetric and p.d matrix.

Quadratic form (RVR') is also positive definite, whose inverse is non-negative.

As a result R.H.S is non-negative so we can conclude.

$$(\text{Var}(\hat{\beta}) > \text{Var}(\hat{\beta}^*) \text{ or } \text{Var}(\hat{\beta}^*) < \text{Var}(\hat{\beta}))$$

It means that restricted estimator is more efficient as compared to unrestricted estimator.)

Extraneous Unbiased Estimator;

GLR model is $Y = X\beta + \epsilon$

In GLR model we are known some unbiased estimators of parameters, using this information we have estimate remaining parameters

$$Y = X_1\beta_1 + X_2\beta_2 + \epsilon$$

Where β_1 contains such parameters, whose unbiased estimators are available. Say unbiased estimator $\hat{\beta}_1^*$ is available for β_1 .

$$\text{i.e. } E(\hat{\beta}_1^*) = \beta_1$$

To estimate restricted estimator of β_2 we should regress $Y - X_1\hat{\beta}_1^*$ on X_2

$$\text{then } \hat{\beta}_2^* = (X_2'X_2)^{-1}X_2'(Y - X_1\hat{\beta}_1^*)$$

Sampling Properties of Restricted Estimator

1) It is linear in observations.

2) Unbiased

$$\hat{\beta}_2^* = (X_2'X_2)^{-1}X_2'(Y - X_1\hat{\beta}_1^*)$$

$$= (X_2'X_2)^{-1}X_2'(X_1\beta_1 + X_2\beta_2 + \epsilon - X_1\hat{\beta}_1^*)$$

$$\hat{\beta}_2^* = (X_2'X_2)^{-1}X_2'X_1\beta_1 + (X_2'X_2)^{-1}X_2'X_2\beta_2 + (X_2'X_2)^{-1}X_2'\epsilon - (X_2'X_2)^{-1}X_2'X_1\hat{\beta}_1^*$$

$$\text{But } \hat{\beta}_1^* = \beta_1 + d, \quad E(d) = 0$$

$$\hat{\beta}_2^* = \beta_2 + (X_2'X_2)^{-1}X_2'X_1\beta_1 + (X_2'X_2)^{-1}X_2'\epsilon - (X_2'X_2)^{-1}X_2'X_1[\beta_1 + d]$$

$$\hat{\beta}_2^* = \beta_2 + (X_2'X_2)^{-1}X_2'X_1\beta_1 + (X_2'X_2)^{-1}X_2'\epsilon - (X_2'X_2)^{-1}X_2'X_1\beta_1 - (X_2'X_2)^{-1}X_2'X_1d$$

$$\beta_2^* = \beta_2 + (X_2' X_2)^{-1} X_2' (\epsilon - X_1 \alpha) \quad \text{--- ①}$$

$$E(\beta_2^*) = \beta_2$$

Hence β_2^* is an unbiased restricted estimator of β_2 .

(3)

$$\begin{aligned} \text{Var}(\beta_2^*) &= E[(\beta_2^* - E(\beta_2^*))(\beta_2^* - E(\beta_2^*))'] \\ &= E[(\beta_2^* - \beta_2)(\beta_2^* - \beta_2)'] \quad \text{--- ②} \end{aligned}$$

$$\text{From ① } \beta_2^* - \beta_2 = (X_2' X_2)^{-1} X_2' (\epsilon - X_1 \alpha)$$

$$\begin{aligned} \text{Var}(\beta_2^*) &= E[(X_2' X_2)^{-1} X_2' (\epsilon - X_1 \alpha)] [(X_2' X_2)^{-1} X_2' (\epsilon - X_1 \alpha)]' \\ &= E[(X_2' X_2)^{-1} X_2' (\epsilon - X_1 \alpha)(\epsilon - X_1 \alpha)' X_2 (X_2' X_2)^{-1}] \\ \text{Var}(\beta_2^*) &= (X_2' X_2)^{-1} X_2' E[\epsilon \epsilon' - \epsilon \alpha' X_1 - X_1 \alpha \epsilon' + X_1 \alpha \alpha' X_1'] X_2 (X_2' X_2)^{-1} \end{aligned}$$

We know $E \epsilon \epsilon' = \sigma^2 I$, we assume $E \epsilon \alpha' = 0$

$$\begin{aligned} \alpha &= \beta_1^* - \beta_1 \\ E(\alpha \alpha') &= E[\beta_1^* - \beta_1][\beta_1^* - \beta_1]' = \text{Var}(\beta_1^*) = V_1^* \end{aligned}$$

$$\begin{aligned} \text{Now } \text{Var}(\beta_2^*) &= (X_2' X_2)^{-1} X_2' [\sigma^2 I - 0 - 0 + X_1 V_1^* X_1'] X_2 (X_2' X_2)^{-1} \\ &= (X_2' X_2)^{-1} (X_2' X_2) (X_2' X_2)^{-1} \sigma^2 + (X_2' X_2)^{-1} X_2' X_1 V_1^* X_1' X_2 (X_2' X_2)^{-1} \end{aligned}$$

$$V_2^* = \text{Var}(\beta_2^*) = (X_2' X_2)^{-1} \sigma^2 + B_{12} V_1^* B_{12}'$$

$$\text{where } B_{12} = (X_2' X_2)^{-1} X_2' X_1$$

We can show that variance of unrestricted estimator $\hat{\beta}_2$ is

$$V_2 = \sigma^2 (X_2' X_2)^{-1} + (X_2' X_2)^{-1} X_2' X_1 V_1 X_1' X_2 (X_2' X_2)^{-1}$$

$$\text{and } V_1 = \text{Var}[\hat{\beta}_1] = \sigma^2 [X_1' X_1 - X_1' X_2 (X_2' X_2)^{-1} X_2' X_1]^{-1} = E^{-1} \sigma^2$$

$$V_2 = \sigma^2 (X_2' X_2)^{-1} + (X_2' X_2)^{-1} X_2'$$

$$V_2 = \sigma^2 (X_2' X_2)^{-1} + B_{12} V_1 B_{12}'$$

Now

$$V_2 - V_2^* = B_{12} (V_1 - V_1^*) B_{12}'$$

If it means that restricted estimator is more efficient if K_{12} is +ve, which is only possible if restricted estimator of β_1 is

efficient to unrestricted estimator of β_1 . It is commonly true, thus the restricted estimator β_1^* is efficient as compared to unrestricted estimator.

✓



MULTICOLLINEARITY TOPIC: 2

The term multicollinearity due to Rager Frisch means the existence of exact or inexact linear relationship among some or all explanatory variables of a regression model. [When there is exact linear relationship then we cannot estimate the parameters uniquely]. [The more difficult problem arises, when there is inexact, partial or imperfect dependence among explanatory variables. In this situation we can estimate the parameters but with very large and unstable variances.

There are several levels of multicollinearity depending upon the degree of dependence between the explanatory variables]

Case 1

Perfect Multicollinearity; Exact, Complete,

[Extreme case of multicollinearity occurs when two or more explanatory variables are completely dependent in the sense that the values of 1 variable can be obtained by a linear combination of the values of another variable i.e. relationship of the form

$$X_{1i} = c + d X_{3i}$$

then correlation Co-efficient r_{13} is unity]

Question We are given the GLR model as

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon$$

Also we know that $X_3 = m + p X_4$. Can we estimate the parameters if not give the reasons?

Sol: In deviation form our model reduces to

$$y = \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \epsilon$$

$$x_3 = p x_4 \quad \text{--- (1)}$$

Normal equations for β_3, β_4 are

$$\sum x_3 y = \hat{\beta}_2 \sum x_2 x_3 + \hat{\beta}_3 \sum x_3^2 + \hat{\beta}_4 \sum x_3 x_4 \quad \text{--- (2)}$$

$$\sum x_4 y = \hat{\beta}_2 \sum x_2 x_4 + \hat{\beta}_3 \sum x_3 x_4 + \hat{\beta}_4 \sum x_4^2 \quad \text{--- (3)}$$

Normal eq. for β_2 is

$$\sum x_2 y = \hat{\beta}_2 \sum x_2^2 + \hat{\beta}_3 \sum x_2 x_3 + \hat{\beta}_4 \sum x_2 x_4 \quad \text{--- (4)}$$

putting $x_3 = p x_4$ in ② and ③

we get ② $\Rightarrow \sum p x_4 y = \hat{\beta}_2 \sum x_2 p x_4 + \hat{\beta}_3 \sum p^2 x_4^2 + \hat{\beta}_4 \sum p x_4^2$
 $\sum x_4 y = \hat{\beta}_2 \sum x_2 x_4 + \hat{\beta}_3 p \sum x_4^2 + \hat{\beta}_4 \sum x_4^2 \quad \text{--- ⑤}$

③ $\Rightarrow \sum x_4 y = \hat{\beta}_2 \sum x_2 x_4 + \hat{\beta}_3 p \sum x_4^2 + \hat{\beta}_4 \sum x_4^2 \quad \text{--- ⑥}$

It means that normal equations for β_3, β_4 are identical, so we have 2 normal equations but 3 parameters to be estimated, therefore we can not estimate the parameters β_3 and β_4 simultaneously.]

[The problem can be solved by taking more observations hopefully that with the addition of few more observations the dependence between the explanatory variables is broken. If taking of more observations does not result in a source to that problem then the 2nd possible case is try to obtain extraneous information about β_3 and β_4].

Case II

PARTIAL OR INCOMPLETE M.C.O.

Generally there is no exact linear relationship among the explanatory variables, but rather ^{some} stochastic relationship among the all or some explanatory variables i.e., they are intercorrelated but not perfectly means that their correlation coefficient is high but not 1.

[Whenever there is slight indication that two or more than 2 explanatory variable are very closely linearly dependant. It is necessary to verify the fact by computing the simple correlation or multiple correlation. If these are numerically close to 1 then there is multicollinearity]

Methods To ^{effect of} Avoid Multicollinearity,

There is no simple or clearest methods to avoid or to minimize the effect of M.C. However the Econometrician suggest the points.

(1)

Care in the initial choice of variables is implied, if it is known that several variables measure the same factor then one of these should not be used without good reason.

- (2) If prior information cannot be used to limit the no. of variables, ~~the~~ selection procedure may be used. Unfortunately such procedures are less effective in the presence of M.col.
- (3) Increase the sample size with the hope that by increasing sample size the dependency between the explanatory variable is broken.
- (4) Certain M.V techniques such as principal components or factor analysis may sometimes be used to help in the selection of independent variables.
- (5) Combine the variables which are linearly dependent, continue this process unless all the explanatory variables used in the model become linearly independent.
- (6) Use extraneous information for parameters.
- (7) In some situations theory or other considerations may dictate that correlated variables must be included in the model for such situations biased estimation procedures such as Ridge regression method can be used.

Polynomial Regression

Consider

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 \dots$$

where X is first degree polynomial.

X^2 is second degree polynomial.

X^3 is third degree polynomial and so on.

We can also express this model in another form as

$$Y = \beta_0 + \beta_1 \phi_1(x) + \beta_2 \phi_2(x) + \dots + \beta_k \phi_k(x) + \epsilon$$

$\phi_i(x)$ = i th degree polynomial in x .

$$X = \begin{bmatrix} 1 & \phi_1(x_1) & \phi_2(x_1) & \dots & \phi_k(x_1) \\ 1 & \phi_1(x_2) & \phi_2(x_2) & \dots & \phi_k(x_2) \\ 1 & \phi_1(x_3) & \phi_2(x_3) & \dots & \phi_k(x_3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \phi_1(x_n) & \phi_2(x_n) & \dots & \phi_k(x_n) \end{bmatrix}$$

LSE is $\hat{\beta} = (X'X)^{-1}X'Y$

$$(X'X) = \begin{bmatrix} n & \sum \phi_1(x) & \sum \phi_2(x) & \dots & \sum \phi_k(x) \\ \sum \phi_1^2(x) & \sum \phi_1(x)\phi_2(x) & \dots & \sum \phi_1(x)\phi_k(x) \\ \sum \phi_2^2(x) & \dots & \sum \phi_2(x)\phi_k(x) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum \phi_k^2(x) & \dots & \dots & \dots & \sum \phi_k(x)\phi_k(x) \end{bmatrix}, \quad X'Y = \begin{bmatrix} \sum Y \\ \sum \phi_1(x)Y \\ \sum \phi_2(x)Y \\ \vdots \\ \sum \phi_k(x)Y \end{bmatrix}$$

Special Form of Polynomial Regression;

ORTHOGONAL REGRESSION ANALYSIS;

A regression analysis with uncorrelated estimators is called orthogonal regression. In this case regressors are not random variables, but constants which are at choice in experimental works, we may now face a new type of problem. How should the elements of X be chosen such that the estimators are uncorrelated.

We can observe that the analysis is orthogonal if and only if

$(X'X)^{-1}$ is diagonal, which is only possible if $(X'X)$ is diagonal.

Thus to obtain an orthogonal analysis we must choose the elements of X such that $(X'X)$ is diagonal. It follows when

$$\sum_{j=1}^n \phi_c(x_j) = 0 \quad \forall c = 1, 2, \dots, K$$

$$\text{and } \sum_{j=1}^n \phi_c(x_j) \phi_d(x_j) = 0 \quad \text{for } c \neq d$$

We have to decide for the best degree of regression for given n pairs of values (x_i, y_i) .

If we fit the first degree eq $Y = \beta_0 + \beta_1 X$ and test it for the coefficients, if this is significant then we pass on the 2nd degree polynomial. We continue this process till for the first time we get two consecutive insignificant regression.

If we get p th degree regression insignificant, then $(p-1)$ th degree regression is the best one.

Regression coefficients for each degree regression equation are not the same, it means we have to make fresh calculations at each stage, which is time consuming and cumbersome procedure. To avoid this difficulty orthogonal polynomials are used.

Under the conditions of orthogonal polynomials, we have

$$X'X = \phi' \phi = \begin{bmatrix} n & 0 & 0 & \dots & 0 & \dots & 0 \\ 0 & \sum \phi_1^2(x) & 0 & \dots & 0 & \dots & 0 \\ 0 & 0 & \sum \phi_2^2(x) & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & \dots & \sum \phi_k^2(x) \end{bmatrix}$$

$$(X'X)^{-1} = \begin{bmatrix} \frac{1}{n} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{\sum \phi_1^2(x)} & 0 & \dots & 0 \\ 0 & 0 & \frac{1}{\sum \phi_2^2(x)} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{1}{\sum \phi_K^2(x)} \end{bmatrix}, \quad X'Y = \begin{bmatrix} \sum Y \\ \sum \phi_1(x)Y \\ \sum \phi_2(x)Y \\ \vdots \\ \sum \phi_K(x)Y \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} \frac{\sum Y}{\sum 1} \\ \frac{\sum \phi_1(x)Y}{\sum \phi_1^2(x)} \\ \frac{\sum \phi_2(x)Y}{\sum \phi_2^2(x)} \\ \vdots \\ \frac{\sum \phi_K(x)Y}{\sum \phi_K^2(x)} \end{bmatrix}$$

Simply the ^{degree} i th regression coefficient is

$$\hat{\beta}_i = \frac{\sum \phi_i(x)Y}{\sum \phi_i^2(x)} \quad \forall \quad i = 0, 1, 2, \dots, K$$

An unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{[Y'Y - \hat{\beta}'X'Y]}{n - (K+1)} \quad Y'Y = \sum Y^2$$

$$\text{Reg SS (unadjusted)} = \hat{\beta}_0 \sum Y + \hat{\beta}_1 \sum \phi_1(x)Y + \hat{\beta}_2 \sum \phi_2(x)Y + \dots + \hat{\beta}_K \sum \phi_K(x)Y$$

$$\text{Reg SS (corrected)} = \hat{\beta}_0 \sum Y + \hat{\beta}_1 \sum \phi_1(x)Y + \hat{\beta}_2 \sum \phi_2(x)Y + \dots + \hat{\beta}_K \sum \phi_K(x)Y - C$$

$$C = \frac{(\sum Y)^2}{n} = \hat{\beta}_0 \sum Y$$

So

$$\text{Reg SS}_G = \sum_{i=1}^K \left(\hat{\beta}_i \sum_{j=1}^n \phi_i(x_j) Y_j \right)$$

Since $\phi_i(x)$ are orthogonal Polynomials and as a result the estimators are also orthogonal.

$\hat{\beta}_1 \sum \phi_1(x) Y$ is Reg SS for 1st degree Polynomial.

$\hat{\beta}_2 \sum \phi_2(x) Y$ is Reg SS for 2nd degree Polynomial.

Soon.

We can test each Regression coefficient of different degree Polynomial step by step.

ANOVA TABLE

S.O.V	d.f	S.S	M.S	F
Total	$n-1$	$Y'Y - cf = \sum (Y - \bar{Y})^2 = S_y^2$	$\frac{\sum Y^2 - cf}{n-1}$	
Linear Reg	1	$\hat{\beta}_1 \sum \phi_1(x) Y$	$\hat{\beta}_1 \sum \phi_1(x) Y$	?
Residual	$n-2$	$TSS - \hat{\beta}_1 \sum \phi_1(x) Y = SSE_1$	$SSE_1 / n-2$	
2nd degree Reg	1	$\hat{\beta}_2 \sum \phi_2(x) Y$		?
Residual	$n-3$	$SSE_2 = SSE_1 - \hat{\beta}_2 \sum \phi_2(x) Y$	$SSE_2 / n-3$	
3rd degree Reg	1	$\hat{\beta}_3 \sum \phi_3(x) Y$		?
		PUACP		

To decide for the best fit continue this process until for the first time we get insignificant result. It is better to have two consecutive insignificant coefficients. If p th degree reg coefficient is insignificant, then $(p-1)$ th degree polynomial is the best fit.

Construction of Orthogonal Polynomials

The polynomials $\phi_i(x_j)$ and $\phi_h(x_j)$ are said to be orthogonal if sum of products vanishes.

$$\text{i.e. } \sum_{j=1}^n \phi_i(x_j) \phi_h(x_j) = 0 \quad \text{--- (1)} \quad i \neq h, 0, 1, \dots, k$$

where

$$\phi_0(x) = C_{00} \quad \text{--- I}$$

$$\phi_1(x) = C_{10} + C_{11}x \quad \text{--- II}$$

$$\phi_2(x) = C_{20} + C_{21}x + C_{22}x^2 \quad \text{--- III}$$

$$\vdots$$

$$\phi_i(x) = C_{i0} + C_{i1}x + C_{i2}x^2 + \dots + C_{ii}x^i \quad \text{--- (i+1)}$$

In general we can write

$$\phi_i(x) = \sum_{c=0}^i C_{ic} x^c \quad \text{--- (2)}$$

There are (i+1) coefficients in Eq (2). Hence in all orthogonal polynomials there are Σ

$$\sum_{c=0}^k (i+1) = \frac{(k+1)(k+2)}{2} \quad \text{Coefficients to be determined}$$

$\phi_0(x)$

Polyn

Coeff

$\phi_0(x)$

1

$\phi_1(x)$

2

$\vdots$

$\phi_k(x)$

k+1

$$\left| \begin{aligned} \frac{n}{C_2} &= \frac{n(n-1)}{2} \\ \frac{k+1}{C_2} &= \frac{k(k+1)}{2} \end{aligned} \right|$$

There will be $\frac{n+1}{2}$ pairs i.e. $\frac{k(k+1)}{2}$ pairs

Now imposing restrictions on coefficients, which contains $\frac{k(k+1)}{2}$ conditions

Thus the unknowns to be determined are

$$\frac{(k+1)(k+2)}{2} - \frac{k(k+1)}{2} = \frac{(k+1)}{2} [k+2 - k] = k+1$$

For a unique solution we must left (k+1) more constraints. So we apply the condition that

$$C_{ii} = 1 \text{ for all } i$$

under these conditions we have directly

$$\phi_0(x) = C_{00} = 1$$

Now from equation ①

$$\sum_{j=1}^n \left[\left(\sum_{t=0}^c c_{it} x_j^t \right) \left(\sum_{s=0}^h c_{hs} x_j^s \right) \right] = 0$$

$$c_{it} \neq 0 \neq c_{hs}$$

$$\sum_{t=0}^c c_{it} \sum_{s=0}^h c_{hs} \sum_{j=1}^n x_j^{t+s} = 0$$

$$\sum_{t=0}^c c_{it} \sum_{s=0}^h c_{hs} \mu_{t+s} = 0 \quad \text{--- ③}$$

$$\frac{\sum x}{n} = \mu_1$$

$$\frac{\sum x^2}{n} = \mu_2$$

$$\frac{\sum x^{t+s}}{n} = \mu_{t+s}$$

where μ_p is the p th moment of the set of x values

$$\text{i.e. } \mu_p = \sum x_j^p$$

Hence μ_{t+s} is the $(t+s)$ th order moment about origin.

Since Eq ③ holds for all $i = 0, 1, 2, \dots, K-1$, we must have

$$\sum_{s=0}^h c_{hs} \mu_{t+s} = 0 \quad \text{for all } t = 0, 1, \dots, K-1 \quad \text{④}$$

$\therefore c_{hs} = \text{non zero coefficient}$

in writing the determinant

$$|M_K| = \begin{vmatrix} \mu_0 & \mu_1 & \mu_2 & \dots & \mu_{K-1} & \mu_K \\ \mu_1 & \mu_2 & \mu_3 & \dots & \mu_K & \mu_{K+1} \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ \mu_{K-1} & \mu_K & \dots & \dots & \mu_{2K-1} \\ \mu_K & \mu_{K+1} & \dots & \dots & \mu_{2K} \end{vmatrix}$$

$(K+1)(K+1)$

PUACP

This is a determinant of $(K+1)$ order and $|M_K^{uv}|$ is minor of the element in u th row and v th column of $|M_K|$.

Then solution of Eq ④ is given by

$$c_{hs} = \frac{|M_K^{K+1, s+1}|}{|M_{K-1}|} \quad \text{--- ⑤}$$

for the minor of $K+1$ row
& $s+1$ column.

Thus $\phi_k(x) = \sum_{i=0}^k C_{ki} x^i$ and Eq ⑤ implies

$$\phi_k(x) = \frac{1}{|M_{k-1}|} \begin{vmatrix} \mu'_0 & \mu'_1 & \mu'_2 & \dots & \mu'_k \\ \mu'_1 & \mu'_2 & \mu'_3 & \dots & \mu'_{k+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mu'_{k-1} & \mu'_k & \mu'_{k+1} & \dots & \mu'_{2k-1} \\ 1 & x & x^2 & \dots & x^k \end{vmatrix}$$

Equation ⑥ can be used to evaluate polynomial for any k .

For $k=1$

$$\phi_1(x) = \frac{1}{|M_0|} \begin{vmatrix} \mu'_0 & \mu'_1 \\ 1 & x \end{vmatrix} = \frac{1}{\mu'_0} \begin{vmatrix} \mu'_0 & \mu'_1 \\ 1 & x \end{vmatrix}$$

$$= (\mu'_0 x - \mu'_1) \cdot \frac{1}{\mu'_0}$$

$$\phi_1(x) = x - \frac{\mu'_1}{\mu'_0}$$

$$\boxed{\phi_1(x) = x}$$

$$\begin{aligned} \mu'_p &= \frac{\sum x^p}{n} \\ \mu'_0 &= 1 \\ \mu'_1 &= \frac{\sum x}{n} = 0 \end{aligned}$$

For $k=2$

$$\phi_2(x) = \frac{1}{|M_1|} \begin{vmatrix} \mu'_0 & \mu'_1 & \mu'_2 \\ \mu'_1 & \mu'_2 & \mu'_3 \\ 1 & x & x^2 \end{vmatrix} \quad \text{where } |M_1| = \begin{vmatrix} \mu'_0 & \mu'_1 \\ \mu'_1 & \mu'_2 \end{vmatrix} = \mu'_2$$

Denominator is obtained by deleting last row and last column.

$$\phi_2(x) = \frac{1}{\mu'_2} \left[\mu'_0(\mu'_2 x^2 - \mu'_3 x) - \mu'_1(\mu'_1 x^2 - \mu'_3) + \mu'_2(\mu'_1 x - \mu'_2) \right]$$

$$\boxed{\phi_2(x) = x^2 - \frac{\mu'_3}{\mu'_2} x - \mu'_2}$$

$$\text{or } \phi_2(x) = x^2 + \frac{(\mu'_2 \mu'_1 - \mu'_3)}{\mu'_2} x + \frac{\mu'_1 \mu'_3 - \mu'_2^2}{\mu'_2}$$

For $K=3$

$$\phi_3(x) = \frac{1}{|M_3|} \begin{vmatrix} \mu'_0 & \mu'_1 & \mu'_2 & \mu'_3 \\ \mu_1 & \mu_2 & \mu_3 & \mu_4 \\ \mu_2 & \mu_3 & \mu_4 & \mu_5 \\ 1 & x & x^2 & x^3 \end{vmatrix}$$

where

$$|M_3| = \begin{vmatrix} \mu'_0 & \mu'_1 & \mu'_2 \\ \mu_1 & \mu_2 & \mu_3 \\ \mu_2 & \mu_3 & \mu_4 \end{vmatrix}$$

$$\phi_3(x) = x^3 - \frac{\mu_4}{\mu_2} x$$

Equally Spaced Values of Independent Variable (x)

The most important applications of orthogonal polynomials in regression analysis are two situations where the regressor takes values at equal intervals. This is often happens in time series data. If we have n such equally spaced values of x , we can measure these values from their mean and take the natural interval as unit. Thus obtaining the values as working values of x .

1930 1 -6

1940 2 -5

1950 3 -4

1960 4 -3

-2

-1

0

1

2

...

$$-\frac{(n-1)}{2}$$

$$-\frac{(n-3)}{2}$$

$$\frac{n-5}{2}$$

$$\frac{n-3}{2}$$

$$\frac{n-1}{2}$$

$$\mu'_1 = 0, \sum x = 0$$

or

$$\mu'_{2k+1} = 0 \quad \sum x^{2k+1} = 0 \quad (k \in \mathbb{Z})$$

For this case the moments of odd order vanish and the moments of even order are

$$\mu_2 = \frac{n^2-1}{12} = \mu'_2 - (\mu'_1)^2$$

$$\sum x^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\bar{x} = \frac{n(n+1)}{2}$$

$$\mu_4 = \mu_2 \left(\frac{3n^2-7}{20} \right)$$

$$\mu_6 = \mu_2 (3n^4 - 18n^2 + 31) / 112$$

By substituting these values in $\phi_k(x)$

$$\phi_k(x) = \frac{1}{|M_{k-1}|} \begin{vmatrix} \mu'_0 & \mu'_1 & \dots & \mu'_k \\ \mu'_1 & \mu'_2 & \dots & \mu'_{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu'_{k-1} & \mu'_k & \dots & \mu'_{2k-1} \\ 1 & x & \dots & x^k \end{vmatrix}$$

We have the Polynomials, when regression is equally spaced which are as follows

$$\phi_0(x) = 1$$

$$\phi_1(x) = x = (x - \bar{x})$$

$$\phi_2(x) = x^2 - \frac{(n^2-1)}{12} = (x - \bar{x})^2 - \frac{(n^2-1)}{12}$$

$$\begin{aligned} \phi_3(x) &= x^3 - \frac{(3n^2-7)}{20} x \\ &= (x - \bar{x})^3 - \frac{(3n^2-7)}{20} (x - \bar{x}) \end{aligned}$$

$$\begin{aligned} \phi_4(x) &= x^4 - \frac{(3n^2-13)}{14} x^2 + \frac{3(n^2-1)(n^2-4)}{560} \\ &= (x - \bar{x})^4 - \frac{(3n^2-13)}{14} (x - \bar{x})^2 + \frac{3(n^2-1)(n^2-4)}{560} \end{aligned}$$

$$\phi_5(x) = x^5 - \frac{5}{18} (n^2-7) x^3 + \frac{1}{1008} (15n^4 - 230n^2 + 407) x$$

$$\phi_6(x) = x^6 - \frac{5}{44} (3n^2-31) x^4 + \frac{1}{176} (5n^4 - 110n^2 + 329) x^2$$

$$- \frac{5}{14874} (n^2-1)(n^2-9)(n^2-25)$$

There exists a recursion relationship for equally spaced x values

$$\phi_{z+1} = \phi_z - \frac{z^2(n^2 - z^2)}{4(z^2 - 1)} \phi_{z-1} \quad \text{for all } z \geq 1$$

By assumption $\phi_0 = 1$, $\phi_1 = x - \bar{x}$

For $z=1$

$$\phi_2 = \phi_1^2 - \frac{(n^2 - 1)}{12} \phi_0 = (x - \bar{x})^2 - \frac{(n^2 - 1)}{12}$$

The tables of orthogonal Polynomials are constructed for different degree Polynomials and for different sample sizes.

Fisher has computed the values upto 5th degree Polynomial for

$$n = 3, 4, \dots, 75.$$

Allan has constructed orthogonal Polynomials $\phi_c(x)$ for $c = 7, 8, 9, 10$.

In the tables of Fisher and Yates values of orthogonal polynomials are given in integral form such that

$$\phi_c(x) = \lambda \cdot \phi_c(x)$$

where λ is some real number which convert the value of $\phi_c(x)$ into integer.

Only the lower half values are given, upper half values are obtained by symmetry if degree of Polynomial is even and skew symmetric if degree of Polynomial is odd.

i.e. for $\phi_1, \phi_3, \phi_5, \dots$, we change the sign

for $\phi_2, \phi_4, \dots$, we will not change the sign.

Example; Fit an appropriate degree of Polynomial for the following data using the orthogonal Polynomials.

Sol;

$$n = 12$$

x	y	ϕ_1	ϕ_2	ϕ_3	ϕ_4	ϕ_5
1	3.0	-11	55	-33	33	-33
2	9.1	-9	25	3	-27	57
3	13.0	-7	1	21	-33	21
4	14.6	-5	-17	25	-13	-29
5	16.7	-3	-29	19	12	-44
6	21.9	-1	-35	7	28	-20
7	27.1	1	-35	-7	28	20
8	25.9	3	-29	-19	12	44
9	25.1	5	-17	-25	-13	29
10	25.6	7	1	-21	-33	-21
11	28.0	9	25	-3	-27	-57
12	26.6	11	55	33	-33	33
λ	2	3	$2/3$	$7/24$	$3/20$	
$\sum \phi_i^2$	572	120/2	5148	8008	15912	

$$\sum y_i^2 = 5415.23, \sum y_i = 236.7, \bar{x} = 6.5$$

$$\sum \phi_1 y = 602.1, \sum \phi_2 y = -1025.7$$

$$\sum \phi_3 y = -19.5, \sum \phi_4 y = 71.7$$

$$\sum \phi_5 y = 246.9$$

$$\begin{aligned} TSS &= \sum y_i^2 - \frac{(\sum y_i)^2}{n} \\ &= 5415.23 - \frac{(236.7)^2}{12} \\ &= 746.32 \end{aligned}$$

$$\hat{\beta} = (\phi' \phi)^{-1} \phi' y$$

$$\hat{\beta}_1 = \frac{\sum \phi_1 Y}{\sum \phi_1^2} = \frac{\lambda \sum \phi_1' Y}{\sum \phi_1^2}$$

$$\hat{\beta}_1 = \frac{\lambda_1 \sum \phi_1' Y}{\sum \phi_1^2}$$

$$\hat{\beta}_1 = \frac{2(602.1)}{572} = 2.105$$

$$\hat{\beta}_2 = \frac{\lambda_2 \sum \phi_2' Y}{\sum \phi_2^2} = \frac{3(-1025.7)}{12012}$$

$$\hat{\beta}_2 = -0.2562$$

$$\hat{\beta}_3 = \frac{\lambda_3 \sum \phi_3' Y}{\sum \phi_3^2} = \frac{2/3(-19.5)}{5148}$$

$$\hat{\beta}_3 = -0.0025$$

$$\hat{\beta}_4 = \frac{\lambda_4 \sum \phi_4' Y}{\sum \phi_4^2} = \frac{7/24(71.7)}{8008} = 0.0026$$

$$\begin{aligned} \text{S.S. due to } \hat{\beta}_1 &= \hat{\beta}_1 \sum \phi_1 Y = \hat{\beta}_1 \frac{1}{\lambda_1} \sum \phi_1' Y & \text{or } \hat{\beta}_1^2 \sum \phi_1^2 \\ &= \frac{2.105}{2} (602.1) = 633.71 \end{aligned}$$

$$\begin{aligned} \text{S.S. due to } \hat{\beta}_2 &= \hat{\beta}_2 \frac{1}{\lambda_2} \sum \phi_2' Y \\ &= \frac{-0.2562}{3} (-1025.7) = 87.5948 \end{aligned}$$

$$\begin{aligned} \text{S.S. due to } \hat{\beta}_3 &= \hat{\beta}_3 \frac{1}{\lambda_3} \sum \phi_3' Y \\ &= \frac{-0.0025}{2/3} (-19.5) = 0.07313 \end{aligned}$$

$$\begin{aligned} \text{S.S. due to } \hat{\beta}_4 &= \hat{\beta}_4 \frac{1}{\lambda_4} \sum \phi_4' Y \\ &= \frac{0.0026}{7/24} (71.7) = 0.6392 \end{aligned}$$

ANOVA

S.O.V	d.f	S.S	M.S	F _{cal}	F _{Tab} $\alpha=0.05$	Conclusion
Total	11	746.32				
Linear Reg	1	633.71	633.71	56.27	$F_{(1,10)} = 4.97$	Significant
Residual	10	112.61	11.261			
2nd degree	1	87.5948	87.5948	31.51	$F_{(1,9)} = 5.12$	Significant
Residual	9	25.0152	2.7795			
3rd degree	1	0.07313	0.07313	0.023	$F_{(1,8)} = 5.32$	Insignificant
Residual	8	24.9421	3.1178			
4th degree	1	0.6392	0.6392	0.184	$F_{(1,7)} = 5.59$	Insignificant
Residual	7	24.3029	3.4718			

Because ~~2nd~~ third and fourth degree regression are insignificant, therefore 2nd degree regression is the best fit.

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \phi_1(x) + \hat{\beta}_2 \phi_2(x)$$

$$\hat{\beta}_0 = \bar{Y} = 19.725$$

$$\hat{Y} = 19.725 + 2.105(x - \bar{X}) + (-0.2562) \left[(x - \bar{X})^2 - \frac{(n^2 - 1)}{12} \right]$$

$$\begin{aligned} \hat{Y} &= 19.725 + 2.105X - 13.6825 - 0.2562 \left[X^2 - 13X + 42.25 - 11.92 \right] \\ &= 6.0425 + 2.105X - 0.2562X^2 + 3.3306X - 10.82445 + 3.0539 \end{aligned}$$

$$\hat{Y} = -1.7280 + 5.4356X - 0.2562X^2$$

Example; Fit a second degree equation by using orthogonal polynomials of the following data.

$$X : 1 \quad -2 \quad 2 \quad 3$$

$$Y : -2 \quad 4 \quad -8 \quad -16$$

Solution; Degree of equation is specified, X values are not equally spaced. We assume

$$Y = \beta_0 + \beta_1 \phi_1 + \beta_2 \phi_2 + \epsilon$$

In general

$$\hat{\beta}_k = \begin{vmatrix} \mu'_0 & \mu'_1 & \mu'_2 & \dots & \mu'_k \\ \mu'_1 & \mu'_2 & \mu'_3 & \dots & \mu'_{k+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mu'_{k-1} & \mu'_k & \mu'_{k+1} & \dots & \mu'_{2k-1} \\ \mu'_0 & \mu'_1 & \mu'_2 & \dots & \mu'_k \end{vmatrix} \div \begin{vmatrix} \mu'_0 & \mu'_1 & \mu'_2 & \dots & \mu'_k \\ \mu'_1 & \mu'_2 & \mu'_3 & \dots & \mu'_{k+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mu'_k & \mu'_{k+1} & \mu'_{k+2} & \dots & \mu'_{2k} \end{vmatrix}$$

Where $\mu'_j = \sum X^j$

$$\mu'_0 = n = 4 \quad , \quad \mu'_1 = \sum X = 4$$

$$\mu'_2 = \sum X^2 = 18 \quad , \quad \mu'_3 = \sum X^3 = 28$$

$$\mu'_4 = \sum X^4 = 114$$

Now

$$\hat{\beta}_0 = \bar{Y} = \frac{\sum Y}{n} = \frac{-22}{4} = -5.5 = -11/2$$

$$\hat{\beta}_1 = \begin{vmatrix} \mu'_0 & \mu'_1 \\ \mu'_1 & \mu'_2 \end{vmatrix} \div \begin{vmatrix} \mu'_0 & \mu'_1 \\ \mu'_1 & \mu'_2 \end{vmatrix}$$

where

$$\mu_{ij} = \sum Y X^j \quad \text{are bivariate moments}$$

$$\mu_{10} = \sum Y = -22$$

$$\mu_{11} = \sum YX = -74$$

$$\mu_{12} = \sum YX^2 = -162$$

$$\hat{\beta}_1 = \begin{vmatrix} 4 & 4 \\ -22 & -14 \end{vmatrix} \div \begin{vmatrix} 4 & 4 \\ 4 & 18 \end{vmatrix}$$

$$= (-296 + 88) \div (72 - 16) = \frac{-208}{56}$$

$$\hat{\beta}_1 = \frac{-26}{7}$$

$$\hat{\beta}_2 = \begin{vmatrix} \mu_0 & \mu_1 & \mu_2 \\ \mu_1 & \mu_2 & \mu_3 \\ \mu_0 & \mu_1 & \mu_2 \end{vmatrix} \div \begin{vmatrix} \mu_0' & \mu_1' & \mu_2' \\ \mu_1' & \mu_2' & \mu_3' \\ \mu_2' & \mu_3' & \mu_4' \end{vmatrix}$$

$$= \begin{vmatrix} 4 & 4 & 18 \\ 4 & 18 & 28 \\ -22 & -14 & -162 \end{vmatrix} \div \begin{vmatrix} 4 & 4 & 18 \\ 4 & 18 & 28 \\ 18 & 28 & 114 \end{vmatrix}$$

$$= [4(-2916 + 2072) - 4(-648 + 616) + 18(-296 + 396)] \div$$

$$[4(2052 - 724) - 4(456 - 504) + 18(112 - 324)]$$

$$\hat{\beta}_2 = -1448 \div 1448 = -1$$

$$\phi_1(x) = \begin{vmatrix} \mu_0' & \mu_1' \\ 1 & x \end{vmatrix} \div \mu_0' = \begin{vmatrix} 4 & 4 \\ 1 & x \end{vmatrix} \div 4 = (x-1)$$

$$\phi_2(x) = \begin{vmatrix} \mu_0' & \mu_1' & \mu_2' \\ \mu_1' & \mu_2' & \mu_3' \\ 1 & x & x^2 \end{vmatrix} \div \begin{vmatrix} \mu_0' & \mu_1' \\ \mu_1' & \mu_2' \end{vmatrix}$$

$$= \begin{vmatrix} 4 & 4 & 18 \\ 4 & 18 & 28 \\ 1 & x & x^2 \end{vmatrix} \div \begin{vmatrix} 4 & 4 \\ 4 & 18 \end{vmatrix}$$

$$= [4(18x^2 - 28x) - 4(4x^2 - 28) + 18(4x - 18)] \div (72 - 16)$$

$$\begin{aligned}\phi_2(x) &= (72x^2 - 112x - 16x^2 + 112 + 72x - 324) \div 56 \\ &= (56x^2 - 40x - 212) \div 56\end{aligned}$$

X	Y	$\phi_1 = x-1$	$\phi_2 = (56x^2 - 40x - 212)/56$	$\hat{y} = 2 - 3x - x^2$
1	-2	0	-197/56	-2
-2	4	-3	92/56	4
2	-8	1	-68/56	-8
3	-16	2	172/56	-16

$$\hat{y} = -\frac{11}{2} + \left(-\frac{26}{7}\right) \phi_1 + (-1) \phi_2$$

$$= -\frac{11}{2} - \frac{26}{7}(x-1) - \frac{1}{56}(56x^2 - 40x - 212)$$

$$= -\frac{11}{2} - \frac{26}{7}x + \frac{26}{7} - \frac{1}{56}[56x^2 - 40x - 212]$$

$$= -\frac{11}{2} + \frac{26}{7} - \left(1456x + 392x^2 - 280x - 1484\right) \times \frac{1}{7 \times 56}$$

$$= \frac{-77+52}{14} - \left(1176x + 392x^2 - 1484\right) \frac{1}{(7)(56)}$$

$$= \frac{-25}{14} - (42x + 14x^2 - 53) \frac{1}{14}$$

$$= \frac{1}{14}(28 - 42x - 14x^2)$$

$$\hat{y} = 2 - 3x - x^2$$

Ridge Regression

[When there is linear relationship among the parameters then X is not of full rank. $(X'X)$ is singular then we cannot find $(X'X)^{-1}$ and this is multicollinearity problem. In Ridge regression we change $(X'X)$ as $(X'X + KI)$ adding a quantity on the main diagonal such that $X'X + KI$ becomes non singular.

The L.S estimation of parameters in linear regression model produces unbiased estimates, but when there exists multicollinearity, then we cannot estimate the parameters, however there are estimating procedures which produce estimates that are not unbiased, such biased estimators may have a very small variance. Hence we may combine estimates which are more reliable i.e. on the average they may be nearer the population parameter value than the minimum variance unbiased estimate.

(Ridge regression is a procedure for obtaining bias estimates of regression coefficients which is particularly useful for situations where the variances of the regression coefficients are inflated by the existence of multicollinearity. In Ridge regression a reduction in the variance inflation factor is "bought" by introducing a bias)

The L.S.E of a vector of regression coefficients is given as

$$\hat{\beta} = (X'X)^{-1} X'Y$$

(The Ridge regression estimate is provided by the following formula)

$$\hat{\beta}^* = (X'X + KI)^{-1} X'Y$$

Where KI represents a scalar multiplication of the identity matrix by a small positive number K . It means a small positive constant is added to each element on the diagonal of matrix $(X'X)$. This automatically reduces variance inflation factors, since the resulting individual pairwise "correlations" have larger denominators.

A problem now arises: what should be the value of K ? The most frequently proposed method is to provide ridge

regression estimates for a sequence of values of k , plot the resulting coefficients against k (this is known as a "ridge plot"), and pick the k -values where the plot seems to "settle down".

Consider

$$\beta^* = (X'X)^{-1} X'Y \quad \text{--- (1)}$$

$$= (X'X + KI)^{-1} X'Y$$

$$= (X'X + KI)^{-1} X' (X\beta + \epsilon)$$

$$E(\beta^*) = (X'X + KI)^{-1} X'X\beta + (X'X + KI)^{-1} X'\epsilon$$

$$E(\beta^*) = (X'X + KI)^{-1} X'X\beta \quad \text{--- (2)}$$

$$\checkmark E(\beta^*) \neq \beta$$

Hence β^* is a biased estimator of β and bias is

$$\text{Bias} = E(\beta^*) - \beta$$

$$= (X'X + KI)^{-1} X'X\beta - \beta$$

$$= [(X'X + KI)^{-1} X'X - I] \beta$$

$$= [(X'X + KI)^{-1} (X'X)^{-1} - I] \beta$$

$$\text{Bias} = [\{(X'X)^{-1} (X'X + KI)^{-1} - I\} \beta] \quad \checkmark$$

Now since

$$\checkmark (I + AB)^{-1} = I - A(I + BA)^{-1}B \quad \text{--- (3)}$$

$$\text{Bias} = [\{I - \frac{I + (X'X)^{-1}KI}{I + \frac{K}{B}A} - I\} \beta]$$

$$= [I - K(I + (X'X)^{-1}K)^{-1}(X'X)^{-1}] - I \beta$$

$$\text{Bias} = -K(I + (X'X)^{-1}K)^{-1}(X'X)^{-1}\beta$$

As bias is negative so Ridge regression estimate is under estimate.

Taking difference of (1) and (2)

$$\beta^* - E(\beta^*) = (X'X + KI)^{-1} X' (X\beta + \epsilon) - (X'X + KI)^{-1} X'X\beta$$

$$\checkmark \text{Var}(\beta^*) = E(\beta^* - E(\beta^*))(\beta^* - E(\beta^*))'$$

$$A^{-1}B^{-1} = (BA)^{-1}$$

$$A = K$$

$$B = (X'X)^{-1}$$

$$\text{Var}(\hat{\beta}^*) = E \left[\begin{aligned} & (X'X + KI)^{-1} X' (X\beta + \epsilon) - (X'X + KI)^{-1} X' X \beta \\ & [(X'X + KI)^{-1} X' (X\beta + \epsilon) - (X'X + KI)^{-1} X' X \beta] \end{aligned} \right]$$

$$= E \left[\begin{aligned} & (X'X + KI)^{-1} X' X \beta + (X'X + KI)^{-1} X' \epsilon - (X'X + KI)^{-1} X' X \beta \\ & [(X'X + KI)^{-1} X' X \beta + (X'X + KI)^{-1} X' \epsilon - (X'X + KI)^{-1} X' X \beta] \end{aligned} \right]$$

$$= E \left[(X'X + KI)^{-1} X' \epsilon \epsilon' X (X'X + KI)^{-1} \right]$$

$$= (X'X + KI)^{-1} X' \sigma^2 X (X'X + KI)^{-1}$$

$$= (X'X + KI)^{-1} X' X (X'X + KI)^{-1} \sigma^2$$

$$\underline{\text{Var}(\hat{\beta})} - \underline{\text{Var}(\hat{\beta}^*)} = (X'X)^{-1} \sigma^2 - \sigma^2 (X'X + KI)^{-1} X' X (X'X + KI)^{-1}$$

$$(I + AB)^{-1} = I - A(I + BA)^{-1}B \quad \text{--- (A)}$$

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